

TOPO-BATHY LIDAR & IMAGERY SURVEY REPORT

Efate, Tanna & Malekula
Vanuatu

CONTRACT No.: SPC 20-293

PREPARED FOR:
THE PACIFIC COMMUNITY (SPC)

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1. INTRODUCTION

Woolpert Inc. was contracted by the The Pacific Community (SPC) to provide airborne topographic and bathymetric lidar acquisition and processing services, along with aerial imagery, for selected areas within Vanuatu, specifically the islands of Efate, Tanna, and Malekula.

All data were acquired using Woolpert's Leica HawkEye 4X system to meet the requirements of the project.

Details of the survey, data processing, QC and product creation are provided in detail within this report.

1.1. SURVEY AREA

The survey extents provided by SPC for Vanuatu covered approximately 1958 square kilometers. Survey areas covered the islands of Efate, Tanna and Malkeula. At the beginning of the project, Woolpert worked with the SPC to adjust the boundaries provided in the statement of work to target areas where good data could most likely be obtained.

The project area was split into survey blocks, allowing flight lines to be planned in the most efficient manner. Lidar data were collected to support 100% coverage.

Table 1: Survey Blocks

Area	Area (sqkm)	Survey Block(s)
Efate	795	BL15, BL16, BL17, BL18, BL19, BL20, BL21, BL22
Tanna	265	BL01, BL02, BL03
Malekula	898	BL06, BL07, BL08, BL09, BL10, BL12



Figure 1: Efate Survey Area

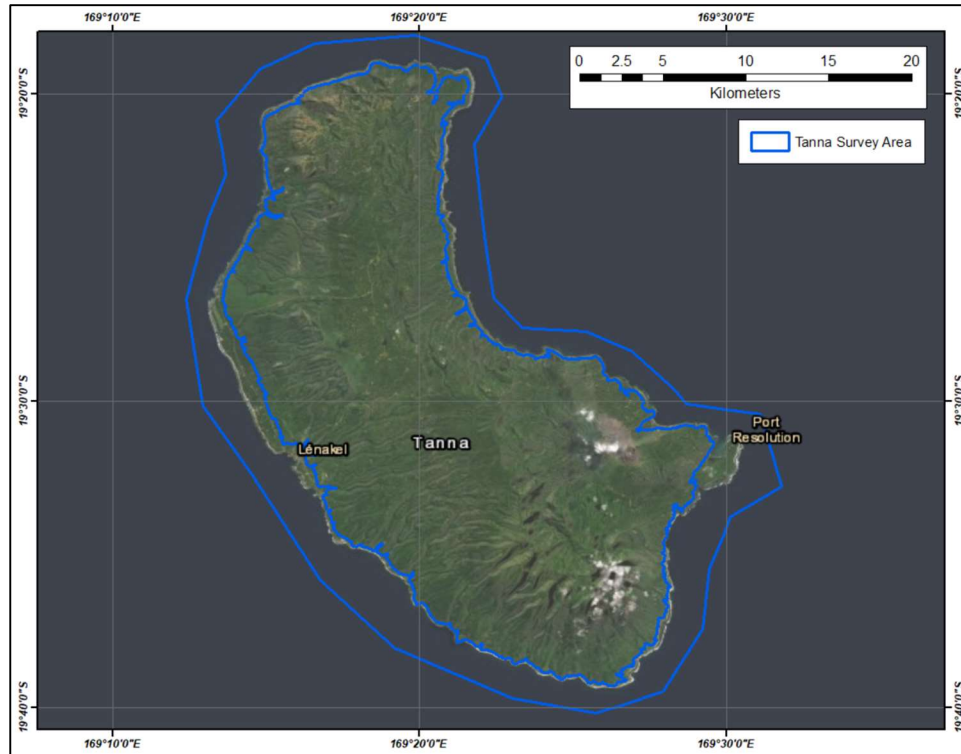


Figure 2: Tanna Survey Area



Figure 3: Malekula Survey Area

2. LIDAR DATA ACQUISITION

All lidar data were acquired using a Chiroptera 4X (CH4X) sensor, with an additional Leica 40kHz deep bathymetric channel. The combination of these sensors is referred to as a Leica HawkEye 4X (HE4X) system. The CH4X sensor was mounted in a Leica PAV100 gyro-stabilized mount integrated with a NovAtel SPAN GNSS and LCI-100C IMU. The deep channel sensor head was mounted over a second hatch, with an additional IMU. Real time navigation and GNSS/IMU data logging was provided by Leica FlightPro software. Lidar data were logged on the Airborne Hydrography, AB (AHAB) operator console.

The aircraft was integrated with the sensor prior to this project. The aircraft arrived in Port Vila, Vanuatu on 06 November 2021, where the crew underwent a 2-week quarantine period. A calibration flight was conducted at Port Vila International Airport (PVI/NVTV) on 28 November 2021. Project data were acquired over 46 flights from 23 November 2021 to 04 March 2022.

2.1. MOBILIZATION

The HE4X sensor was installed in a Reims-Cessna 406 (ZK-XLF) owned and operated by Kiwi Air (Figure 4).



Figure 4: Mobilized Aircraft

2.1.1. AIRCRAFT OFFSET SURVEY

Physical mounting offsets between the GNSS antenna, IMU and gyro-stabilized mount were determined through a combination of manual measurements and iterative processing in NovAtel Inertial Explorer software.

Manual measurements were taken from the GNSS antenna to the reference point on the CH4X sensor and the deep channel sensor head. These measurements are combined with the known offsets between the reference point, IMU, and in the case of the CH4X sensor head, the rotation center of the gyro-stabilized mount. This preliminary value was then used to seed the post-processing software which, through an iterative computation, uses the dynamic accelerations and rotations during flight to refine the offsets. Once the solution converges, the final offsets are entered into the flight management software and used in subsequent post-processing of the GNSS/IMU data for final trajectories.

Final offsets, shown in the Leica reference frame, are presented in Table 2.

Table 2: Aircraft Offsets

Sensor Head	Lever Arm	X (forward)	Y (right)	Z (down)
CHII (Topo and Shallow Channel)	Reference to GNSS Antenna L1 Phase Center	0.604 m	-0.056 m	-1.226 m
	Reference to IMU	-0.003 m	-0.006 m	-0.296 m
	Reference to IMU Rotation	0 °	0 °	90 °
Deep Channel	IMU to GNSS Antenna L1 Phase Center	1.234	-0.134	-0.825
	Reference to IMU Rotation	0°	90°	-90°

2.2. CALIBRATION

Field calibration of the HE4X system is carried out to eliminate systematic errors by calculating corrections for boresight errors, scanner angle errors, remaining IMU angle errors and any necessary internal timing errors. To verify or compute the field calibration, the following lines are flown (Figure 5):

- 2 x Line A over mixed terrain with flat or gentle slopes and features such as peaked roof buildings (1 x each direction)
- 1 x Line B offset +50% from Line A in one direction
- 1 x Line C offset -50% from Line A in the same direction as Line B
- 2 x Line D orthogonal to previous lines (1 x each direction)

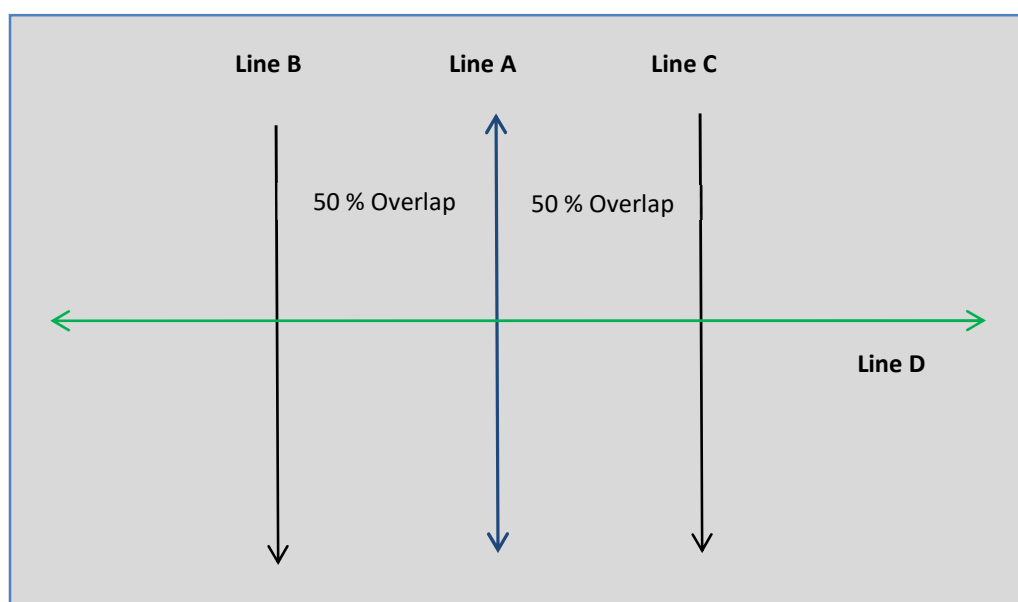


Figure 5: Schematic of HE4X Calibration Lines

A set of calibration lines were acquired at 800m, 500m, and 400m altitude. All sets of lines are used to calibrate and verify the topographic lidar, while the 500m and 400m lines are used for the bathymetric lidar.

Calibration values are calculated using the automatic calibration routine within the Leica Lidar Survey Studio (LSS) software. This utility first identifies patches or areas of gentle slope within the overlap region of all the lines to use for calibration. Patch selection prevents areas of vegetation, side of cars or buildings, from being used in the calibration process. Next, the utility compares the front side and back side of the elliptical scan within the same line, as well as comparing all lines to each other, to identify suitable calibration parameters such that data within the patches match. The procedure is iterative and continues until the best possible solution is computed.

Calibration for each channel (topo, shallow and deep) is done independently. Topo channel calibration was computed using the 500m altitude lines. The 800m and 400m lines were then used for verification. Calibration of

the shallow channel and deep channel were computed using 500m altitude data. Any lower altitude data were used for verification.

At each step of the calibration process, quality assurance is conducted to ensure values being calculated are valid. This is done using the Leica LSS Quality Control Utility. Two types of checks are done; firstly, the front scan is compared to the back scan for every line. Secondly, a single line is chosen as a baseline and is compared to every other line. We would expect the average errors from both of these checks to be small. In addition, the data is visually reviewed. In particular, features are studied to ensure lines from different directions show structures in the same position, in other words, verifying horizontal accuracy is maintained. These tests all provide assurance of relative accuracy.

Results from the calibration verification checks are provided in Table 3 below. Results are good and indicate that calibration was successful. Calibration values computed were used for the entire project.

Table 3: Calibration QA Results

Test		Topo 800m	Topo 500m	Topo 400m	Shallow 500m	Shallow 400m	Deep 500m	Deep 400m
Front to Back Scan Comparison	Average Error (m)	-0.0063	-0.0002	0.0025	0.0002	-0.0019	0.0046	0.0120
	Std. Dev. of Error	0.0006	0.0007	0.0022	0.0008	0.0014	0.0017	0.0025
Line to Line Comparison	Average Error (m)	-0.0116	0.0003	0.0057	-0.0002	0.0042	0.0011	0.0010
	Std. Dev. of Error	0.0042	0.0012	0.0025	0.0020	0.0030	0.0040	0.0045

Woolpert acquired a detailed set of ground truth data over the apron at the Port Vila International Airport. The ground truth was acquired using Trimble R10 GNSS receivers, Real Time Kinematic (RTK) and Post-Processed Kinematic (PPK) survey techniques.

Ground truth is not used within the automatic calibration routine; however, a comparison to the lidar data was used to verify absolute accuracy. Results presented in Table 4 show data is well within required accuracy specifications.

Table 4: Calibration Ground Truth Comparisons

	Topo	Shallow	Deep
Average dz (m)	0.0003	-0.0015	0.0262
St Dev (m)	0.0307	0.0094	0.0104

2.3. SURVEY OPERATIONS

For ease of operations and data management, the Vanuatu survey area was split into survey blocks. Actual flight lines flown, including start and end date and unique line ID, are provided in the flight line database included with the project deliverables in SHP format.

A summary of the daily operations is shown in Table 5. Detailed Flight Logs for each day are provided in Appendix A. Operations for the project were based out of Port Vila International (PVI/NVVV).

Twenty-three cross lines were acquired across the during the survey for quality purposes. Crosslines were planned perpendicular to main scheme survey lines and used to verify the relative accuracy of the data where bottom coverage allowed.

Table 5: Summary of Daily Operations

Flight	Engine Time	Air Time	Flown km %		Reflown km %		Comments
2021-11-23A	2:36:00	1:50:00	140.0	1.3%			Efate
2021-11-26A	5:21:00	4:26:00	399.1	3.7%	60.4	0.6%	Efate
2021-11-28B	3:51:00	3:22:00	263.5	2.4%	39.0	0.4%	Calibration, Efate
2021-11-29A	6:26:00	5:57:00	447.4	4.1%	168.4	1.6%	Efate
2021-11-30A	3:14:00	2:53:00	370.3	3.4%	0.0	0.0%	Efate
2021-11-30B	5:43:00	5:18:00	295.5	2.7%	61.3	0.6%	Tanna
2021-12-01A	3:49:00	3:28:00	483.8	4.5%	38.9	0.4%	Efate
2021-12-03A	6:03:00	5:31:00	336.7	3.1%	88.6	0.8%	Tanna
2021-12-04A	5:02:00	4:37:00	376.7	3.5%	82.0	0.8%	Tanna, Efate
2021-12-05A	2:40:00	2:15:00	18.8	0.2%	14.0	0.1%	Efate
2021-12-07A	6:30:00	5:49:00	298.1	2.8%	103.2	1.0%	Efate
2021-12-08A	6:03:00	5:31:00	433.1	4.0%	121.9	1.1%	Tanna
2021-12-15A	6:10:00	5:44:00	512.5	4.7%	97.4	0.9%	Tanna
2021-12-16A	3:12:00	2:48:00	135.7	1.3%	91.5	0.8%	Efate
2021-12-17A	4:13:00	3:43:00	357.3	3.3%	27.9	0.3%	Efate
2021-12-17B	6:00:00	5:34:00	497.4	4.6%			Malekula
2021-12-19A	6:27:00	6:01:00	617.8	5.7%	98.6	0.9%	Efate
2021-12-20A	4:27:00	4:04:00	402.9	3.7%	1.6	0.0%	Malekula
2021-12-21A	5:17:00	4:55:00	431.7	4.0%			Malekula
2021-12-21B	1:38:00	0:49:00	72.8	0.7%			Efate
2021-12-26A	5:24:00	5:00:00	608.5	5.6%			Malekula
2021-12-27A	3:31:00	3:09:00	311.0	2.9%			Malekula
2021-12-27B	5:07:00	4:40:00	444.0	4.1%	46.9	0.4%	Malekula
2021-12-28A	7:00:00	5:50:00	750.4	6.9%	0.9	0.0%	Efate
2021-12-29A	4:36:00	4:10:00	362.1	3.3%	118.4	1.1%	Malekula
2022-01-03A	5:01:00	4:35:00	219.8	2.0%	15.2	0.1%	Tanna, Efate
2022-01-04A	1:26:00	1:03:00	107.1	1.0%			Efate
2022-01-04B	2:10:00	1:41:00	7.2	0.1%	39.6	0.4%	Malekula
2022-01-06A	5:19:00	4:46:00	206.1	1.9%	29.9	0.3%	Tanna
2022-01-07A	5:16:00	4:52:00	300.2	2.8%	1.6	0.0%	Malekula
2022-01-07B	4:18:00	3:54:00	150.9	1.4%	11.7	0.1%	Efate
2022-01-08A	2:26:00	2:02:00	86.6	0.8%	20.6	0.2%	Efate
2022-01-09A	5:17:00	4:53:00	564.0	5.2%	31.0	0.3%	Malekula
2022-01-10A	3:40:00	3:16:00	292.9	2.7%			Malekula
2022-01-10B	3:52:00	3:21:00	322.5	3.0%	79.2	0.7%	Malekula
2022-01-11A	4:27:00	4:01:00	84.9	0.8%	80.3	0.7%	Tanna

Flight	Engine Time	Air Time	Flown km %		Reflown km %		Comments
2022-01-13A	3:19:00	2:28:00	31.7	0.3%	32.8	0.3%	Efate
2022-01-16A	2:53:00	2:22:00	40.6	0.4%	2.3	0.0%	Tanna, Efate
2022-01-21A	6:00:00	5:20:00	228.1	2.1%	6.3	0.1%	Malekula
2022-01-21B	4:14:00	3:47:00	182.1	1.7%	34.3	0.3%	Close-Out Calibration, Tanna, Efate
2022-01-22A	4:58:00	4:34:00	240.0	2.2%			Malekula
2022-02-16A	4:43:00	4:18:00	232.6	2.1%	20.7	0.2%	Malekula, Efate
2022-02-17A	3:32:00	3:10:00	233.1	2.2%	25.8	0.2%	Malekula, Efate
2022-02-17B	4:15:00	3:53:00	395.8	3.7%			Malekula
2022-02-22A	2:48:00	2:20:00	40.0	0.4%	14.3	0.1%	Malekula
2022-03-04A	2:18:00	1:46:00	21.7	0.2%	3.7	0.0%	Malekula
TOTAL	207:45:00	183:48:00	13355	123.4%	1710	15.8%	

2.3.1. THE HAWKEYE 4X

All lidar data were acquired using a HE4X sensor. The HE4X is a latest generation topographic and bathymetric lidar sensor. The system provides denser data than previous traditional bathymetric lidar systems. It is unique in its ability to acquire bathymetric lidar, topographic lidar and 4-band digital camera imagery simultaneously.

The HE4X provided 500 kHz topographic data, an effective 140 kHz shallow bathymetric data and 40 kHz deep bathymetric data. 4-band 80 MP digital camera imagery was also collected simultaneously with the sensor's RCD-30 camera.

The bathymetric and topographic lasers are independent and do not share an optical chain or receivers, so they are optimized for their specific function. As with any bathymetric lidar, maximum depth penetration is a function of water clarity and seabed reflectivity. The HE4X is designed to penetrate to 3 times the secchi depth. This is also represented as $D_{max} = 4/K$, where K is the diffuse attenuation coefficient, and assuming K is between 0.1 and 0.3, a normal sea state and 15% seabed reflectance.

Both the topographic and bathymetric sub-systems use a palmer scanner to produce an elliptical scan pattern of laser points with a degree of incidence ranging from +/-14° (front and back) to +/-20° (sides), providing a 40° field of view. This has the benefit of providing multiple look angles on a single pass and helps to eliminate shadowing effects. This can be of particular use in urban areas, where all sides of a building are illuminated, or for bathymetric features such as the sides of narrow water channels, or features on the seafloor such as smaller objects and wrecks. It also assists with penetration in the surf zone where the back scan passes the same ground location a couple of seconds after the front scan, allowing the areas of whitewater to shift.

The bathymetric laser is a diode pumped class 4 laser which operates in the green spectrum. Full waveform data is acquired for every pulse. The topographic laser operates in the infra-red spectrum at 1064nm. Up to 4 returns per pulse are acquired from each lidar.

For this project, the flight parameters shown in Table 6 were used to provide 100% coverage. Flight parameters used exceed the requirements for the survey to meet IHO Order 1B.

Table 6: HE4X Survey Flight Parameters

	Topo-Bathy Flight Lines	Topo Only FlightLines
Topo PRF (kHz)	500	440
Topo Points per m ²	>12	>10
Shallow Bathy PRF (kHz)	140	
Shallow Bathy Points per m ²	4.5	1.3
Deep Bathy PRF (kHz)	40	N/A
Deep Bathy Points per m ²	1.6	N/A
Swath Width (m)	349	450
Flight Line Sidelap (%)	20	
Altitude (m)	480	620
Survey Speed (knots)	125	

During acquisition, flight lines are shown on a pilot's display, and the aircraft is controlled by the pilot at all times. The HE4X system includes a NovAtel SPAN GNSS system with an LCI-100C IMU for aircraft position and orientation. One IMU is in the main Chiroptera sensor head, which includes the topo channel, shallow channel and RCD30 camera. Data from this IMU are also used in real-time by the PAV100 gyro-stabilized mount to compensate for deviations in pitch and roll. A second IMU is contained within the deep channel sensor head, installed over a second hatch in the aircraft. This head does not include a gyro-stabilized mount. Aircraft bank angles were restricted to 25° to avoid any potential GNSS dropouts. No flights were planned if the PDOP was expected to go above 3.0.

Data were monitored for quality during acquisition using the Operators Console running on the AHAB collection computer. The operator monitored system status of the scanners and receivers, waveforms, camera images, data coverage, flight lines and the health of the navigation system.

All data were recorded to a removable solid-state hard disk. At the end of the flight the hard disk was removed and taken to the field office where data were copied onto backup disks for transmittal back to the main processing office. Preliminary data were reviewed daily in the field for quality and coverage.

2.3.2. POSITIONING

Position and orientation data were acquired in the aircraft using a NovAtel SPAN with LCI-100C IMU. All data were post-processed using NovAtel Inertial Explorer software to provide a tightly coupled position and orientation trajectory solution.

Efate data were processed using single-base Post Processed Kinematic (PPK) techniques. Woolpert occupied an existing control point (AAMPri) north of the Port Vila International Airport terminal apron established by AAM in a 2012. Final updated coordinates for the GNSS base station were computed by CSRS-PPP/AUSPOS with daily results averaged to finalize the coordinates relative to the ITRF2014 Epoch 2021.9 datum.

Table 7: AAMPri Coordinates in ITRF2014 (2021.9)

GNSS Base Station	Latitude	Longitude	Height (m)
AAMPri	17° 41' 59.79551" S	168° 19' 07.74891" E	82.680

Tanna and Malekula data were acquired using Precise Point Positioning (PPP) techniques and no GNSS base stations were used. Logs for the base station and trajectory processing are provided in Appendix B

3. SURVEY CHECK POINTS

For quality control purposes a daily QC flight line was acquired over the same location every day, when flying from Port Vila. Topographic control points data were collected with a Trimble R-10 GNSS receiver using RTK GNSS techniques over the area covered by the daily lidar QC line.

Further discussion of the processing and results will be presented in the Final Project Report.

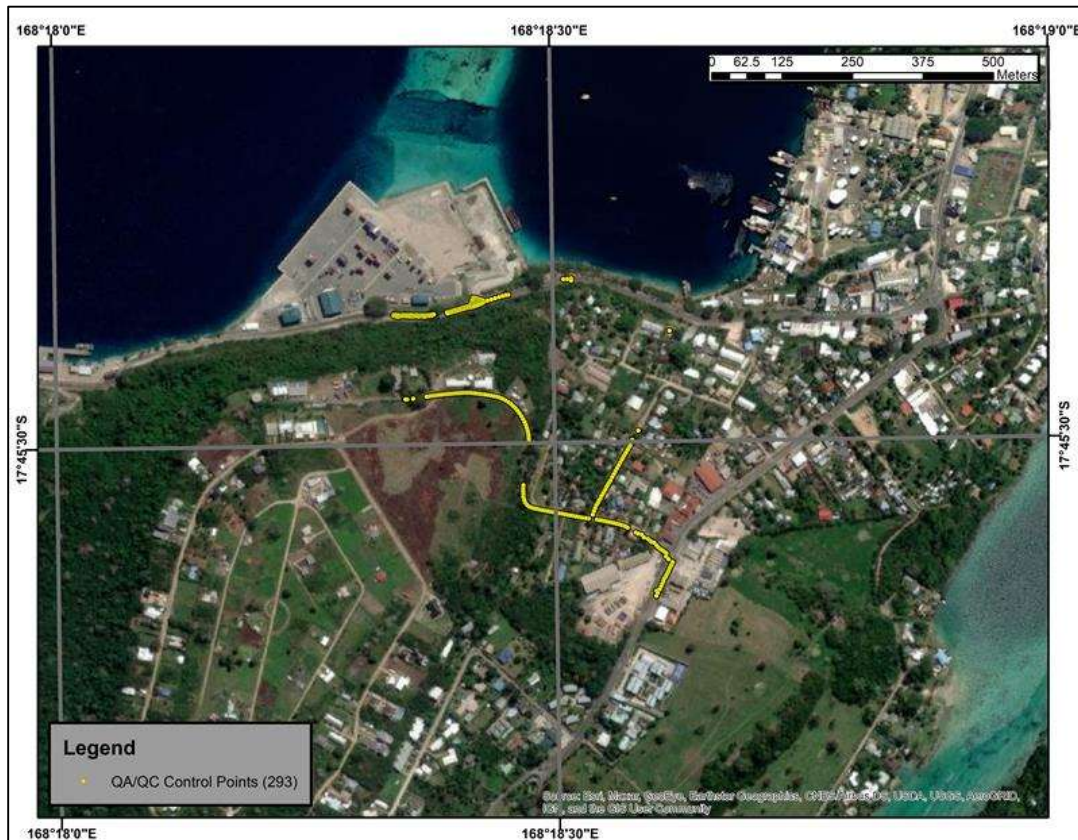


Figure 6: Topographic Ground Control Points in Daily QC Line Location

APPENDIX A : FLIGHT LOGS

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL17, BL18, BL19, BL20	DATE:	23 November 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Efate	CLOUDS:	Clouds @ 2000ft
BASE STATION:	AAMPri	WIND:	10-15 kts @ 80
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-03
ENGINE START:	3:16	ENGINE OFF:	5:52
TAKEOFF:	3:53	LANDING:	5:43
		ENGINE TIME:	02:36
		AIR TIME	01:50

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		03:42:00				Static start out AAM Pri base
		03:53:00				Takeoff
		04:14:25				DS: BL18_20211123_041425
000_FL61	1861	04:14:25	04:18:00	480	500 13	
001_FL70	1897	04:21:17	04:23:12	480	500 13	
002_FL69	1896	04:27:20	04:29:51	480	500 13	
003_FL68	1895	04:35:46	04:38:12	480	500 13	
		04:41:50				DS: BL17_20211123_044150
000_FL39	1795	04:41:50	04:44:36	480	500 13	
		04:53:11				DS: BL20_20211123_045311
000_FL40	2095	04:53:11	04:56:20	480	500 13	
		05:01:45				DS: BL19_20211123_050145
000_FL52	1995	05:01:45	05:05:35	480	500 13	
001_FL14	1914	05:09:21	05:13:07	480	500 13	
002_FL15	1915	05:14:55	05:18:40	480	500 13	
003_FL25	1925	05:21:55	05:29:16	480	500 13	
004_FL26	1926	05:30:52	05:37:48	480	500 13	Don't use approx. last 3rd of line
		05:43:00				Landing
		05:45:00				Static close out AAM Pri base

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / 500m, 800m, BL19	DATE:	26 November 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Efate, TongaVanuatu	CLOUDS:	Clouds @ 2000ft
BASE STATION:	AAMPri	WIND:	5-10 kts @ 80
LIDAR DRIVE:	HE4X-07	RCD DRIVE:	RCD-05
ENGINE START:	1:24	ENGINE OFF:	6:45
TAKEOFF:	2:08	LANDING:	6:34
		ENGINE TIME:	05:21
		AIR TIME	04:26

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		02:00:00				Static start out AAM Pri base
		02:08:00				Takeoff
		02:14:00				Start cal at 800m
		02:14:00				Start Block 19
		02:15:32				DS: 800m_20211126_021532
000_FL1	6381	02:15:32	02:17:45	800	330 30	BAD: Didn't finish cal
001_FL2	6382	02:20:14	02:22:26	800	330 30	BAD: Didn't finish cal
002_FL3	6383	02:25:18	02:27:13	800	330 30	BAD: Didn't finish cal
003_FL4	6384	02:29:43	02:31:34	800	330 30	BAD: Didn't finish cal
004_FL5	6385	02:33:29	02:35:27	800	330 30	BAD: Didn't finish cal
		02:50:00				DS: 800m_20211126_025000
000_FL1	6381	02:50:00	02:52:27	800	330 30	BAD: Didn't finish cal
001_FL1	6381	03:00:40	03:02:29	800	330 30	BAD: Didn't finish cal
		03:06:51				DS: 500m_20211126_030651
000_FL1	6351	03:06:51	03:08:40	490	500 13	BAD: Didn't finish cal
001_FL2	6352	03:10:46	03:12:40	490	500 13	BAD: Didn't finish cal
002_FL3	6353	03:21:14	03:22:43	490	500 13	BAD: Didn't finish cal
		03:25:00				Aborted cal due to weather
		03:26:00				Reboot complete system
		03:39:49				DS: BL19_20211126_033949
000_FL16	1916	03:39:49	03:45:29	480	500 13	
001_FL17	1917	03:47:49	03:53:21	480	500 13	
002_FL18	1918	03:55:41	04:01:25	480	500 13	
003_FL19	1919	04:03:37	04:10:03	480	500 13	
004_FL20	1920	04:12:49	04:19:30	480	500 13	
005_FL21	1921	04:22:04	04:28:39	480	500 13	
006_FL22	1922	04:30:54	04:37:43	480	500 13	
007_FL23	1923	04:39:56	04:46:50	480	500 13	
008_FL24	1924	04:49:21	04:56:22	480	500 13	
009_FL26	1926	04:58:40	05:05:32	480	500 13	Refly from 21Nov21
010_FL27	1927	05:08:24	05:15:43	480	500 13	

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / 500m, 800m, BL19	DATE:	26 November 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Efate, TongaVanuatu	CLOUDS:	Clouds @ 2000ft
BASE STATION:	AAMPri	WIND:	5-10 kts @ 80
LIDAR DRIVE:	HE4X-07	RCD DRIVE:	RCD-05
ENGINE START:	1:24	ENGINE OFF:	6:45
TAKEOFF:	2:08	LANDING:	6:34
		ENGINE TIME:	05:21
		AIR TIME	04:26

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
011_FL28	1928	05:17:54	05:24:47	480	500 13	
012_FL29	1929	05:27:29	05:34:51	480	500 13	
013_FL30	1930	05:37:33	05:44:25	480	500 13	
014_FL31	1931	05:47:19	05:48:14	480	500 13	BAD: Plane too high. Reflew
015_FL31	1931	05:51:46	05:59:00	480	500 13	
016_FL32	1932	06:01:40	06:08:47	480	500 13	
017_FL33	1933	06:11:38	06:19:14	480	500 13	
018_FL34	1934	06:21:46	06:29:16	480	500 13	
		06:30:00				RTB
		06:34:00				Landing
		06:40:00				Static close out AAM Pri base

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / 600m	DATE:	27 November 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	TongaVanuatu	CLOUDS:	Clouds @ 1800ft
BASE STATION:	AAMPri	WIND:	5-10 kts @ 240
LIDAR DRIVE:	HE4X-07	RCD DRIVE:	RCD-04
ENGINE START:	1:18	ENGINE OFF:	3:11
TAKEOFF:	1:53	LANDING:	2:59
		ENGINE TIME:	01:53
		AIR TIME	01:06

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		01:42:00				Static start out AAM Pri base
		01:53:00				Takeoff
		02:14:00				Start cal at 800m
		02:15:39				DS: 600m_20211127_021539
000_FL1	6361	02:15:39	02:15:45	600	440 13	BAD: Didn't finish cal
001_FL2	6362	02:19:44	02:21:29	600	440 13	BAD: Didn't finish cal
002_FL1	6361	02:28:27	02:28:42	600	440 13	BAD: Accidental laser trigger
003_FL1	6361	02:28:51	02:28:54	600	440 13	BAD: Accidental laser trigger
004_FL1	6361	02:29:31	02:29:34	600	440 13	BAD: Accidental laser trigger
005_FL1	6361	02:32:26	02:34:06	600	440 13	BAD: Accidental laser trigger
006_FL3	6363	02:37:29	02:39:02	600	440 13	BAD: Didn't finish cal
007_FL5	6365	02:42:59	02:44:30	600	440 13	BAD: Didn't finish cal
008_FL6	6366	02:48:33	02:50:07	600	440 13	BAD: Didn't finish cal
009_FL4	6364	02:53:23	02:54:53	600	440 13	BAD: Didn't finish cal
		02:56:00				Aborted cal due to weather and RTB
		02:59:00				Landing
		03:01:00				Static close out AAM Pri base

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / 400m, 500m, 800m, QC61, QC62	DATE:	28 November 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	TongaVanuatu	CLOUDS:	Clouds @ 3000ft
BASE STATION:	AAMPri	WIND:	5-10 kts @ 60
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-03
ENGINE START:	3:02	ENGINE OFF:	6:22
TAKEOFF:	3:18	LANDING:	6:14
		ENGINE TIME:	03:20
		AIR TIME	02:56

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		01:42:00				Static start out AAM Pri base
		02:14:00				Start cal at 800m
		03:18:00				Takeoff
		03:22:14				DS: 800m_20211128_032214
000_FL6	6386	03:22:14	03:22:47	800	330	BAD: Chiller not on
		03:28:00				Reboot due to temps and chiller was off a
		03:31:22				DS: 800m_20211128_033122
000_FL5	6385	03:31:22	03:33:11	800	330	30
001_FL5	6385	03:35:27	03:37:15	800	330	30 BAD: Flew wrong direction
002_FL6	6386	03:39:42	03:41:33	800	330	30
003_FL4	6384	03:45:23	03:47:12	800	330	30
004_FL3	6383	03:49:51	03:51:39	800	330	30
005_FL2	6382	03:55:05	03:57:11	800	330	30
006_FL1	6381	04:01:56	04:02:36	800	330	30 BAD: 1st 1/3 flown low, clouds
007_FL1	6381	04:06:22	04:08:33	800	330	30 BAD: 1st 1/3 flown low, clouds
008_FL1	6381	04:13:02	04:14:04	800	330	30 BAD: Off line after 2/3 flown
009_FL1	6381	04:18:32	04:19:49	800	330	30 BAD: Eye safety triggered
010_FL1	6381	04:23:24	04:25:39	800	330	30 Low at start of line
		04:28:44				DS: 500m_20211128_042844
000_FL1	6351	04:28:44	04:30:35	490	500	13
001_FL2	6352	04:33:09	04:34:59	490	500	13
002_FL3	6353	04:37:44	04:39:17	490	500	13
003_FL4	6354	04:41:51	04:43:22	490	500	13 BAD: Clouds likely present
004_FL5	6355	04:52:52	04:54:27	490	500	13 BAD: Rain at end
005_FL6	6356	04:59:25	05:01:05	490	500	13 Rain at end
006_FL4	6354	05:03:24	05:04:55	490	500	13 BAD: Rain at beg. Use if first doesn't work
		05:09:08				DS: 400m_20211128_050908
000_FL1	6341	05:09:08	05:11:05	400	300	10
001_FL2	6342	05:13:16	05:15:04	400	300	10
002_FL3	6343	05:18:06	05:19:38	400	300	10
003_FL4	6344	05:22:03	05:23:30	400	300	10

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / 400m, 500m, 800m, QC61, QC62	DATE:	28 November 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	TongaVanuatu	CLOUDS:	Clouds @ 3000ft
BASE STATION:	AAMPri	WIND:	5-10 kts @ 60
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-03
ENGINE START:	3:02	ENGINE OFF:	6:22
TAKEOFF:	3:18	LANDING:	6:14
		ENGINE TIME:	03:20
		AIR TIME	02:56

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
004_FL5	6345	05:26:02	05:27:30	400	300 10	
005_FL6	6346	05:31:33	05:33:01	400	300 10	
		05:37:14				DS: 500m_20211128_053714
000_FL4	6354	05:37:14	05:38:42	490	500 13	Low at first
001_FL5	6355	05:41:20	05:42:24	490	500 13	Eye safe at end
		05:55:00				RTB
		05:56:00				Check out of base AAM Pri
		06:00:38				DS: QC61_20211128_060038
000_FL1	6101	06:00:38	06:02:06	480	500 13	BAD: Wrong sensor settings
		06:05:41				DS: QC62_20211128_060541
000_FL1	6201	06:05:41	06:07:31	480	500 13	
		06:09:51				DS: QC61_20211128_060951
000_FL1	6101	06:09:51	06:11:39	480	500 13	
		06:14:00				Landing
		06:15:00				Static close out AAM Pri base

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / 400m, 500m, 800m, BL15, BL20	DATE:	28 November 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.

MISSION ID:	Efate, TongaVanuatu	CLOUDS:	
BASE STATION:	AAMPri	WIND:	@
LIDAR DRIVE:		RCD DRIVE:	

ENGINE START:	20:18	ENGINE OFF:	0:09	ENGINE TIME:	03:51
TAKEOFF:	20:34	LANDING:	23:56	AIR TIME	03:22

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		20:25:00				Static start out AAM Pri base
		20:34:00				Takeoff
		20:36:00				Start cal at 800m
		20:37:53				DS: 800m_20211128_203753
000_FL1	6381	20:37:53	20:39:58	800	330 30	
001_FL2	6382	20:42:53	20:43:08	800	330 30	BAD: Eye safety
002_FL2	6382	20:47:24	20:49:28	800	330 30	
003_FL3	6383	20:52:34	20:54:19	800	330 30	
004_FL4	6384	20:57:10	20:57:40	800	330 30	
005_FL4	6384	21:01:34	21:03:28	800	330 30	
006_FL5	6385	21:05:45	21:07:26	800	330 30	
007_FL6	6386	21:12:25	21:14:06	800	330 30	
		21:17:12				DS: 500m_20211128_211712
000_FL1	6351	21:17:12	21:18:55	490	500 13	
001_FL2	6352	21:21:27	21:23:10	490	500 13	
002_FL3	6353	21:26:02	21:27:26	490	500 13	
003_FL4	6354	21:29:57	21:31:22	490	500 13	
004_FL5	6355	21:33:30	21:34:58	490	500 13	
005_FL6	6356	21:39:49	21:41:14	490	500 13	
		21:44:53				DS: 400m_20211128_214453
000_FL1	6341	21:44:53	21:46:36	400	300 10	
001_FL2	6342	21:48:56	21:50:38	400	300 10	
002_FL3	6343	21:53:42	21:55:02	400	300 10	
003_FL4	6344	21:57:30	21:58:45	400	300 10	
004_FL5	6345	22:00:45	22:02:03	400	300 10	
005_FL6	6346	22:08:18	22:09:39	400	300 10	Terminal apron was wet
		22:12:00				Check out of base AAM Pri
		22:16:24				DS: QC61_20211128_221624
000_FL1	6101	22:16:24	22:17:52	480	500 13	BAD:
001_FL1	6101	22:18:15	22:18:17	480	500 13	BAD:
		22:28:35				DS: BL20_20211128_222835

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / 400m, 500m, 800m, BL15, BL20	DATE:	28 November 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.

MISSION ID:	Efate, TongaVanuatu	CLOUDS:	
BASE STATION:	AAMPri	WIND:	@
LIDAR DRIVE:		RCD DRIVE:	

ENGINE START:	20:18	ENGINE OFF:	0:09	ENGINE TIME:	03:51
TAKEOFF:	20:34	LANDING:	23:56	AIR TIME	03:22

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
000_FL39	2039	22:28:35	22:34:42	480	500 13	
001_FL10	2010	22:38:42	22:44:32	480	500 13	
002_FL11	2011	22:47:17	22:53:43	480	500 13	
003_FL12	2012	22:56:05	22:59:30	480	500 13	
004_FL12	2012	23:00:27	23:02:22	480	500 13	
005_FL9	2009	23:04:59	23:11:11	480	500 13	
		23:14:33				DS: BL15_20211128_231433
000_FL1	1501	23:14:33	23:15:48	480	500 13	
001_FL2	1502	23:18:39	23:20:19	480	500 13	
002_FL3	1503	23:22:47	23:24:38	480	500 13	
003_FL4	1504	23:27:35	23:29:38	480	500 13	
004_FL9	1509	23:31:50	23:33:29	480	500 13	Eye safety at end of line
		23:39:42				DS: BL20_20211128_233942
000_FL12	2012	23:39:42	23:45:37	480	500 13	Eye safety at end of line
		23:54:00				RTB
		23:56:00				Landing
		23:58:00				Static close out AAM Pri base

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL15, BL19, BL20, BL21, QC61	DATE:	29 November 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Efate, TongaVanuatu	CLOUDS:	Clouds @ 2500ft
BASE STATION:	AAMPri	WIND:	10-15 kts @ 60
LIDAR DRIVE:	HE4X-06	RCD DRIVE:	RCD-04
ENGINE START:	19:36	ENGINE OFF:	2:02
TAKEOFF:	19:56	LANDING:	1:53
		ENGINE TIME:	06:26
		AIR TIME	05:57

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		19:47:00				Static start out AAM Pri base
		19:56:00				Takeoff
		19:59:24				DS: QC61_20211129_195924
000_FL1	6101	19:59:24	20:01:09	480	500 13	
		20:10:02				DS: BL21_20211129_201002
000_FL44	2144	20:10:02	20:11:29	620	440 14	
001_FL43	2143	20:14:10	20:15:39	620	440 14	
002_FL42	2142	20:18:02	20:19:39	620	440 14	
003_FL41	2141	20:22:16	20:23:52	620	440 14	
004_FL40	2140	20:26:14	20:27:49	620	440 14	
005_FL39	2139	20:30:38	20:33:24	620	440 14	
006_FL38	2138	20:36:09	20:39:02	620	440 14	
007_FL37	2137	20:41:45	20:44:37	620	440 14	
		20:55:35				DS: BL15_20211129_205535
000_FL11	1511	20:55:35	20:57:47	480	440 14	BAD: Wrong sensor settings
001_FL10	1510	21:00:18	21:02:21	480	440 14	BAD: Wrong sensor settings
002_FL9	1509	21:05:06	21:07:07	480	500 13	
003_FL11	1511	21:09:41	21:11:45	480	500 13	
004_FL10	1510	21:14:15	21:16:20	480	500 13	
005_FL8	1508	21:18:58	21:20:56	480	500 13	
006_FL7	1507	21:23:17	21:25:09	480	500 13	45-50m depths at west end
007_FL6	1506	21:28:10	21:30:03	480	500 13	
008_FL5	1505	21:32:45	21:34:40	480	500 13	
009_FL20	1520	21:37:55	21:38:57	480	500 13	Eye safety at end
010_FL19	1519	21:42:25	21:43:21	480	500 13	Late start
011_FL18	1518	21:46:18	21:47:07	480	500 13	
012_FL18	1518	21:47:19	21:47:55	480	500 13	
013_FL17	1517	21:51:03	21:52:17	480	500 13	Late start
014_FL16	1516	21:55:10	21:57:02	480	500 13	
015_FL15	1515	21:59:36	21:59:46	480	500 13	BAD: Clouds
016_FL15	1515	22:03:16	22:04:12	480	500 13	Eye safety at end

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL15, BL19, BL20, BL21, QC61	DATE:	29 November 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Efate, TongaVanuatu	CLOUDS:	Clouds @ 2500ft
BASE STATION:	AAMPri	WIND:	10-15 kts @ 60
LIDAR DRIVE:	HE4X-06	RCD DRIVE:	RCD-04
ENGINE START:	19:36	ENGINE OFF:	2:02
TAKEOFF:	19:56	LANDING:	1:53
		ENGINE TIME:	06:26
		AIR TIME	05:57

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
017_FL15	1515	22:07:57	22:08:48	480	500 13	
018_FL15	1515	22:08:57	22:09:32	480	500 13	Eye safety at end
019_FL14	1514	22:12:12	22:13:05	480	500 13	BAD: Clouds
020_FL14	1514	22:13:29	22:14:01	480	500 13	BAD: Clouds
		22:15:55				DS: BL20_20211129_221555
000_FL8	2008	22:15:55	22:17:31	480	500 13	BAD: Clouds
001_FL8	2008	22:17:35	22:17:51	480	500 13	BAD: Clouds
002_FL8	2008	22:24:27	22:30:07	480	500 13	
003_FL7	2007	22:32:12	22:38:06	480	500 13	
004_FL6	2006	22:40:05	22:45:55	480	500 13	
005_FL5	2005	22:48:09	22:53:40	480	500 13	
006_FL4	2004	22:55:58	23:01:39	480	500 13	
007_FL3	2003	23:04:06	23:09:08	480	500 13	
008_FL2	2002	23:10:55	23:16:08	480	500 13	
009_FL1	2001	23:19:14	23:24:01	480	500 13	
		23:41:19				DS: BL19_20211129_234119
000_FL16	1916	23:41:19	23:47:00	480	500 13	
001_FL13	1913	23:49:57	23:53:29	480	500 13	
002_FL12	1912	23:55:29	23:59:14	480	500 13	
003_FL11	1911	00:01:01	00:04:34	480	500 13	
004_FL10	1910	00:06:32	00:10:11	480	500 13	
005_FL9	1909	00:13:12	00:17:41	480	500 13	
006_FL8	1908	00:20:03	00:24:17	480	500 13	
007_FL7	1907	00:26:36	00:31:06	480	500 13	
008_FL6	1906	00:32:59	00:37:26	480	500 13	
009_FL5	1905	00:39:34	00:43:56	480	500 13	
010_FL4	1904	00:48:16	00:52:30	480	500 13	
011_FL3	1903	00:54:21	00:58:40	480	500 13	
012_FL24	1924	01:02:47	01:09:53	480	500 13	
013_FL26	1926	01:11:55	01:18:41	480	500 13	Low start
		01:21:04				DS: BL20_20211130_012104

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL15, BL19, BL20, BL21, QC61	DATE:	29 November 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Efate, TongaVanuatu	CLOUDS:	Clouds @ 2500ft
BASE STATION:	AAMPri	WIND:	10-15 kts @ 60
LIDAR DRIVE:	HE4X-06	RCD DRIVE:	RCD-04
ENGINE START:	19:36	ENGINE OFF:	2:02
TAKEOFF:	19:56	LANDING:	1:53
		ENGINE TIME:	06:26
		AIR TIME	05:57

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
000_FL13	2013	01:21:04	01:27:48	480	500 13	
		01:31:07				DS: BL21_20211130_013107
000_FL43	2143	01:31:07	01:32:39	620	440 14	
001_FL36	2136	01:35:21	01:35:35	620	440 14	BAD: Clouds
002_FL36	2136	01:41:05	01:41:08	620	440 13	BAD: Clouds
		01:41:24				DS: BL20_20211130_014124
000_FL14	2014	01:41:24	01:48:05	480	500 13	
		01:50:00				RTB
		01:53:00				Landing
		01:54:00				Static close out AAM Pri base

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL19	DATE:	30 November 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Efate	CLOUDS:	Clouds @ 2500ft
BASE STATION:	AAMPri	WIND:	5-10 kts @ 80
LIDAR DRIVE:	HE4X-08	RCD DRIVE:	RCD-04
ENGINE START:	2:56	ENGINE OFF:	6:10
TAKEOFF:	3:08	LANDING:	6:01
		ENGINE TIME:	03:14
		AIR TIME	02:53

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		03:08:00				Takeoff
		03:18:00				CZ3X Crash and full reboot
		03:28:00				Check-in over AAM Pri base
		03:31:11				DS: BL19_20211130_033111
000_FL2	1902	03:31:11	03:35:11	480	500 13	
001_FL1	1901	03:37:37	03:39:30	480	500 13	
002_FL26	1926	03:45:16	03:52:38	480	500 13	
003_FL17	1917	03:56:43	04:02:23	480	500 13	
004_FL18	1918	04:04:42	04:10:58	480	500 13	
005_FL19	1919	04:15:55	04:22:24	480	500 13	
006_FL20	1920	04:24:33	04:31:18	480	500 13	
007_FL21	1921	04:33:40	04:40:31	480	500 13	
008_FL22	1922	04:42:35	04:49:38	480	500 13	
009_FL23	1923	04:51:51	04:58:42	480	500 13	
010_FL28	1928	05:00:53	05:08:03	480	500 13	
011_FL29	1929	05:10:22	05:17:18	480	500 13	
012_FL30	1930	05:19:42	05:27:09	480	500 13	
013_FL31	1931	05:29:26	05:36:30	480	500 13	
014_FL32	1932	05:39:38	05:47:13	480	500 13	
015_FL33	1933	05:49:27	05:56:53	480	500 13	
		05:57:00				RTB
		06:01:00				Landing
		06:03:00				Static close out AAM Pri base

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL02, QC61, QC62	DATE:	30 November 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Tanna, TongaVanuatu	CLOUDS:	Clouds @ 2500ft
BASE STATION:	AAMPri	WIND:	5-10 kts @ 70
LIDAR DRIVE:	HE4X-07	RCD DRIVE:	RCD-03
ENGINE START:	20:00	ENGINE OFF:	1:43
TAKEOFF:	20:16	LANDING:	1:34
		ENGINE TIME:	05:43
		AIR TIME	05:18

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		20:16:00				Takeoff
		20:26:13				DS: QC61_20211130_202613
000_FL1	6101	20:26:13	20:28:01	480	500 13	Flew low, clouds
		20:29:00				Start Tanna transit
		21:11:53				DS: BL02_20211130_211153
000_FL67	0297	21:11:53	21:14:02	480	500 13	
001_FL52	0252	21:17:12	21:18:32	480	500 13	
002_FL51	0251	21:20:42	21:22:08	480	500 13	
003_FL50	0250	21:24:22	21:25:44	480	500 13	
004_FL49	0249	21:28:25	21:29:50	480	500 13	
005_FL48	0248	21:32:36	21:34:19	480	500 13	
006_FL47	0247	21:36:33	21:39:38	480	500 13	
007_FL46	0246	21:42:32	21:45:31	480	500 13	
008_FL45	0245	21:47:59	21:51:10	480	500 13	
009_FL44	0244	21:53:36	21:56:58	480	500 13	
010_FL43	0243	21:59:40	22:03:20	480	500 13	
011_FL42	0242	22:05:39	22:06:14	480	500 13	BAD: Eye saftey triggered
012_FL33	0233	22:09:08	22:09:22	480	500 13	BAD: Sensor in improper state
013_FL33	0233	22:13:12	22:16:45	480	500 13	Stopped early, clouds
014_FL66	0296	22:19:14	22:21:23	480	500 13	
015_FL42	0242	22:24:14	22:27:46	480	500 13	
016_FL41	0241	22:30:47	22:34:17	480	500 13	
017_FL40	0240	22:36:30	22:40:07	480	500 13	
018_FL39	0239	22:42:40	22:46:15	480	500 13	
019_FL38	0238	22:48:30	22:52:05	480	500 13	
020_FL37	0237	22:54:54	22:58:26	480	500 13	
021_FL36	0236	23:00:50	23:04:29	480	500 13	
022_FL35	0235	23:06:44	23:10:27	480	500 13	
023_FL34	0234	23:12:53	23:16:28	480	500 13	
024_FL68	0298	23:19:08	23:20:53	480	500 13	
025_FL53	0253	23:26:29	23:28:37	480	500 13	

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL02, QC61, QC62	DATE:	30 November 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Tanna, TongaVanuatu	CLOUDS:	Clouds @ 2500ft
BASE STATION:	AAMPri	WIND:	5-10 kts @ 70
LIDAR DRIVE:	HE4X-07	RCD DRIVE:	RCD-03
ENGINE START:	20:00	ENGINE OFF:	1:43
TAKEOFF:	20:16	LANDING:	1:34
		ENGINE TIME:	05:43
		AIR TIME	05:18

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
026_FL54	0254	23:30:49	23:33:15	480	500 13	
027_FL55	0255	23:35:43	23:38:31	480	500 13	
028_FL56	0256	23:40:30	23:43:32	480	500 13	
029_FL57	0257	23:45:36	23:46:25	480	500 13	BAD: Eye saftey triggered
030_FL57	0257	23:50:35	23:53:49	480	500 13	
031_FL58	0258	23:55:57	23:59:49	480	500 13	
032_FL59	0259	00:02:38	00:06:00	480	500 13	
033_FL60	0260	00:08:57	00:12:58	480	500 13	
034_FL61	0261	00:15:17	00:18:44	480	500 13	
035_FL62	0262	00:20:56	00:24:43	480	500 13	Eye safety at very end
036_FL63	0263	00:28:26	00:30:40	480	500 13	Delayed start, clouds
037_FL64	0264	00:33:44	00:34:57	480	500 13	BAD: Eye saftey triggered
		00:50:00				RTB
		01:29:35				DS: QC62_20211201_012935
000_FL1	6201	01:29:35	01:31:26	480	500 13	
		01:34:00				Landing
		01:36:00				Static close out AAM Pri base

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL19, BL20	DATE:	1 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Efate	CLOUDS:	Clouds @ 2000ft
BASE STATION:	AAMPri	WIND:	10-15 kts @ 120
LIDAR DRIVE:	HE4X-06	RCD DRIVE:	RCD-03
ENGINE START:	2:06	ENGINE OFF:	5:55
TAKEOFF:	2:22	LANDING:	5:50
		ENGINE TIME:	03:49
		AIR TIME	03:28

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		02:22:00				Takeoff
		02:29:57				DS: BL19_20211201_022957
000_FL34	1934	02:29:58	02:37:27	480	500 13	
001_FL35	1935	02:39:36	02:47:12	480	500 13	
002_FL36	1936	02:53:28	03:00:55	480	500 13	
		03:03:49				DS: BL20_20211201_030349
000_FL15	2015	03:03:49	03:10:44	480	500 13	
001_FL16	2016	03:13:18	03:20:25	480	500 13	
		03:21:20				DS: BL19_20211201_032120
000_FL37	1937	03:21:20	03:28:55	480	500 13	
001_FL38	1938	03:31:50	03:38:31	480	500 13	Late start, clouds
002_FL39	1939	03:40:55	03:48:44	480	500 13	
003_FL38	1938	03:50:38	03:52:42	480	500 13	Reflyeast end, filled cloud gap
004_FL40	1940	03:57:43	04:05:21	480	500 13	
005_FL41	1941	04:08:08	04:16:16	480	500 13	
006_FL42	1942	04:18:09	04:26:04	480	500 13	
007_FL43	1943	04:28:38	04:36:46	480	500 13	
008_FL44	1944	04:38:46	04:46:31	480	500 13	
009_FL45	1945	04:48:26	04:55:50	480	500 13	
010_FL46	1946	04:58:46	05:05:57	480	500 13	
011_FL47	1947	05:08:33	05:15:12	480	500 13	
012_FL48	1948	05:17:43	05:23:35	480	500 13	
013_FL49	1949	05:26:40	05:31:52	480	500 13	
014_FL50	1950	05:34:13	05:38:44	480	500 13	
015_FL51	1951	05:41:31	05:45:28	480	500 13	
		05:46:00				RTB
		05:50:00				Landing
		05:51:00				Static close out AAM Pri base

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL01, BL02, QC61, QC62	DATE:	3 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Tanna, TongaVanuatu	CLOUDS:	Clouds @ 2000ft
BASE STATION:	AAMPri	WIND:	15 kts @ 100
LIDAR DRIVE:	HE4X-03	RCD DRIVE:	RCD-04
ENGINE START:	20:16	ENGINE OFF:	2:19
TAKEOFF:	20:30	LANDING:	2:01
		ENGINE TIME:	06:03
		AIR TIME	05:31

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		20:30:00				Takeoff
		20:44:45				DS: QC61_20211203_204445
000_FL1	6101	20:44:45	20:46:41	480	500 13	
		21:25:29				DS: BL02_20211203_212529
000_FL64	0264	21:25:29	21:26:42	480	500 13	Eye safe, terrain
001_FL64	0264	21:31:04	21:33:40	480	500 13	
002_FL63	0263	21:37:25	21:37:35	480	500 13	BAD: Late start
003_FL63	0263	21:41:28	21:45:10	480	500 13	
004_FL62	0262	21:47:36	21:51:23	480	500 13	
005_FL61	0261	21:53:35	21:57:37	480	500 13	
006_FL59	0259	21:59:43	22:03:38	480	500 13	
		22:07:39				DS: BL01_20211203_220739
000_FL1	0101	22:07:39	22:09:32	480	500 13	
001_FL2	0102	22:11:40	22:13:33	480	500 13	
002_FL3	0103	22:16:19	22:18:24	480	500 13	
003_FL4	0104	22:20:33	22:22:47	480	500 13	
004_FL5	0105	22:25:12	22:27:29	480	500 13	
005_FL6	0106	22:29:40	22:32:03	480	500 13	
006_FL7	0107	22:34:45	22:37:19	480	500 13	
007_FL8	0108	22:40:05	22:42:30	480	500 13	
008_FL9	0109	22:45:06	22:47:34	480	500 13	
009_FL65	0195	22:49:45	22:51:29	480	500 13	
010_FL11	0111	22:54:08	22:55:11	480	500 13	
011_FL12	0112	22:57:52	22:59:02	480	500 13	
012_FL13	0113	23:01:26	23:02:39	480	500 13	
013_FL10	0110	23:05:23	23:06:31	480	500 13	
014_FL14	0114	23:09:45	23:12:01	480	500 13	
015_FL15	0115	23:14:16	23:17:23	480	500 13	
016_FL16	0116	23:19:36	23:22:49	480	500 13	
017_FL17	0117	23:24:38	23:28:09	480	500 13	
018_FL18	0118	23:30:36	23:34:31	480	500 13	

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL01, BL02, QC61, QC62	DATE:	3 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.

MISSION ID:	Tanna, TongaVanuatu	CLOUDS:	Clouds @ 2000ft
BASE STATION:	AAMPri	WIND:	15 kts @ 100
LIDAR DRIVE:	HE4X-03	RCD DRIVE:	RCD-04

ENGINE START:	20:16	ENGINE OFF:	2:19	ENGINE TIME:	06:03
TAKEOFF:	20:30	LANDING:	2:01	AIR TIME	05:31

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO		REMARKS
					PRF	PWR	
019_FL19	0119	23:36:27	23:40:24	480	500	13	
020_FL20	0120	23:42:47	23:46:51	480	500	13	
021_FL21	0121	23:49:18	23:53:31	480	500	13	
022_FL22	0122	23:59:03	00:03:21	480	500	13	
023_FL23	0123	00:05:29	00:09:43	480	500	13	
024_FL24	0124	00:11:19	00:15:33	480	500	13	
025_FL25	0125	00:17:40	00:21:54	480	500	13	
026_FL26	0126	00:23:41	00:27:46	480	500	13	
027_FL27	0127	00:29:51	00:33:55	480	500	13	
028_FL28	0128	00:37:06	00:40:55	480	500	13	
029_FL66	0196	00:43:57	00:45:56	480	500	13	
030_FL44	0144	00:51:16	00:55:13	480	500	13	
031_FL43	0143	00:58:27	00:58:34	480	500	13	BAD: Accidental laser fire
032_FL43	0143	01:04:37	01:08:30	480	500	13	
033_FL42	0142	01:10:10	01:14:16	480	500	13	
034_FL41	0141	01:16:25	01:20:30	480	500	13	
		01:25:00					RTB
		01:58:08					DS: QC62_20211204_015808
000_FL1	6201	01:58:08	01:59:52	480	500	13	
		02:01:00					Landing
		02:05:00					Static close out AAM Pri base

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL01, BL02, BL20, QC62	DATE:	4 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.

MISSION ID:	Efate, Tanna, TongaVanuatu	CLOUDS:	Clouds @ 2000ft
BASE STATION:	AAMPri	WIND:	10 kts @ 60
LIDAR DRIVE:	HE4X-07	RCD DRIVE:	RCD-05

ENGINE START:	19:53	ENGINE OFF:	0:55	ENGINE TIME:	05:02
TAKEOFF:	20:09	LANDING:	0:46	AIR TIME	04:37

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		20:09:00				Takeoff
		20:23:00				DS: QC62_20211204_202300
000_FL1	6201	20:23:00	20:24:49	480	500 13	
		21:04:27				DS: BL02_20211204_210427
000_FL33	0233	21:04:27	21:05:13	480	500 13	BAD:
001_FL33	0233	21:07:39	21:13:33	480	500 13	
		21:22:42				DS: BL01_20211204_212242
000_FL40	0140	21:22:42	21:25:31	480	500 13	BAD:
		22:05:12				DS: QC62_20211204_220512
000_FL1	6201	22:05:12	22:07:04	480	500 13	
		22:09:00				Check-in AAM Pri base
		22:15:22				DS: BL20_20211204_221522
000_FL39	2039	22:15:22	22:22:54	480	500 13	
001_FL38	2038	22:24:59	22:33:00	480	500 13	
002_FL37	2037	22:35:00	22:43:05	480	500 13	
003_FL36	2036	22:45:08	22:53:24	480	500 13	
004_FL35	2035	22:55:38	23:04:00	480	500 13	
005_FL34	2034	23:06:23	23:14:57	480	500 13	
006_FL33	2033	23:16:41	23:25:08	480	500 13	
007_FL32	2032	23:26:52	23:35:31	480	500 13	
008_FL31	2031	23:41:14	23:49:36	480	500 13	
009_FL30	2030	23:51:50	00:00:17	480	500 13	
010_FL29	2029	00:02:54	00:11:00	480	500 13	
011_FL28	2028	00:12:55	00:21:18	480	500 13	
012_FL17	2017	00:22:46	00:29:41	480	500 13	
013_FL16	2016	00:31:31	00:35:08	480	500 13	BAD:
		00:41:00				RTB
		00:46:00				Landing
		00:48:00				Static close out AAM Pri base

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL18	DATE:	5 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.

MISSION ID:	Efate	CLOUDS:	Clouds @ 1500ft
BASE STATION:	AAMPri	WIND:	5 kts @ 40
LIDAR DRIVE:	HE4X-08	RCD DRIVE:	RCD-05

ENGINE START:	1:32	ENGINE OFF:	4:12	ENGINE TIME:	02:40
TAKEOFF:	1:48	LANDING:	4:03	AIR TIME	02:15

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR		REMARKS
		01:48:00					Takeoff
		02:46:00					Check-in AAM Pri base
		03:29:00					Check-in AAM Pri base
		03:33:51					DS: BL18_20211205_033351
000_FL52	1852	03:33:51	03:35:26	480	500	13	
001_FL53	1853	03:37:24	03:39:03	480	500	13	
002_FL54	1854	03:41:08	03:42:51	480	500	13	
003_FL55	1855	03:45:13	03:47:25	480	500	13	
004_FL67	1867	03:51:21	03:55:00	480	500	13	BAD: Clouds
		04:03:00					Landing
		04:05:00					Static close out AAM Pri base

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL15, BL16, BL21, BL22	DATE:	7 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Efate	CLOUDS:	Clouds @ 2000ft
BASE STATION:	AAMPri	WIND:	10 kts @ 80
LIDAR DRIVE:	HE4X-06	RCD DRIVE:	RCD-04
ENGINE START:	23:21	ENGINE OFF:	5:51
TAKEOFF:	23:51	LANDING:	5:40
		ENGINE TIME:	06:30
		AIR TIME	05:49

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		23:51:00				Takeoff
		00:15:12				DS: BL15_20211208_001512
000_FL31	1531	00:15:12	00:17:49	480	500 13	
001_FL30	1530	00:21:03	00:23:51	480	500 13	
002_FL29	1529	00:26:01	00:29:04	480	500 13	
003_FL12	1512	00:31:04	00:32:42	480	500 13	
004_FL13	1513	00:34:47	00:36:39	480	500 13	
005_FL14	1514	00:38:52	00:40:54	480	500 13	
006_FL15	1515	00:43:27	00:44:33	480	500 13	
007_FL16	1516	00:47:15	00:49:10	480	500 13	
008_FL17	1517	00:51:26	00:53:22	480	500 13	
009_FL15	1515	00:55:37	00:57:39	480	500 13	
010_FL28	1528	00:59:42	01:02:46	480	500 13	
011_FL27	1527	01:05:16	01:08:21	480	500 13	
012_FL26	1526	01:10:34	01:14:12	480	500 13	
013_FL25	1525	01:16:33	01:20:07	480	500 13	
014_FL24	1524	01:22:24	01:25:51	480	500 13	
015_FL23	1523	01:27:49	01:30:31	480	500 13	BAD: Eye safe
016_FL18	1518	01:35:42	01:35:50	480	500 13	BAD: Eye safe
017_FL18	1518	01:40:12	01:41:57	480	500 13	
018_FL19	1519	01:44:28	01:46:04	480	500 13	
019_FL20	1520	01:48:11	01:49:01	480	500 13	BAD: Eye safe
020_FL23	1523	01:53:26	01:56:49	480	500 13	
021_FL22	1522	01:59:02	02:01:45	480	500 13	
022_FL22	1522	02:07:33	02:08:28	480	500 13	
023_FL21	1521	02:12:12	02:15:11	480	500 13	
024_FL20	1520	02:17:26	02:18:18	480	500 13	BAD: Eye safe
		02:25:28				DS: BL16_20211208_022528
000_FL34	1634	02:25:28	02:26:44	480	500 13	
001_FL33	1633	02:28:54	02:30:17	480	500 13	
002_FL32	1632	02:32:06	02:33:26	480	500 13	

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL15, BL16, BL21, BL22	DATE:	7 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Efate	CLOUDS:	Clouds @ 2000ft
BASE STATION:	AAMPri	WIND:	10 kts @ 80
LIDAR DRIVE:	HE4X-06	RCD DRIVE:	RCD-04
ENGINE START:	23:21	ENGINE OFF:	5:51
TAKEOFF:	23:51	LANDING:	5:40
		ENGINE TIME:	06:30
		AIR TIME	05:49

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
003_FL31	1631	02:35:54	02:37:21	480	500 13	
004_FL30	1630	02:39:36	02:41:08	480	500 13	
005_FL29	1629	02:43:26	02:44:48	480	500 13	
006_FL28	1628	02:47:03	02:48:42	480	500 13	
007_FL27	1627	02:50:46	02:52:19	480	500 13	
008_FL26	1626	02:54:32	02:56:15	480	500 13	
009_FL25	1625	02:58:00	02:59:34	480	500 13	
010_FL24	1624	03:01:49	03:03:29	480	500 13	
011_FL23	1623	03:05:22	03:07:09	480	500 13	
		03:10:54				DS: BL15_20211208_031054
000_FL20	1520	03:10:54	03:11:51	480	500 13	
001_FL20	1520	03:15:11	03:16:46	480	500 13	
		03:19:21				DS: BL21_20211208_031921
000_FL1	2101	03:19:21	03:20:52	620	440 14	
001_FL2	2102	03:22:56	03:24:32	620	440 14	
002_FL3	2103	03:26:32	03:28:26	620	440 14	
003_FL4	2104	03:30:17	03:32:05	620	440 14	
004_FL5	2105	03:34:04	03:35:03	620	440 14	BAD: Eye safe
005_FL6	2106	03:41:21	03:43:12	620	440 14	
006_FL5	2105	03:45:01	03:46:51	620	440 14	
007_FL7	2107	03:48:49	03:50:38	620	440 14	
008_FL8	2108	03:52:17	03:54:05	620	440 14	
009_FL9	2109	03:56:08	03:57:59	620	440 14	
010_FL10	2110	04:00:22	04:02:08	620	440 14	
011_FL11	2111	04:04:07	04:05:55	620	440 14	
012_FL12	2112	04:07:42	04:09:31	620	440 14	
013_FL13	2113	04:11:51	04:13:36	620	440 14	
014_FL24	2124	04:16:51	04:18:31	620	440 14	
015_FL25	2125	04:20:28	04:22:18	620	440 14	
016_FL26	2126	04:24:00	04:26:05	620	440 14	
017_FL27	2127	04:28:37	04:30:14	620	440 14	

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL15, BL16, BL21, BL22	DATE:	7 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Efate	CLOUDS:	Clouds @ 2000ft
BASE STATION:	AAMPri	WIND:	10 kts @ 80
LIDAR DRIVE:	HE4X-06	RCD DRIVE:	RCD-04
ENGINE START:	23:21	ENGINE OFF:	5:51
TAKEOFF:	23:51	LANDING:	5:40
		ENGINE TIME:	06:30
		AIR TIME	05:49

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
018_FL28	2128	04:32:47	04:34:27	620	440 14	
019_FL29	2129	04:36:47	04:38:55	620	440 14	
020_FL30	2130	04:41:06	04:43:28	620	440 14	
021_FL31	2131	04:45:25	04:48:04	620	440 14	
		04:55:40				DS: BL22_20211208_045540
000_FL21	2221	04:55:40	04:57:04	620	440 14	
001_FL20	2220	04:58:48	05:00:11	620	440 14	
002_FL19	2219	05:01:51	05:03:19	620	440 14	
003_FL14	2214	05:05:03	05:06:26	620	440 14	
004_FL15	2215	05:08:28	05:09:54	620	440 14	
005_FL16	2216	05:11:30	05:13:01	620	440 14	
006_FL17	2217	05:14:57	05:16:28	620	440 14	
007_FL18	2218	05:18:16	05:19:45	620	440 14	
008_FL13	2213	05:21:48	05:23:38	620	440 14	Laser on during turn, trim
009_FL12	2212	05:25:07	05:27:03	620	440 14	Laser on during turn, trim
010_FL11	2211	05:28:53	05:30:49	620	440 14	Laser on during turn, trim
011_FL10	2210	05:32:21	05:34:20	620	440 14	Laser on during turn, trim
		05:40:00				Landing
		05:42:00				Static close out AAM Pri base

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL01, BL02, QC61, QC62	DATE:	8 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Tanna, TongaVanuatu	CLOUDS:	Clouds @ 2500ft
BASE STATION:	AAMPri	WIND:	15-20 kts @ 40
LIDAR DRIVE:	HE4X-07	RCD DRIVE:	RCD-03
ENGINE START:	22:32	ENGINE OFF:	4:35
TAKEOFF:	22:53	LANDING:	4:24
		ENGINE TIME:	06:03
		AIR TIME	05:31

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		22:53:00				Takeoff
		22:57:23				DS: QC61_20211208_225723
000_FL1	6101	22:57:23	22:59:06	480	500 13	
		23:43:27				DS: BL01_20211208_234327
000_FL63	0163	23:43:27	23:47:25	480	500 13	
		23:51:25				DS: BL02_20211208_235125
000_FL65	0295	23:51:25	23:53:06	480	500 13	
001_FL1	0201	23:56:53	00:00:54	480	500 13	
002_FL2	0202	00:03:24	00:07:15	480	500 13	
003_FL3	0203	00:09:21	00:13:34	480	500 13	
004_FL4	0204	00:15:33	00:19:41	480	500 13	
005_FL5	0205	00:21:52	00:26:13	480	500 13	
006_FL6	0206	00:28:12	00:29:04	480	500 13	BAD:
007_FL6	0206	00:33:12	00:33:37	480	500 13	BAD:
008_FL6	0206	00:38:23	00:42:29	480	500 13	
009_FL7	0207	00:45:26	00:49:29	480	500 13	
010_FL8	0208	00:52:06	00:55:59	480	500 13	
011_FL9	0209	00:58:10	01:02:06	480	500 13	
012_FL10	0210	01:04:53	01:08:40	480	500 13	
013_FL11	0211	01:11:18	01:14:51	480	500 13	
014_FL12	0212	01:17:29	01:20:47	480	500 13	
015_FL13	0213	01:22:56	01:26:02	480	500 13	
		01:34:42				DS: BL02_20211209_013442
000_FL14	0214	01:34:42	01:39:04	480	500 13	
001_FL15	0215	01:41:01	01:45:45	480	500 13	
002_FL16	0216	01:47:57	01:53:21	480	500 13	
003_FL17	0217	01:55:02	02:00:40	480	500 13	
004_FL18	0218	02:02:37	02:08:47	480	500 13	
005_FL19	0219	02:10:36	02:16:46	480	500 13	
006_FL20	0220	02:18:46	02:25:10	480	500 13	
007_FL21	0221	02:27:22	02:33:40	480	500 13	

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL01, BL02, QC61, QC62	DATE:	8 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Tanna, TongaVanuatu	CLOUDS:	Clouds @ 2500ft
BASE STATION:	AAMPri	WIND:	15-20 kts @ 40
LIDAR DRIVE:	HE4X-07	RCD DRIVE:	RCD-03
ENGINE START:	22:32	ENGINE OFF:	4:35
TAKEOFF:	22:53	LANDING:	4:24
		ENGINE TIME:	06:03
		AIR TIME	05:31

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
008_FL22	0222	02:35:54	02:42:19	480	500 13	
009_FL23	0223	02:57:35	03:04:03	480	500 13	
010_FL24	0224	03:06:04	03:12:37	480	500 13	
011_FL25	0225	03:14:54	03:21:32	480	500 13	
012_FL26	0226	03:23:30	03:30:04	480	500 13	
013_FL27	0227	03:32:06	03:38:31	480	500 13	
014_FL28	0228	03:40:22	03:46:44	480	500 13	
		03:47:00				RTB
		04:20:47				DS: QC62_20211209_042047
000_FL1	6201	04:20:47	04:22:30	480	500 13	
		04:24:00				Landing
		04:26:00				Static close out AAM Pri base

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL01, BL02, QC61, QC62	DATE:	15 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Tanna, TongaVanuatu	CLOUDS:	Clouds @ 3500ft
BASE STATION:	AAMPri	WIND:	5-10 kts @ 180
LIDAR DRIVE:	HE4X-06	RCD DRIVE:	RCD-03
ENGINE START:	20:00	ENGINE OFF:	2:10
TAKEOFF:	20:18	LANDING:	2:02
		ENGINE TIME:	06:10
		AIR TIME	05:44

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		20:18:00				Takeoff
		20:24:43				DS: QC61_20211215_202443
000_FL1	6101	20:24:43	20:26:26	480	500 13	
		21:06:17				DS: BL01_20211215_210617
000_FL67	0197	21:06:17	21:08:35	480	500 13	
001_FL40	0140	21:12:37	21:16:51	480	500 13	
002_FL39	0139	21:19:09	21:23:27	480	500 13	
003_FL38	0138	21:25:03	21:29:23	480	500 13	
004_FL37	0137	21:31:25	21:36:04	480	500 13	
005_FL36	0136	21:37:51	21:42:28	480	500 13	
006_FL35	0135	21:44:09	21:48:49	480	500 13	
007_FL34	0134	21:50:24	21:55:02	480	500 13	
008_FL33	0133	22:00:14	22:03:58	480	500 13	
009_FL32	0132	22:05:35	22:08:57	480	500 13	
010_FL31	0131	22:11:02	22:14:32	480	500 13	
011_FL30	0130	22:16:21	22:19:29	480	500 13	
012_FL29	0129	22:21:45	22:24:43	480	500 13	
013_FL45	0145	22:26:53	22:29:47	480	500 13	
014_FL46	0146	22:31:36	22:34:31	480	500 13	
015_FL47	0147	22:36:55	22:40:14	480	500 13	
016_FL48	0148	22:42:04	22:45:19	480	500 13	
017_FL49	0149	22:47:39	22:52:50	480	500 13	
018_FL50	0150	22:54:49	23:00:13	480	500 13	
019_FL51	0151	23:02:22	23:07:53	480	500 13	
020_FL52	0152	23:09:42	23:15:06	480	500 13	
021_FL53	0153	23:17:24	23:22:56	480	500 13	
022_FL54	0154	23:24:56	23:30:27	480	500 13	
023_FL55	0155	23:32:28	23:37:58	480	500 13	
024_FL56	0156	23:39:58	23:45:32	480	500 13	
025_FL57	0157	23:47:57	23:53:20	480	500 13	
026_FL58	0158	23:55:18	00:00:34	480	500 13	

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL01, BL02, QC61, QC62	DATE:	15 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.

MISSION ID:	Tanna, TongaVanuatu	CLOUDS:	Clouds @ 3500ft
BASE STATION:	AAMPri	WIND:	5-10 kts @ 180
LIDAR DRIVE:	HE4X-06	RCD DRIVE:	RCD-03

ENGINE START:	20:00	ENGINE OFF:	2:10	ENGINE TIME:	06:10
TAKEOFF:	20:18	LANDING:	2:02	AIR TIME	05:44

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
027_FL59	0159	00:03:20	00:08:25	480	500 13	
028_FL60	0160	00:10:42	00:15:35	480	500 13	
029_FL61	0161	00:17:43	00:22:31	480	500 13	
030_FL62	0162	00:24:24	00:29:01	480	500 13	
031_FL64	0164	00:33:32	00:34:44	480	500 13	
032_FL63	0163	00:36:53	00:40:52	480	500 13	
		00:42:20				DS: BL02_20211216_004220
000_FL5	0205	00:42:20	00:46:22	480	500 13	
001_FL4	0204	00:48:48	00:53:08	480	500 13	
002_FL3	0203	00:55:01	00:59:08	480	500 13	
003_FL2	0202	01:01:21	01:05:26	480	500 13	
004_FL1	0201	01:07:46	01:11:36	480	500 13	
005_FL29	0229	01:16:19	01:22:35	480	500 13	
		01:26:00				RTB
		01:58:18				DS: QC62_20211216_015818
000_FL1	6201	01:58:18	02:00:03	480	500 13	
		02:02:00				Landing
		02:04:00				Static close out AAM Pri base

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL15, BL21, BL22	DATE:	16 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.

MISSION ID:	Efate	CLOUDS:	Clouds @ 3000ft
BASE STATION:	AAMPri	WIND:	5-10 kts @ 180
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-03

ENGINE START:	3:37	ENGINE OFF:	6:49	ENGINE TIME:	03:12
TAKEOFF:	3:52	LANDING:	6:40	AIR TIME	02:48

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR		REMARKS
		03:52:00					Takeoff
		03:59:00					Deep Channel Pause
		03:59:07					DS: BL22_20211216_035907
000_FL1	2201	03:59:07	04:00:35	620	440	14	
001_FL2	2202	04:02:48	04:04:22	620	440	14	
002_FL3	2203	04:06:21	04:07:59	620	440	14	
003_FL4	2204	04:09:57	04:11:46	620	440	14	
004_FL5	2205	04:14:03	04:15:51	620	440	14	
005_FL6	2206	04:17:46	04:19:43	620	440	14	
006_FL7	2207	04:22:17	04:24:18	620	440	14	
007_FL8	2208	04:26:07	04:28:05	620	440	14	
008_FL9	2209	04:29:53	04:31:47	620	440	14	
		04:36:26					DS: BL21_20211216_043626
000_FL41	2141	04:36:26	04:38:10	620	440	14	
001_FL42	2142	04:40:51	04:42:23	620	440	14	
002_FL43	2143	04:44:36	04:46:09	620	440	14	
003_FL32	2132	04:49:13	04:52:03	620	440	14	
004_FL33	2133	04:54:09	04:56:56	620	440	14	
005_FL34	2134	04:59:03	05:01:54	620	440	14	
006_FL35	2135	05:04:14	05:07:12	620	440	14	
007_FL36	2136	05:09:04	05:12:04	620	440	14	
008_FL14	2114	05:14:19	05:16:51	620	440	14	
009_FL15	2115	05:19:08	05:21:54	620	440	14	
010_FL16	2116	05:23:33	05:26:02	620	440	14	
011_FL17	2117	05:28:31	05:31:02	620	440	14	
012_FL18	2118	05:33:16	05:35:43	620	440	14	
013_FL19	2119	05:39:05	05:40:27	620	440	14	
014_FL20	2120	05:42:20	05:43:38	620	440	14	
015_FL21	2121	05:46:12	05:47:32	620	440	14	

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL15, BL21, BL22	DATE:	16 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.

MISSION ID:	Efate	CLOUDS:	Clouds @ 3000ft
BASE STATION:	AAMPri	WIND:	5-10 kts @ 180
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-03

ENGINE START:	3:37	ENGINE OFF:	6:49	ENGINE TIME:	03:12
TAKEOFF:	3:52	LANDING:	6:40	AIR TIME	02:48

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR		REMARKS
016_FL22	2122	05:49:32	05:50:54	620	440	14	
017_FL23	2123	05:53:04	05:54:22	620	440	14	
		05:57:20					DS: BL15_20211216_055720
000_FL1	1501	05:57:20	05:58:36	480	500	13	BAD: Deep channel still paused
		05:59:00					Deep Channel Unpause
001_FL1	1501	06:00:44	06:02:00	480	500	13	
002_FL2	1502	06:03:44	06:05:28	480	500	13	
003_FL3	1503	06:07:29	06:09:34	480	500	13	
004_FL4	1504	06:11:23	06:13:30	480	500	13	
005_FL20	1520	06:16:17	06:17:27	480	500	13	BAD: Eye safe
006_FL21	1521	06:20:16	06:23:10	480	500	13	
007_FL22	1522	06:24:52	06:27:27	480	500	13	BAD: Eye safe
008_FL22	1522	06:29:48	06:33:14	480	500	13	
		06:35:00					RTB
		06:40:00					Landing
		06:41:00					Static close out AAM Pri base

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL16, BL20	DATE:	17 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Efate	CLOUDS:	Clear
BASE STATION:	AAMPri	WIND:	15 kts @ 130
LIDAR DRIVE:	HE4X-07	RCD DRIVE:	RCD-05
ENGINE START:	2:30	ENGINE OFF:	6:43
TAKEOFF:	2:52	LANDING:	6:35
		ENGINE TIME:	04:13
		AIR TIME	03:43

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		02:52:00				Takeoff
		03:00:29				DS: BL20_20211217_030029
000_FL12	2012	03:00:29	03:06:57	480	500 13	BAD:
		03:23:59				DS: BL20_20211217_032359
000_FL16	2016	03:23:59	03:30:45	480	500 13	
001_FL18	2018	03:32:59	03:40:29	480	500 13	
002_FL19	2019	03:42:31	03:49:51	480	500 13	
003_FL20	2020	03:52:08	03:59:53	480	500 13	
004_FL21	2021	04:02:41	04:10:18	480	500 13	
005_FL22	2022	04:12:43	04:20:24	480	500 13	
006_FL23	2023	04:22:37	04:30:17	480	500 13	
007_FL24	2024	04:32:26	04:40:14	480	500 13	
008_FL25	2025	04:42:17	04:50:04	480	500 13	
009_FL26	2026	04:52:02	04:59:53	480	500 13	
010_FL27	2027	05:02:21	05:10:10	480	500 13	
		05:15:27				DS: BL16_20211217_051527
000_FL22	1622	05:15:27	05:17:13	480	500 13	
001_FL21	1621	05:19:14	05:20:59	480	500 13	
002_FL20	1620	05:23:29	05:25:15	480	500 13	
003_FL19	1619	05:27:24	05:28:56	480	500 13	
004_FL18	1618	05:31:06	05:32:40	480	500 13	
005_FL17	1617	05:34:26	05:36:08	480	500 13	
006_FL16	1616	05:38:23	05:40:11	480	500 13	
007_FL15	1615	05:41:50	05:43:36	480	500 13	
008_FL1	1601	05:46:21	05:47:38	480	500 13	
009_FL2	1602	05:49:32	05:50:52	480	500 13	
010_FL3	1603	05:52:52	05:54:07	480	500 13	
011_FL4	1604	05:55:38	05:56:55	480	500 13	
012_FL5	1605	05:59:02	06:00:16	480	500 13	
013_FL6	1606	06:01:52	06:03:09	480	500 13	
014_FL7	1607	06:05:11	06:06:30	480	500 13	

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL16, BL20	DATE:	17 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Efate	CLOUDS:	Clear
BASE STATION:	AAMPri	WIND:	15 kts @ 130
LIDAR DRIVE:	HE4X-07	RCD DRIVE:	RCD-05
ENGINE START:	2:30	ENGINE OFF:	6:43
TAKEOFF:	2:52	LANDING:	6:35
		ENGINE TIME:	04:13
		AIR TIME	03:43

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR		REMARKS
015_FL8	1608	06:08:22	06:09:38	480	500	13	
016_FL9	1609	06:11:49	06:13:00	480	500	13	
017_FL10	1610	06:14:52	06:16:01	480	500	13	
018_FL11	1611	06:18:08	06:19:08	480	500	13	
019_FL12	1612	06:21:07	06:22:03	480	500	13	
020_FL13	1613	06:24:16	06:25:04	480	500	13	
021_FL14	1614	06:26:59	06:27:51	480	500	13	
		06:29:00					RTB
		06:35:00					Landing
		06:37:00					Static close out AAM Pri base

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL10, QC61, QC62	DATE:	17 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.

MISSION ID:	Malekula, TongaVanuatu	CLOUDS:	Clouds @ 3000ft
BASE STATION:	AAMPri	WIND:	10-15 kts @ 60
LIDAR DRIVE:	HE4X-06	RCD DRIVE:	RCD-03

ENGINE START:	20:55	ENGINE OFF:	2:55	ENGINE TIME:	06:00
TAKEOFF:	21:14	LANDING:	2:48	AIR TIME	05:34

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR		REMARKS
		21:14:00					Takeoff
		21:18:44					DS: QC61_20211217_211844
000_FL1	6101	21:18:44	21:20:26	480	500	13	
		22:00:28					DS: BL10_20211217_220028
000_FL66	1066	22:00:28	22:01:45	480	500	13	
001_FL65	1065	22:03:33	22:04:47	480	500	13	
002_FL62	1062	22:07:06	22:09:51	480	500	13	
003_FL63	1063	22:12:40	22:13:57	480	500	13	
004_FL64	1064	22:16:11	22:17:25	480	500	13	
005_FL61	1061	22:19:09	22:22:26	480	500	13	
006_FL60	1060	22:24:15	22:27:37	480	500	13	
007_FL59	1059	22:29:48	22:33:31	480	500	13	
008_FL58	1058	22:35:58	22:39:35	480	500	13	
009_FL57	1057	22:41:40	22:45:37	480	500	13	
010_FL56	1056	22:48:11	22:51:59	480	500	13	
011_FL55	1055	22:54:19	22:58:14	480	500	13	
012_FL54	1054	23:00:37	23:04:19	480	500	13	
013_FL53	1053	23:07:01	23:10:50	480	500	13	
014_FL52	1052	23:13:26	23:16:57	480	500	13	
015_FL51	1051	23:18:51	23:22:35	480	500	13	
016_FL50	1050	23:25:17	23:28:51	480	500	13	
017_FL49	1049	23:31:16	23:34:43	480	500	13	
018_FL48	1048	23:37:33	23:40:59	480	500	13	
019_FL47	1047	23:43:20	23:46:49	480	500	13	
020_FL46	1046	23:49:23	23:52:40	480	500	13	
021_FL45	1045	23:55:40	23:59:04	480	500	13	
022_FL69	1097	00:02:51	00:04:54	480	500	13	
023_FL44	1044	00:05:45	00:07:54	480	500	13	
024_FL43	1043	00:10:10	00:13:49	480	500	13	

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL10, QC61, QC62	DATE:	17 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.

MISSION ID:	Malekula, TongaVanuatu	CLOUDS:	Clouds @ 3000ft
BASE STATION:	AAMPri	WIND:	10-15 kts @ 60
LIDAR DRIVE:	HE4X-06	RCD DRIVE:	RCD-03

ENGINE START:	20:55	ENGINE OFF:	2:55	ENGINE TIME:	06:00
TAKEOFF:	21:14	LANDING:	2:48	AIR TIME	05:34

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR		REMARKS
025_FL42	1042	00:15:24	00:19:00	480	500	13	
026_FL41	1041	00:21:29	00:25:18	480	500	13	
027_FL40	1040	00:26:55	00:30:26	480	500	13	
028_FL39	1039	00:32:18	00:36:05	480	500	13	
029_FL38	1038	00:37:58	00:41:25	480	500	13	
030_FL37	1037	00:43:37	00:47:21	480	500	13	
031_FL36	1036	00:48:58	00:52:28	480	500	13	
032_FL35	1035	00:54:34	00:58:20	480	500	13	
033_FL34	1034	01:01:27	01:04:51	480	500	13	
034_FL33	1033	01:07:21	01:10:49	480	500	13	
035_FL32	1032	01:13:02	01:16:19	480	500	13	
036_FL31	1031	01:18:38	01:22:05	480	500	13	
037_FL30	1030	01:24:27	01:27:41	480	500	13	
038_FL29	1029	01:30:11	01:33:22	480	500	13	
039_FL28	1028	01:35:26	01:38:36	480	500	13	
040_FL27	1027	01:40:55	01:44:10	480	500	13	
041_FL26	1026	01:45:49	01:48:52	480	500	13	
042_FL25	1025	01:50:45	01:53:54	480	500	13	
043_FL24	1024	01:55:45	01:58:40	480	500	13	
044_FL68	1096	02:00:52	02:03:01	480	500	13	
		02:42:46					DS: QC62_20211218_024246
000_FL1	6201	02:42:46	02:44:27	480	500	13	
		02:46:00					RTB
		02:48:00					Landing
		02:51:00					Static close out AAM Pri base

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL17, BL18, QC61, QC62	DATE:	19 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Efate, TongaVanuatu	CLOUDS:	Clouds @ 2000ft
BASE STATION:	AAMPri	WIND:	5-10 kts @ 280
LIDAR DRIVE:	HE4X-07	RCD DRIVE:	RCD-04
ENGINE START:	20:57	ENGINE OFF:	3:24
TAKEOFF:	21:14	LANDING:	3:15
		ENGINE TIME:	06:27
		AIR TIME	06:01

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		21:14:00				Takeoff
		21:21:42				DS: QC61_20211219_212142
000_FL1	6101	21:21:42	21:22:30	480	500 13	BAD:
		21:26:00				DS: QC62_20211219_212600
000_FL1	6201	21:26:00	21:27:50	480	500 13	
		21:36:50				DS: BL17_20211219_213650
000_FL1	1701	21:36:50	21:40:43	480	500 13	BAD: Eye Safe
001_FL1	1701	21:40:48	21:43:00	480	500 13	BAD: Eye Safe
		21:47:49				DS: BL18_20211219_214749
000_FL11	1811	21:47:49	21:53:45	480	500 13	
001_FL26	1826	21:55:09	22:00:14	480	500 13	
002_FL25	1825	22:01:57	22:06:56	480	500 13	
003_FL24	1824	22:09:05	22:14:20	480	500 13	
004_FL23	1823	22:16:08	22:21:16	480	500 13	
005_FL22	1822	22:23:50	22:29:24	480	500 13	
006_FL21	1821	22:30:57	22:36:06	480	500 13	
007_FL20	1820	22:38:03	22:43:37	480	500 13	
008_FL19	1819	22:45:32	22:51:02	480	500 13	
009_FL18	1818	22:53:00	22:58:54	480	500 13	
010_FL17	1817	23:00:34	23:06:14	480	500 13	
011_FL16	1816	23:08:26	23:14:27	480	500 13	
012_FL15	1815	23:16:17	23:22:07	480	500 13	
013_FL14	1814	23:24:08	23:30:09	480	500 13	
014_FL13	1813	23:31:42	23:37:35	480	500 13	
015_FL12	1812	23:39:44	23:46:08	480	500 13	
016_FL10	1810	23:47:44	23:53:43	480	500 13	
017_FL9	1809	23:55:33	00:01:50	480	500 13	
018_FL8	1808	00:04:18	00:09:33	480	500 13	
019_FL7	1807	00:11:40	00:15:23	480	500 13	
020_FL6	1806	00:16:55	00:20:16	480	500 13	
021_FL5	1805	00:21:58	00:22:24	480	500 13	BAD: Eye safe

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL17, BL18, QC61, QC62	DATE:	19 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Efate, TongaVanuatu	CLOUDS:	Clouds @ 2000ft
BASE STATION:	AAMPri	WIND:	5-10 kts @ 280
LIDAR DRIVE:	HE4X-07	RCD DRIVE:	RCD-04
ENGINE START:	20:57	ENGINE OFF:	3:24
TAKEOFF:	21:14	LANDING:	3:15
		ENGINE TIME:	06:27
		AIR TIME	06:01

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
022_FL1	1801	00:25:33	00:28:22	480	500 13	
023_FL4	1804	00:29:49	00:32:52	480	500 13	
024_FL3	1803	00:34:40	00:37:45	480	500 13	
025_FL2	1802	00:39:30	00:42:20	480	500 13	
026_FL5	1805	00:44:19	00:47:40	480	500 13	
027_FL68	1895	00:51:07	00:53:28	480	500 13	
028_FL52	1852	00:57:50	00:59:25	480	500 13	
029_FL53	1853	01:00:53	01:02:31	480	500 13	
030_FL54	1854	01:05:21	01:07:13	480	500 13	
031_FL55	1855	01:08:32	01:10:44	480	500 13	
032_FL56	1856	01:12:30	01:14:56	480	500 13	
033_FL70	1897	01:17:51	01:19:48	480	500 13	
034_FL57	1857	01:22:59	01:26:12	480	500 13	
035_FL58	1858	01:28:20	01:32:11	480	500 13	
036_FL59	1859	01:34:30	01:38:09	480	500 13	
037_FL60	1860	01:40:44	01:44:45	480	500 13	
038_FL61	1861	01:46:48	01:50:21	480	500 13	
039_FL62	1862	01:52:12	01:56:03	480	500 13	
040_FL63	1863	01:58:13	02:01:54	480	500 13	
041_FL64	1864	02:04:08	02:07:46	480	500 13	
042_FL65	1865	02:09:47	02:13:21	480	500 13	
043_FL66	1866	02:15:55	02:19:52	480	500 13	
044_FL67	1867	02:21:45	02:25:21	480	500 13	
		02:34:05				DS: BL17_20211220_023405
000_FL38	1738	02:34:05	02:35:15	480	500 13	
001_FL37	1737	02:37:09	02:38:27	480	500 13	
002_FL36	1736	02:40:46	02:42:15	480	500 13	
003_FL35	1735	02:43:59	02:45:39	480	500 13	
004_FL34	1734	02:47:38	02:49:28	480	500 13	
005_FL33	1733	02:51:32	02:53:24	480	500 13	
006_FL32	1732	02:55:14	02:57:27	480	500 13	

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL17, BL18, QC61, QC62	DATE:	19 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Efate, TongaVanuatu	CLOUDS:	Clouds @ 2000ft
BASE STATION:	AAMPri	WIND:	5-10 kts @ 280
LIDAR DRIVE:	HE4X-07	RCD DRIVE:	RCD-04
ENGINE START:	20:57	ENGINE OFF:	3:24
TAKEOFF:	21:14	LANDING:	3:15
		ENGINE TIME:	06:27
		AIR TIME	06:01

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR		REMARKS
007_FL31	1731	02:58:56	03:01:15	480	500	13	
008_FL30	1730	03:03:31	03:06:03	480	500	13	
		03:06:00					RTB
		03:15:00					Landing
		03:17:00					Static close out AAM Pri base

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL10, QC61	DATE:	20 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Malekula, TongaVanuatu	CLOUDS:	Clear
BASE STATION:	PPP	WIND:	5-10 kts @ 340
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-05
ENGINE START:	19:29	ENGINE OFF:	23:56
TAKEOFF:	19:46	LANDING:	23:50
		ENGINE TIME:	04:27
		AIR TIME	04:04

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		19:46:00				Takeoff
		19:50:52				DS: QC61_20211220_195052
000_FL1	6101	19:50:52	19:52:35	480	500 13	
		20:32:05				DS: BL10_20211220_203205
000_FL23	1023	20:32:05	20:36:11	480	500 13	
001_FL22	1022	20:38:19	20:38:21	480	13	BAD: OC Crash
		20:54:39				DS: BL10_20211220_205439
000_FL20	1020	20:54:39	20:58:42	480	500 13	
001_FL21	1021	21:00:48	21:00:58	480	500 13	BAD: Safety Shutoff
002_FL21	1021	21:03:00	21:03:15	480	500 13	BAD: Safety Shutoff
003_FL17	1017	21:03:58	21:08:08	480	500 13	
004_FL21	1021	21:09:34	21:13:38	480	500 13	
005_FL19	1019	21:15:43	21:19:59	480	500 13	
006_FL18	1018	21:22:06	21:26:10	480	500 13	
007_FL16	1016	21:27:59	21:32:23	480	500 13	
008_FL15	1015	21:34:13	21:39:39	480	500 13	
009_FL14	1014	21:41:55	21:47:42	480	500 13	
010_FL13	1013	21:49:40	21:55:18	480	500 13	
011_FL12	1012	21:57:16	22:03:11	480	500 13	
012_FL11	1011	22:05:09	22:10:50	480	500 13	
013_FL10	1010	22:12:35	22:18:26	480	500 13	
014_FL9	1009	22:20:42	22:26:42	480	500 13	
015_FL8	1008	22:28:51	22:34:55	480	500 13	
016_FL7	1007	22:37:04	22:43:03	480	500 13	
017_FL6	1006	22:45:16	22:51:18	480	500 13	
018_FL5	1005	22:53:37	22:59:24	480	500 13	
019_FL4	1004	23:02:02	23:07:25	480	500 13	
020_FL2	1002	23:15:21	23:19:43	480	500 13	
021_FL3	1003	23:21:16	23:25:58	480	500 13	
022_FL1	1001	23:28:08	23:32:02	480	500 13	
023_FL67	1095	23:35:25	23:37:44	480	500 13	

PROJECT NAME:		2021-81155 SPC TongaVanuatu		BASE AIRPORT:		Port Vila (VLI)	
LOCATION / AREA:		Tonga and Vanuatu / BL10, QC61		DATE:		20 December 2021	
AIRCRAFT:		ZK-XLF		PILOT:		Jabin L.	
SYSTEM:		Hawkeye 4X		OPERATOR:		Brent H.	
MISSION ID:		Malekula, TongaVanuatu		CLOUDS:		Clear	
BASE STATION:		PPP		WIND:		5-10 kts @ 340	
LIDAR DRIVE:		HE4X-05		RCD DRIVE:		RCD-05	
ENGINE START:		19:29	ENGINE OFF:	23:56	ENGINE TIME:		04:27
TAKEOFF:		19:46	LANDING:	23:50	AIR TIME		04:04
FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR		REMARKS
		23:50:00					Landing
		23:52:00					Static close out at Santo Airport

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL08, BL09, QC61	DATE:	21 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Malekula, TongaVanuatu	CLOUDS:	Clouds @ 1800ft
BASE STATION:	PPP	WIND:	5-10 kts @ 230
LIDAR DRIVE:	HE4X-06	RCD DRIVE:	RCD-05
ENGINE START:	0:16	ENGINE OFF:	5:33
TAKEOFF:	0:30	LANDING:	5:25
		ENGINE TIME:	05:17
		AIR TIME	04:55

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		00:30:00				Takeoff
		00:45:34				DS: BL09_20211221_004534
000_FL52	0996	00:45:34	00:47:24	480	500 13	
001_FL34	0934	00:50:05	00:53:35	480	500 13	
002_FL33	0933	00:55:31	00:56:49	480	500 13	
003_FL32	0932	00:58:43	00:59:56	480	500 13	
004_FL31	0931	01:02:30	01:03:44	480	500 13	
005_FL30	0930	01:06:23	01:07:20	480	500 13	
006_FL29	0929	01:09:45	01:10:37	480	500 13	
007_FL28	0903	01:12:51	01:14:01	480	500 13	
008_FL27	0902	01:16:15	01:17:26	480	500 13	
009_FL26	0901	01:19:16	01:20:29	480	500 13	
010_FL35	0935	01:22:30	01:26:28	480	500 13	
011_FL36	0936	01:28:52	01:32:49	480	500 13	
012_FL37	0937	01:34:53	01:38:53	480	500 13	
013_FL38	0938	01:41:24	01:45:24	480	500 13	
014_FL51	0995	01:48:45	01:51:04	480	500 13	
015_FL1	0904	01:53:57	01:56:21	480	500 13	
016_FL2	0905	01:58:04	02:01:33	480	500 13	
017_FL3	0906	02:07:12	02:10:41	480	500 13	
018_FL4	0907	02:13:13	02:16:43	480	500 13	
		02:21:05				DS: BL08_20211221_022105
000_FL29	0829	02:21:05	02:22:04	480	500 13	
001_FL27	0827	02:22:27	02:23:30	480	500 13	
002_FL28	0828	02:25:25	02:26:32	480	500 13	
003_FL30	0830	02:29:33	02:30:46	480	500 13	
004_FL31	0831	02:33:30	02:36:02	480	500 13	
005_FL32	0832	02:38:10	02:40:41	480	500 13	
006_FL33	0833	02:42:50	02:45:17	480	500 13	
007_FL34	0834	02:47:13	02:49:54	480	500 13	
008_FL35	0835	02:52:21	02:55:09	480	500 13	

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL08, BL09, QC61	DATE:	21 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Malekula, TongaVanuatu	CLOUDS:	Clouds @ 1800ft
BASE STATION:	PPP	WIND:	5-10 kts @ 230
LIDAR DRIVE:	HE4X-06	RCD DRIVE:	RCD-05
ENGINE START:	0:16	ENGINE OFF:	5:33
TAKEOFF:	0:30	LANDING:	5:25
		ENGINE TIME:	05:17
		AIR TIME	04:55

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
009_FL36	0836	02:57:01	02:59:44	480	500 13	
010_FL37	0837	03:02:11	03:05:05	480	500 13	
011_FL38	0838	03:07:26	03:10:09	480	500 13	
012_FL39	0839	03:12:11	03:15:06	480	500 13	
013_FL40	0840	03:17:26	03:20:16	480	500 13	
014_FL41	0841	03:22:16	03:25:17	480	500 13	
015_FL42	0842	03:27:05	03:30:09	480	500 13	
016_FL43	0843	03:31:58	03:35:06	480	500 13	
017_FL44	0844	03:37:17	03:40:24	480	500 13	
018_FL45	0845	03:42:29	03:45:27	480	500 13	
019_FL46	0846	03:47:38	03:50:55	480	500 13	
020_FL47	0847	03:53:11	03:56:35	480	500 13	
021_FL48	0848	03:58:53	04:02:18	480	500 13	
022_FL50	0896	04:05:02	04:07:13	480	500 13	
		04:09:52				DS: BL09_20211221_040952
000_FL25	0928	04:09:52	04:11:46	480	500 13	
001_FL24	0927	04:13:15	04:15:53	480	500 13	
		04:17:32				DS: BL08_20211221_041732
000_FL1	0801	04:17:32	04:25:18	480	500 13	
001_FL2	0802	04:27:15	04:34:42	480	500 13	
002_FL3	0803	04:36:20	04:44:03	480	500 13	
003_FL49	0895	04:46:45	04:49:06	480	500 13	
		05:19:26				DS: QC61_20211221_051926
000_FL1	6101	05:19:26	05:21:14	480	500 13	
		05:25:00				Landing
		05:25:00				RTB
		05:27:00				Static close out at Port Vila airport

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL18	DATE:	21 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Efate	CLOUDS:	Clouds @ 1800ft
BASE STATION:	AAMPri	WIND:	10-15 kts @ 20
LIDAR DRIVE:	HE4X-07	RCD DRIVE:	RCD-03
ENGINE START:	19:52	ENGINE OFF:	21:30
TAKEOFF:	20:08	LANDING:	20:57
		ENGINE TIME:	01:38
		AIR TIME	00:49

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR		REMARKS
		20:08:00					Takeoff
		20:17:11					DS: BL18_20211221_201711
000_FL51	1851	20:17:11	20:22:12	480	500	13	
001_FL50	1850	20:24:18	20:29:30	480	500	13	
002_FL49	1849	20:31:30	20:36:40	480	500	13	
003_FL48	1848	20:38:29	20:43:32	480	500	13	
		20:50:00					RTB
		20:57:00					Landing
		20:58:00					Static close out at AAMPri

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL08, BL09, QC62	DATE:	26 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Malekula, TongaVanuatu	CLOUDS:	Clouds @ 2500ft
BASE STATION:	PPP	WIND:	Calm @ 20
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-05
ENGINE START:	20:31	ENGINE OFF:	1:55
TAKEOFF:	20:46	LANDING:	1:46
		ENGINE TIME:	05:24
		AIR TIME	05:00

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		20:46:00				Takeoff
		20:50:33				DS: QC62_20211226_205033
000_FL1	6201	20:50:33	20:52:15	480	500 13	
		21:27:47				DS: BL09_20211226_212747
000_FL39	0939	21:27:47	21:31:43	480	500 13	
001_FL40	0940	21:33:42	21:37:39	480	500 13	
002_FL41	0941	21:40:07	21:44:17	480	500 13	
003_FL42	0942	21:46:20	21:50:25	480	500 13	
004_FL43	0943	21:52:28	21:56:41	480	500 13	
005_FL44	0944	21:58:39	22:02:55	480	500 13	
006_FL45	0945	22:04:59	22:09:24	480	500 13	
007_FL46	0946	22:11:46	22:16:14	480	500 13	
008_FL47	0947	22:18:29	22:22:57	480	500 13	
009_FL48	0948	22:24:49	22:29:20	480	500 13	
010_FL49	0949	22:32:07	22:36:39	480	500 13	
011_FL50	0950	22:38:56	22:43:38	480	500 13	
012_FL5	0908	22:46:34	22:50:11	480	500 13	
013_FL6	0909	22:51:58	22:55:30	480	500 13	
014_FL7	0910	22:57:28	23:01:01	480	500 13	
015_FL8	0911	23:02:40	23:06:20	480	500 13	
016_FL9	0912	23:08:40	23:12:18	480	500 13	
017_FL10	0913	23:14:11	23:17:54	480	500 13	
018_FL11	0914	23:20:17	23:23:56	480	500 13	
019_FL12	0915	23:25:40	23:29:26	480	500 13	
020_FL13	0916	23:31:42	23:35:31	480	500 13	
021_FL14	0917	23:37:24	23:41:09	480	500 13	
022_FL15	0918	23:43:22	23:47:11	480	500 13	
023_FL16	0919	23:48:57	23:52:46	480	500 13	
024_FL17	0920	23:55:03	23:58:58	480	500 13	
025_FL18	0921	00:00:46	00:04:47	480	500 13	
026_FL19	0922	00:07:06	00:11:11	480	500 13	

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL08, BL09, QC62	DATE:	26 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Malekula, TongaVanuatu	CLOUDS:	Clouds @ 2500ft
BASE STATION:	PPP	WIND:	Calm @ 20
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-05
ENGINE START:	20:31	ENGINE OFF:	1:55
TAKEOFF:	20:46	LANDING:	1:46
		ENGINE TIME:	05:24
		AIR TIME	05:00

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
027_FL20	0923	00:13:18	00:17:07	480	500 13	
028_FL21	0924	00:19:24	00:23:10	480	500 13	
029_FL22	0925	00:25:25	00:28:50	480	500 13	
030_FL23	0926	00:31:30	00:34:32	480	500 13	
		00:36:17				DS: BL08_20211227_003617
000_FL4	0804	00:36:17	00:43:51	480	500 13	
001_FL5	0805	00:46:08	00:53:58	480	500 13	
002_FL6	0806	00:55:32	01:03:22	480	500 13	LD Current Clamp Error, keep
003_FL7	0807	01:05:28	01:13:28	480	500 13	LD Current Clamp Error, keep
004_FL8	0808	01:15:02	01:22:55	480	500 13	LD Current Clamp Error, keep
005_FL9	0809	01:24:42	01:32:36	480	500 13	
		01:38:00				RTB
		01:46:00				Landing
		01:48:00				Static close out at Port Vila airport

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL08, QC62	DATE:	27 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Malekula, TongaVanuatu	CLOUDS:	Clouds @ 2500ft
BASE STATION:	PPP	WIND:	5-10 kts @ 160
LIDAR DRIVE:	HE4X-07	RCD DRIVE:	RCD-05
ENGINE START:	2:36	ENGINE OFF:	6:07
TAKEOFF:	2:51	LANDING:	6:00
		ENGINE TIME:	03:31
		AIR TIME	03:09

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		02:51:00				Takeoff
		03:04:37				DS: BL08_20211227_030437
000_FL10	0810	03:04:37	03:12:37	480	500 13	
001_FL11	0811	03:14:41	03:23:03	480	500 13	
002_FL12	0812	03:24:57	03:33:12	480	500 13	
003_FL13	0813	03:35:28	03:43:44	480	500 13	LD Current Clamp Error, keep
		03:54:36				DS: BL08_20211227_035436
000_FL14	0814	03:54:36	04:02:47	480	500 13	
001_FL15	0815	04:04:54	04:13:11	480	500 13	
002_FL16	0816	04:15:30	04:23:34	480	500 13	LD Current Clamp Error, keep
003_FL17	0817	04:25:39	04:30:20	480	500 13	
004_FL18	0818	04:32:42	04:33:52	480	500 13	Eye safe high terrain, keep
005_FL20	0820	04:37:23	04:40:13	480	500 13	
006_FL19	0819	04:42:37	04:45:30	480	500 13	
007_FL21	0821	04:47:28	04:50:24	480	500 13	
008_FL22	0822	04:52:31	04:55:24	480	500 13	
009_FL23	0823	04:57:21	05:00:16	480	500 13	
010_FL24	0824	05:02:18	05:05:21	480	500 13	
011_FL25	0825	05:07:32	05:10:24	480	500 13	
012_FL26	0826	05:12:49	05:13:43	480	500 13	
013_FL18	0818	05:17:04	05:20:58	480	500 13	Eye safe high terrain, keep
		05:54:33				DS: QC62_20211227_055433
000_FL1	6201	05:54:33	05:56:24	480	500 13	
		06:00:00				Landing
		06:02:00				Static close out at Port Vila airport

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL07, QC61, QC62	DATE:	27 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Malekula, TongaVanuatu	CLOUDS:	Clouds @ 3000ft
BASE STATION:	PPP	WIND:	5-10 kts @ 350
LIDAR DRIVE:	HE4X-06	RCD DRIVE:	RCD-04
ENGINE START:	20:00	ENGINE OFF:	1:07
TAKEOFF:	20:18	LANDING:	0:58
		ENGINE TIME:	05:07
		AIR TIME	04:40

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		20:18:00				Takeoff
		20:23:29				DS: QC61_20211227_202329
000_FL1	6101	20:23:29	20:25:30	480	500 13	
		20:55:54				DS: BL07_20211227_205554
000_FL48	0795	20:55:54	20:59:58	480	500 13	
001_FL47	0747	21:05:45	21:10:54	480	500 13	
002_FL46	0746	21:13:03	21:18:05	480	500 13	
003_FL45	0745	21:19:54	21:25:06	480	500 13	
004_FL44	0744	21:27:05	21:32:40	480	500 13	
005_FL43	0743	21:34:47	21:40:35	480	500 13	
006_FL42	0742	21:42:51	21:48:34	480	500 13	
007_FL41	0741	21:50:05	21:56:05	480	500 13	
008_FL40	0740	21:58:11	22:04:32	480	500 13	
009_FL39	0739	22:06:27	22:12:50	480	500 13	
010_FL38	0738	22:14:53	22:21:26	480	500 13	
011_FL37	0737	22:23:09	22:29:52	480	500 13	
012_FL36	0736	22:32:06	22:39:05	480	500 13	
013_FL35	0735	22:40:49	22:47:53	480	500 13	
014_FL34	0734	22:50:14	22:57:43	480	500 13	
015_FL33	0733	22:59:40	23:06:45	480	500 13	
016_FL32	0732	23:09:05	23:16:34	480	500 13	
017_FL31	0731	23:18:29	23:25:39	480	500 13	LD Current Clamp Error
018_FL30	0730	23:27:42	23:35:05	480	500 13	LD Current Clamp Error
019_FL29	0729	23:37:14	23:41:03	480	500 13	BAD: Serial Pipe 2 Error
		23:50:49				DS: BL07_20211227_235049
000_FL29	0729	23:50:49	23:57:37	480	500 13	
001_FL28	0728	00:00:06	00:01:10	480	500 13	BAD: Serial Pipe 2 Error
		00:27:55				DS: QC62_20211228_002755
000_FL1	6201	00:27:55	00:29:08	480	500 13	Serial Pipe 2 Error
		00:54:49				DS: QC62_20211228_005449
000_FL1	6201	00:54:49	00:56:19	480	500 13	Serial Pipe 2 Error

PROJECT NAME:		2021-81155 SPC TongaVanuatu		BASE AIRPORT:		Port Vila (VLI)	
LOCATION / AREA:		Tonga and Vanuatu / BL07, QC61, QC62		DATE:		27 December 2021	
AIRCRAFT:		ZK-XLF		PILOT:		Jabin L.	
SYSTEM:		Hawkeye 4X		OPERATOR:		Brent H.	
MISSION ID:		Malekula, TongaVanuatu		CLOUDS:		Clouds @ 3000ft	
BASE STATION:		PPP		WIND:		5-10 kts @ 350	
LIDAR DRIVE:		HE4X-06		RCD DRIVE:		RCD-04	
ENGINE START:		20:00	ENGINE OFF:	1:07	ENGINE TIME:		05:07
TAKEOFF:		20:18	LANDING:	0:58	AIR TIME		04:40
FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR		REMARKS
		00:58:00					Landing
		01:00:00					Static close out at Port Vila airport

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL17, BL18	DATE:	28 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Efate	CLOUDS:	Clouds @ 3000ft
BASE STATION:	PPP	WIND:	10-15 kts @ 350
LIDAR DRIVE:	HE4X-07	RCD DRIVE:	RCD-03
ENGINE START:	21:35	ENGINE OFF:	4:35
TAKEOFF:	22:38	LANDING:	4:28
		ENGINE TIME:	07:00
		AIR TIME	05:50

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		22:38:00				Takeoff
		23:26:09				DS: BL18_20211228_232609
000_FL27	1827	23:26:09	23:29:15	480	500 13	
001_FL28	1828	23:31:06	23:34:25	480	500 13	
002_FL29	1829	23:36:33	23:40:17	480	500 13	
003_FL30	1830	23:42:13	23:46:10	480	500 13	
004_FL31	1831	23:48:13	23:52:19	480	500 13	
005_FL32	1832	23:54:24	23:59:07	480	500 13	
006_FL33	1833	00:00:46	00:06:08	480	500 13	
007_FL34	1834	00:07:39	00:12:42	480	500 13	
008_FL35	1835	00:14:29	00:19:30	480	500 13	
009_FL36	1836	00:21:37	00:27:21	480	500 13	
010_FL37	1837	00:29:06	00:34:50	480	500 13	
011_FL38	1838	00:36:38	00:42:20	480	500 13	
012_FL39	1839	00:44:02	00:49:48	480	500 13	
013_FL40	1840	00:51:50	00:57:24	480	500 13	
014_FL41	1841	00:59:41	01:05:27	480	500 13	
015_FL42	1842	01:07:06	01:12:35	480	500 13	
016_FL43	1843	01:14:33	01:20:07	480	500 13	
017_FL44	1844	01:21:41	01:27:04	480	500 13	
018_FL45	1845	01:29:00	01:34:27	480	500 13	
019_FL46	1846	01:36:37	01:41:52	480	500 13	
020_FL47	1847	01:43:57	01:49:30	480	500 13	
		01:53:16				DS: BL17_20211229_015316
000_FL29	1729	01:53:16	01:53:28	480	500 13	BAD: Premature start
001_FL29	1729	01:53:34	01:53:38	480	500 13	BAD: Rain on line, aborted
002_FL1	1701	02:00:09	02:06:13	480	500 13	
003_FL2	1702	02:08:36	02:14:46	480	500 13	
004_FL3	1703	02:16:40	02:22:49	480	500 13	
005_FL4	1704	02:24:37	02:30:36	480	500 13	
006_FL5	1705	02:32:51	02:39:05	480	500 13	

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL17, BL18	DATE:	28 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Efate	CLOUDS:	Clouds @ 3000ft
BASE STATION:	PPP	WIND:	10-15 kts @ 350
LIDAR DRIVE:	HE4X-07	RCD DRIVE:	RCD-03
ENGINE START:	21:35	ENGINE OFF:	4:35
TAKEOFF:	22:38	LANDING:	4:28
		ENGINE TIME:	07:00
		AIR TIME	05:50

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR		REMARKS
007_FL6	1706	02:40:53	02:46:43	480	500	13	
008_FL7	1707	02:48:58	02:54:48	480	500	13	
009_FL8	1708	02:56:13	03:01:59	480	500	13	
010_FL9	1709	03:04:05	03:09:52	480	500	13	
011_FL10	1710	03:11:58	03:17:26	480	500	13	
012_FL11	1711	03:19:22	03:24:52	480	500	13	
013_FL12	1712	03:26:39	03:32:17	480	500	13	
014_FL13	1713	03:34:35	03:39:58	480	500	13	
015_FL14	1714	03:41:50	03:47:00	480	500	13	
016_FL15	1715	03:48:54	03:54:14	480	500	13	
017_FL16	1716	03:56:15	04:01:19	480	500	13	
018_FL17	1717	04:03:11	04:08:13	480	500	13	
019_FL18	1718	04:10:05	04:15:04	480	500	13	
020_FL19	1719	04:16:56	04:21:55	480	500	13	
		04:28:00					Landing
		04:29:00					Static close out at Port Vila airport

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL07, QC61	DATE:	29 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Malekula, TongaVanuatu	CLOUDS:	Clouds @ 2000ft
BASE STATION:	PPP	WIND:	5-10 kts @ 140
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-04
ENGINE START:	21:17	ENGINE OFF:	1:53
TAKEOFF:	21:35	LANDING:	1:45
		ENGINE TIME:	04:36
		AIR TIME	04:10

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		21:35:00				Takeoff
		21:38:15				DS: QC61_20211229_213815
000_FL1	6101	21:38:15	21:40:08	480	500	
		22:07:24				DS: BL07_20211229_220724
000_FL31	0731	22:07:24	22:14:28	480	500	13
001_FL30	0730	22:16:39	22:23:49	480	500	13
002_FL28	0728	22:25:19	22:32:13	480	500	13
003_FL27	0727	22:34:22	22:41:29	480	500	13
004_FL26	0726	22:43:24	22:50:01	480	500	13
005_FL25	0725	22:52:05	22:58:40	480	500	13
006_FL24	0724	23:00:24	23:06:56	480	500	13
007_FL23	0723	23:08:59	23:12:17	480	500	13
008_FL1	0701	23:15:21	23:16:34	480	500	13
009_FL2	0702	23:18:15	23:20:11	480	500	13
010_FL3	0703	23:22:12	23:24:30	480	500	13
011_FL4	0704	23:26:14	23:29:02	480	500	13
012_FL5	0705	23:30:47	23:33:49	480	500	13
013_FL6	0706	23:35:58	23:39:41	480	500	13
014_FL7	0707	23:41:40	23:45:28	480	500	13
015_FL8	0708	23:47:36	23:52:09	480	500	13
016_FL9	0709	23:54:15	23:58:42	480	500	13
017_FL10	0710	00:00:58	00:05:55	480	500	13
018_FL11	0711	00:08:17	00:13:09	480	500	13
019_FL12	0712	00:14:37	00:19:51	480	500	13
020_FL13	0713	00:21:39	00:26:53	480	500	13
021_FL14	0714	00:28:42	00:34:25	480	500	13
022_FL15	0715	00:36:46	00:42:19	480	500	13
023_FL16	0716	00:44:49	00:50:49	480	500	13
024_FL17	0717	00:53:02	00:57:21	480	500	13
025_FL18	0718	01:01:16	01:03:35	480	500	13
026_FL17	0717	01:08:13	01:14:13	480	500	13

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL07, QC61	DATE:	29 December 2021
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Malekula, TongaVanuatu	CLOUDS:	Clouds @ 2000ft
BASE STATION:	PPP	WIND:	5-10 kts @ 140
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-04
ENGINE START:	21:17	ENGINE OFF:	1:53
TAKEOFF:	21:35	LANDING:	1:45
		ENGINE TIME:	04:36
		AIR TIME	04:10

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR		REMARKS
		01:40:26					DS: QC61_20211230_014026
000_FL1	6101	01:40:26	01:42:17	480	500	13	
		01:45:00					Landing
		01:47:00					Static close out at Port Vila airport

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL01, BL02, BL17, QC61, QC62	DATE:	3 January 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Efate, Tanna, TongaVanuatu	CLOUDS:	Clouds @ 2000ft
BASE STATION:	AAMPri and PPP	WIND:	10 kts @ 350
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-04
ENGINE START:	19:31	ENGINE OFF:	0:32
TAKEOFF:	19:47	LANDING:	0:22
		ENGINE TIME:	05:01
		AIR TIME	04:35

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		19:47:00				Takeoff
		19:54:05				DS: QC61_20220103_195405
000_FL1	6101	19:54:05	19:54:14	480	500 13	BAD: False start
001_FL1	6101	19:54:59	19:55:05	480	500 13	BAD: False start
002_FL1	6101	19:56:48	19:58:35	480	500 13	
		20:04:26				DS: BL17_20220103_200426
000_FL29	1729	20:04:26	20:10:28	480	500 13	
001_FL28	1728	20:12:07	20:18:12	480	500 13	
002_FL27	1727	20:19:55	20:25:56	480	500 13	
003_FL26	1726	20:27:35	20:33:29	480	500 13	
004_FL25	1725	20:35:09	20:41:15	480	500 13	
005_FL20	1720	20:42:20	20:46:12	480	500 13	BAD: Eye safe
		21:42:30				DS: BL02_20220103_214230
000_FL69	0265	21:42:30	21:44:27	480	500 13	
001_FL70	0266	21:46:36	21:47:56	480	500 13	
002_FL71	0267	21:50:54	21:52:21	480	500 13	
		22:04:44				DS: BL01_20220103_220444
000_FL88	0185	22:04:44	22:05:58	480	500 13	
001_FL87	0184	22:09:13	22:10:40	480	500 13	
002_FL86	0183	22:13:10	22:14:28	480	500 13	
003_FL85	0182	22:17:35	22:19:17	480	500 13	
004_FL84	0181	22:21:21	22:22:57	480	500 13	
005_FL89	0186	22:25:31	22:26:46	480	500 13	
006_FL90	0187	22:28:48	22:30:08	480	500 13	
007_FL91	0188	22:32:36	22:33:50	480	500 13	
008_FL92	0189	22:36:04	22:37:19	480	500 13	
009_FL93	0190	22:39:49	22:41:00	480	500 13	
010_FL94	0191	22:43:02	22:43:54	480	500 13	
011_FL83	0180	22:45:28	22:46:54	480	500 13	
012_FL82	0179	22:49:10	22:50:11	480	500 13	
013_FL81	0178	22:52:46	22:53:46	480	500 13	

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL01, BL02, BL17, QC61, QC62	DATE:	3 January 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Efate, Tanna, TongaVanuatu	CLOUDS:	Clouds @ 2000ft
BASE STATION:	AAMPri and PPP	WIND:	10 kts @ 350
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-04
ENGINE START:	19:31	ENGINE OFF:	0:32
TAKEOFF:	19:47	LANDING:	0:22
		ENGINE TIME:	05:01
		AIR TIME	04:35

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
014_FL80	0177	22:55:09	22:56:41	480	500 13	
015_FL79	0176	22:59:07	23:00:33	480	500 13	
016_FL78	0175	23:01:31	23:02:57	480	500 13	
017_FL77	0174	23:05:00	23:06:23	480	500 13	
018_FL76	0173	23:07:57	23:08:59	480	500 13	
019_FL75	0172	23:12:03	23:13:21	480	500 13	
020_FL70	0167	23:21:42	23:22:41	480	500 13	
021_FL69	0166	23:24:27	23:25:37	480	500 13	
022_FL74	0171	23:26:47	23:28:22	480	500 13	
023_FL73	0170	23:30:37	23:32:12	480	500 13	
024_FL68	0165	23:34:33	23:35:47	480	500 13	
025_FL72	0169	23:37:53	23:39:23	480	500 13	
		00:16:44				DS: QC62_20220104_001644
000_FL1	6201	00:16:44	00:18:24	480	500 13	
		00:22:00				Landing
		00:24:00				Static close out at Port Vila airport

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL17	DATE:	4 January 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Efate	CLOUDS:	Clouds @ 2000ft
BASE STATION:	AAMPri	WIND:	15 kts @ 30
LIDAR DRIVE:	HE4X-06	RCD DRIVE:	RCD-04
ENGINE START:	0:55	ENGINE OFF:	2:21
TAKEOFF:	1:10	LANDING:	2:13
		ENGINE TIME:	01:26
		AIR TIME	01:03

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR		REMARKS
		01:10:00					Takeoff
		01:17:34					DS: BL17_20220104_011734
000_FL24	1724	01:17:34	01:23:06	480	500	13	
001_FL23	1723	01:25:01	01:30:55	480	500	13	
002_FL22	1722	01:32:49	01:38:42	480	500	13	
003_FL21	1721	01:41:00	01:46:56	480	500	13	
004_FL20	1720	01:48:53	01:54:45	480	500	13	
		02:13:00					Landing
		02:15:00					Static close out at Port Vila airport

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL07, BL08, QC61	DATE:	4 January 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Malekula, TongaVanuatu	CLOUDS:	Clouds @ 1500ft
BASE STATION:	AAMPri	WIND:	5-10 kts @ 340
LIDAR DRIVE:	HE4X-07	RCD DRIVE:	RCD-05
ENGINE START:	19:41	ENGINE OFF:	21:51
TAKEOFF:	20:08	LANDING:	21:49
		ENGINE TIME:	02:10
		AIR TIME	01:41

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR		REMARKS
		20:08:00					Takeoff
		20:13:24					DS: QC61_20220104_201324
000_FL1	6101	20:13:24	20:15:18	480	500	13	BAD:
		20:42:20					DS: BL07_20220104_204220
000_FL18	0718	20:42:20	20:45:50	480	500	13	BAD:
		20:52:21					DS: BL08_20220104_205221
000_FL16	0816	20:52:21	20:52:53	480	500	13	BAD:
001_FL16	0816	20:56:37	21:02:15	480	500	13	BAD:
		21:49:00					Landing
		21:50:00					No static as no usable data collected

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL02, BL03, QC61, QC62	DATE:	6 January 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Tanna, TongaVanuatu	CLOUDS:	Clouds @ 2000ft
BASE STATION:	PPP	WIND:	20-25 kts @ 100
LIDAR DRIVE:	HE4X-07	RCD DRIVE:	RCD-05
ENGINE START:	19:33	ENGINE OFF:	0:52
TAKEOFF:	19:54	LANDING:	0:40
		ENGINE TIME:	05:19
		AIR TIME	04:46

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		19:54:00				Takeoff
		19:58:40				DS: QC61_20220106_195840
000_FL1	6101	19:58:40	20:00:25	480	500	
		20:40:24				DS: BL02_20220106_204024
000_FL32	0232	20:40:24	20:46:15	480	500	13
001_FL31	0231	20:48:24	20:54:22	480	500	13
002_FL30	0230	20:56:44	21:03:03	480	500	13
003_FL33	0233	21:05:01	21:10:44	480	500	13
		21:14:49				DS: BL03_20220106_211449
000_FL44	0344	21:14:49	21:17:16	620	440	14
001_FL1	0301	21:19:10	21:20:23	620	440	14
002_FL43	0343	21:23:18	21:25:27	620	440	14
003_FL42	0342	21:27:32	21:28:34	620	440	14
004_FL41	0341	21:29:24	21:30:33	620	440	14
005_FL40	0340	21:33:11	21:34:21	620	440	14
006_FL39	0339	21:36:03	21:37:16	620	440	14
007_FL38	0338	21:39:55	21:41:53	620	440	14
008_FL37	0337	21:43:47	21:44:49	620	440	14
009_FL36	0336	21:46:49	21:47:57	620	440	14
010_FL35	0335	21:49:48	21:51:24	620	440	14
011_FL34	0334	21:53:14	21:54:21	620	440	14
012_FL33	0333	21:56:03	21:57:01	620	440	14
013_FL32	0332	21:59:28	22:00:46	620	440	14
014_FL31	0331	22:03:38	22:05:54	620	440	14
015_FL30	0330	22:07:42	22:08:58	620	440	14
016_FL29	0329	22:11:16	22:12:17	620	440	14
017_FL28	0328	22:14:53	22:16:32	620	440	14
018_FL25	0325	22:16:51	22:18:03	620	440	14
019_FL26	0326	22:20:00	22:21:29	620	440	14
020_FL27	0327	22:24:13	22:25:06	620	440	14
021_FL24	0324	22:27:53	22:28:50	620	440	14

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL02, BL03, QC61, QC62	DATE:	6 January 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.

MISSION ID:	Tanna, TongaVanuatu	CLOUDS:	Clouds @ 2000ft
BASE STATION:	PPP	WIND:	20-25 kts @ 100
LIDAR DRIVE:	HE4X-07	RCD DRIVE:	RCD-05

ENGINE START:	19:33	ENGINE OFF:	0:52	ENGINE TIME:	05:19
TAKEOFF:	19:54	LANDING:	0:40	AIR TIME	04:46

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO		REMARKS
					PRF	PWR	
022_FL21	0321	22:29:10	22:30:29	620	440	14	
023_FL22	0322	22:33:10	22:34:45	620	440	14	
024_FL23	0323	22:37:27	22:38:32	620	440	14	
025_FL20	0320	22:41:07	22:42:46	620	440	14	
026_FL19	0319	22:45:17	22:46:23	620	440	14	
027_FL18	0318	22:50:12	22:51:37	620	440	14	
028_FL17	0317	22:53:48	22:54:44	620	440	14	
029_FL16	0316	22:58:02	22:59:19	620	440	14	
030_FL15	0315	23:01:45	23:03:05	620	440	14	
031_FL12	0312	23:04:46	23:06:13	620	440	14	
032_FL9	0309	23:08:36	23:09:39	620	440	14	
033_FL10	0310	23:10:53	23:11:58	620	440	14	
034_FL11	0311	23:16:34	23:17:25	620	440	14	
035_FL14	0314	23:21:27	23:22:37	620	440	14	
036_FL13	0313	23:24:02	23:25:37	620	440	14	
037_FL6	0306	23:25:54	23:27:14	620	440	14	
038_FL7	0307	23:29:29	23:30:31	620	440	14	
039_FL8	0308	23:31:57	23:33:09	620	440	14	
040_FL3	0303	23:34:36	23:35:37	620	440	14	
041_FL5	0305	23:37:13	23:38:54	620	440	14	
042_FL4	0304	23:40:48	23:42:31	620	440	14	
043_FL2	0302	23:44:49	23:46:36	620	440	14	
		23:57:24					DS: BL02_20220106_235724
000_FL65	0295	23:57:24	23:59:00	480	500	13	
		00:36:06					DS: QC62_20220107_003606
000_FL1	6201	00:36:06	00:37:57	480	500	13	
		00:40:00					Landing
		00:42:00					Static close out at Port Vila airport

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL06, BL07, BL08, QC61, QC62	DATE:	7 January 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Malekula, TongaVanuatu	CLOUDS:	Clouds @ 2500ft
BASE STATION:	PPP	WIND:	5-10 kts @ 90
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-05
ENGINE START:	1:14	ENGINE OFF:	6:30
TAKEOFF:	1:30	LANDING:	6:22
		ENGINE TIME:	05:16
		AIR TIME	04:52

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		01:30:00				Takeoff
		01:36:48				DS: QC61_20220107_013648
000_FL1	6101	01:36:48	01:38:34	480	500 13	
		02:08:44				DS: BL08_20220107_020844
000_FL6	0806	02:08:44	02:16:40	480	500 13	
001_FL7	0807	02:19:00	02:27:00	480	500 13	
002_FL8	0808	02:29:34	02:37:12	480	500 13	
003_FL9	0809	02:39:15	02:47:03	480	500 13	
004_FL13	0813	02:48:54	02:56:48	480	500 13	
005_FL16	0816	02:59:11	03:07:07	480	500 13	
		03:09:43				DS: BL07_20220107_030943
000_FL31	0731	03:09:43	03:16:59	480	500 13	
001_FL30	0730	03:19:12	03:25:56	480	500 13	
002_FL22	0722	03:27:38	03:34:12	480	500 13	
003_FL21	0721	03:36:30	03:42:53	480	500 13	
004_FL20	0720	03:45:09	03:51:45	480	500 13	
005_FL19	0719	03:53:29	03:59:31	480	500 13	
006_FL18	0718	04:02:06	04:08:11	480	500 13	
		04:13:39				DS: BL06_20220107_041339
000_FL52	0695	04:13:39	04:17:09	480	500 13	
001_FL51	0651	04:20:04	04:21:52	480	500 13	
002_FL50	0650	04:24:12	04:26:06	480	500 13	
003_FL49	0649	04:28:10	04:30:12	480	500 13	
004_FL48	0648	04:32:01	04:34:02	480	500 13	
005_FL47	0647	04:36:10	04:38:09	480	500 13	
006_FL46	0646	04:40:23	04:42:22	480	500 13	
007_FL45	0645	04:44:52	04:46:41	480	500 13	
008_FL44	0644	04:48:50	04:49:44	480	500 13	
009_FL43	0643	04:51:53	04:52:49	480	500 13	
010_FL1	0601	04:55:03	04:56:14	480	500 13	
011_FL2	0602	04:58:16	04:59:42	480	500 13	

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL06, BL07, BL08, QC61, QC62	DATE:	7 January 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Malekula, TongaVanuatu	CLOUDS:	Clouds @ 2500ft
BASE STATION:	PPP	WIND:	5-10 kts @ 90
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-05
ENGINE START:	1:14	ENGINE OFF:	6:30
TAKEOFF:	1:30	LANDING:	6:22
		ENGINE TIME:	05:16
		AIR TIME	04:52

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
012_FL3	0603	05:01:46	05:01:55	480	500 13	BAD: Eye safe, terrain
013_FL3	0603	05:05:05	05:06:17	480	500 13	
014_FL4	0604	05:08:22	05:09:52	480	500 13	
015_FL5	0605	05:12:31	05:13:41	480	500 13	
016_FL6	0606	05:15:51	05:17:13	480	500 13	
017_FL7	0607	05:19:33	05:21:21	480	500 13	
018_FL8	0608	05:23:39	05:27:50	480	500 13	
019_FL9	0609	05:30:13	05:34:26	480	500 13	
020_FL10	0610	05:36:27	05:40:51	480	500 13	
021_FL11	0611	05:43:00	05:47:00	480	500 13	
		06:16:14				DS: QC62_20220107_061614
000_FL1	6201	06:16:14	06:18:02	480	500 13	
		06:22:00				Landing
		06:23:00				Static close out at Port Vila airport

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL21, BL22, BL23	DATE:	7 January 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.

MISSION ID:	Efate	CLOUDS:	Clouds @ 3500ft
BASE STATION:	AAMPri	WIND:	15-20 kts @ 200
LIDAR DRIVE:	HE4X-01	RCD DRIVE:	RCD-03

ENGINE START:	19:47	ENGINE OFF:	0:05	ENGINE TIME:	04:18
TAKEOFF:	20:06	LANDING:	0:00	AIR TIME	03:54

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		20:06:00				Takeoff
		20:14:49				DS: BL21_20220107_201449
000_FL71	2169	20:14:49	20:15:36	620	440 14	Deep paused after start
001_FL70	2168	20:17:55	20:18:49	620	440 14	
002_FL69	2167	20:21:07	20:22:01	620	440 14	
003_FL68	2166	20:24:12	20:25:05	620	440 14	
004_FL67	2165	20:27:31	20:28:31	620	440 14	
005_FL66	2164	20:31:06	20:32:06	620	440 14	
006_FL65	2163	20:33:29	20:34:17	620	440 14	
007_FL64	2162	20:36:37	20:37:36	620	440 14	
008_FL63	2161	20:39:42	20:40:38	620	440 14	
009_FL58	2156	20:44:06	20:45:08	620	440 14	
010_FL51	2150	20:45:35	20:46:40	620	440 14	Eye safe at end, terrain
011_FL54	2152	20:49:15	20:50:16	620	440 14	
012_FL61	2159	20:51:10	20:51:53	620	440 14	
013_FL59	2157	20:53:39	20:54:43	620	440 14	
014_FL50	2149	20:55:13	20:56:05	620	440 14	
015_FL62	2160	20:57:59	20:58:12	620	440 14	BAD:
016_FL60	2158	20:59:27	21:00:28	620	440 14	
017_FL55	2153	21:01:32	21:03:54	620	440 14	
018_FL57	2155	21:04:42	21:05:16	620	440 14	BAD:
		21:11:09				DS: BL21_20220107_211109
000_FL57	2155	21:11:09	21:12:41	620	440 14	
001_FL56	2154	21:14:21	21:15:49	620	440 14	
002_FL53	2151	21:17:44	21:18:53	620	440 14	
003_FL45	2145	21:21:10	21:22:40	620	440 14	
004_FL47	2146	21:24:32	21:25:56	620	440 14	Eye safe at end, terrain
005_FL48	2147	21:28:12	21:29:33	620	440 14	
006_FL49	2148	21:31:28	21:32:49	620	440 14	Eye safe at end, terrain
007_FL62	2160	21:35:31	21:36:19	620	440 14	
		21:41:27				DS: BL22_20220107_214127

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL21, BL22, BL23	DATE:	7 January 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Efate	CLOUDS:	Clouds @ 3500ft
BASE STATION:	AAMPri	WIND:	15-20 kts @ 200
LIDAR DRIVE:	HE4X-01	RCD DRIVE:	RCD-03
ENGINE START:	19:47	ENGINE OFF:	0:05
TAKEOFF:	20:06	LANDING:	0:00
		ENGINE TIME:	04:18
		AIR TIME	03:54

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
000_FL30	2230	21:41:27	21:42:21	620	440 14	
001_FL28	2228	21:43:55	21:44:57	620	440 14	
002_FL29	2229	21:46:50	21:47:59	620	440 14	
003_FL25	2225	21:49:44	21:50:48	620	440 14	
004_FL22	2222	21:53:02	21:54:18	620	440 14	
005_FL27	2227	21:56:49	21:57:50	620	440 14	
006_FL23	2223	22:03:01	22:04:17	620	440 14	
007_FL26	2226	22:05:58	22:07:03	620	440 14	
008_FL24	2224	22:08:45	22:10:03	620	440 14	
		22:12:18				DS: BL23_20220107_221218
000_FL1	2301	22:12:18	22:13:24	620	440 14	
001_FL2	2302	22:15:31	22:16:41	620	440 14	
002_FL3	2303	22:18:43	22:19:50	620	440 14	
003_FL23	2323	22:22:30	22:23:31	620	440 14	
004_FL22	2322	22:25:28	22:26:35	620	440 14	
005_FL21	2321	22:28:47	22:30:24	620	440 14	
006_FL20	2320	22:32:42	22:34:07	620	440 14	
007_FL17	2317	22:36:06	22:38:13	620	440 14	
008_FL18	2318	22:40:17	22:42:17	620	440 14	
009_FL19	2319	22:44:44	22:46:24	620	440 14	
010_FL8	2308	22:46:51	22:48:11	620	440 14	
011_FL9	2309	22:50:22	22:51:42	620	440 14	
012_FL10	2310	22:53:55	22:55:21	620	440 14	
013_FL11	2311	22:57:24	22:58:42	620	440 14	
014_FL12	2312	23:00:49	23:02:11	620	440 14	
015_FL13	2313	23:03:54	23:05:09	620	440 14	
016_FL14	2314	23:07:09	23:08:21	620	440 14	
017_FL15	2315	23:10:19	23:11:23	620	440 14	
018_FL16	2316	23:13:30	23:14:32	620	440 14	
019_FL29	2329	23:14:56	23:16:28	620	440 14	
020_FL30	2330	23:17:35	23:19:11	620	440 14	

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL21, BL22, BL23	DATE:	7 January 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Efate	CLOUDS:	Clouds @ 3500ft
BASE STATION:	AAMPri	WIND:	15-20 kts @ 200
LIDAR DRIVE:	HE4X-01	RCD DRIVE:	RCD-03
ENGINE START:	19:47	ENGINE OFF:	0:05
TAKEOFF:	20:06	LANDING:	0:00
		ENGINE TIME:	04:18
		AIR TIME	03:54

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO		REMARKS
					PRF	PWR	
021_FL7	2307	23:21:38	23:22:37	620	440	14	
022_FL28	2328	23:25:23	23:27:08	620	440	14	
023_FL26	2326	23:29:13	23:30:47	620	440	14	
024_FL25	2325	23:32:50	23:34:28	620	440	14	
025_FL27	2327	23:35:45	23:37:03	620	440	14	
026_FL24	2324	23:43:02	23:44:43	620	440	14	
027_FL5	2305	23:46:53	23:48:16	620	440	14	
028_FL6	2306	23:50:25	23:51:16	620	440	14	
029_FL4	2304	23:53:16	23:55:04	620	440	14	
		00:00:00					Landing
		00:02:00					Static close out at Port Vila airport

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL15, BL16, BL17, BL18, BL19	DATE:	8 January 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.

MISSION ID:	Efate	CLOUDS:	Clouds @ 3000ft
BASE STATION:	AAMPri	WIND:	10-15 kts @ 220
LIDAR DRIVE:	HE4X-06	RCD DRIVE:	RCD-03

ENGINE START:	0:35	ENGINE OFF:	3:01	ENGINE TIME:	02:26
TAKEOFF:	0:52	LANDING:	2:54	AIR TIME	02:02

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		00:52:00				Takeoff
		00:59:31				DS: BL19_20220108_005931
000_FL56	1953	00:59:31	01:00:27	480	500 13	
001_FL58	1955	01:01:48	01:02:52	480	500 13	
002_FL57	1954	01:05:06	01:06:10	480	500 13	
003_FL55	1952	01:07:42	01:08:42	480	500 13	
		01:12:02				DS: BL18_20220108_011202
000_FL92	1885	01:12:02	01:13:22	480	500 13	
001_FL90	1884	01:15:58	01:17:19	480	500 13	
002_FL89	1883	01:20:30	01:21:43	480	500 13	
003_FL88	1882	01:24:16	01:25:34	480	500 13	
004_FL79	1876	01:27:48	01:29:24	480	500 13	
005_FL78	1875	01:32:02	01:33:45	480	500 13	
006_FL77	1874	01:36:27	01:38:00	480	500 13	
007_FL87	1881	01:39:55	01:40:58	480	500 13	
008_FL86	1880	01:42:46	01:43:52	480	500 13	
009_FL85	1879	01:46:03	01:47:18	480	500 13	
010_FL76	1873	01:49:24	01:51:26	480	500 13	
011_FL83	1878	01:53:50	01:54:54	480	500 13	
012_FL82	1877	01:57:11	01:58:20	480	500 13	
013_FL71	1868	01:59:09	02:00:33	480	500 13	
014_FL72	1869	02:02:36	02:04:01	480	500 13	
015_FL75	1872	02:06:55	02:08:21	480	500 13	
016_FL74	1871	02:10:59	02:12:20	480	500 13	
017_FL73	1870	02:15:43	02:16:43	480	500 13	
		02:17:48				DS: BL17_20220108_021748
000_FL40	1739	02:17:48	02:19:02	480	500 13	
		02:21:25				DS: BL16_20220108_022125
000_FL36	1636	02:21:25	02:22:57	480	500 13	
001_FL35	1635	02:24:52	02:26:12	480	500 13	
		02:28:58				DS: BL15_20220108_022858

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL15, BL16, BL17, BL18, BL19	DATE:	8 January 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Efate	CLOUDS:	Clouds @ 3000ft
BASE STATION:	AAMPri	WIND:	10-15 kts @ 220
LIDAR DRIVE:	HE4X-06	RCD DRIVE:	RCD-03
ENGINE START:	0:35	ENGINE OFF:	3:01
TAKEOFF:	0:52	LANDING:	2:54
		ENGINE TIME:	02:26
		AIR TIME	02:02

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR		REMARKS
000_FL33	1533	02:28:58	02:31:56	480	500	13	
001_FL32	1532	02:33:26	02:34:24	480	500	13	
002_FL14	1514	02:36:56	02:38:54	480	500	13	
003_FL13	1513	02:40:58	02:42:47	480	500	13	
004_FL12	1512	02:45:11	02:46:51	480	500	13	
		02:54:00					Landing
		02:55:00					Static close out at Port Vila airport

PROJECT NAME: 2021-81155 SPC TongaVanuatu
LOCATION / AREA: Tonga and Vanuatu / BL06, QC62
AIRCRAFT: ZK-XLF
SYSTEM: Hawkeye 4X

BASE AIRPORT: Port Vila (VLI)
DATE: 9 January 2022
PILOT: Jabin L.
OPERATOR: Brent H.

MISSION ID: Malekula, TongaVanuatu
BASE STATION: PPP
LIDAR DRIVE: HE4X-07

CLOUDS: Clouds @ 2000ft
WIND: 10-15 kts @ 130
RCD DRIVE: RCD-05

ENGINE START: 20:12 **ENGINE OFF:** 1:29 **ENGINE TIME:** 05:17
TAKEOFF: 20:31 **LANDING:** 1:24 **AIR TIME** 04:53

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR		REMARKS
		20:31:00					Takeoff
		20:36:07					DS: QC62_20220109_203607
000_FL1	6201	20:36:07	20:37:52	480	500	13	
		21:05:46					DS: BL06_20220109_210546
000_FL12	0612	21:05:46	21:10:20	480	500	13	
001_FL42	0642	21:12:53	21:17:01	480	500	13	
002_FL39	0639	21:19:11	21:24:03	480	500	13	
003_FL36	0636	21:25:38	21:30:47	480	500	13	
004_FL33	0633	21:32:48	21:38:07	480	500	13	
005_FL32	0632	21:40:02	21:45:28	480	500	13	
006_FL31	0631	21:47:25	21:52:59	480	500	13	
007_FL30	0630	21:55:01	22:00:50	480	500	13	
008_FL29	0629	22:03:00	22:08:35	480	500	13	
009_FL28	0628	22:10:33	22:16:13	480	500	13	
010_FL35	0635	22:17:16	22:22:29	480	500	13	
011_FL34	0634	22:24:42	22:30:06	480	500	13	
012_FL27	0627	22:31:46	22:37:28	480	500	13	
013_FL26	0626	22:39:40	22:45:23	480	500	13	
014_FL25	0625	22:48:01	22:53:40	480	500	13	
015_FL24	0624	22:56:11	23:01:46	480	500	13	
016_FL23	0623	23:04:06	23:09:38	480	500	13	
017_FL22	0622	23:12:35	23:17:57	480	500	13	
018_FL21	0621	23:21:14	23:26:46	480	500	13	
019_FL20	0620	23:28:57	23:34:14	480	500	13	
020_FL19	0619	23:36:42	23:41:57	480	500	13	
021_FL18	0618	23:44:22	23:49:36	480	500	13	
022_FL17	0617	23:51:39	23:54:02	480	500	13	BAD: Eye Safe
023_FL38	0638	23:57:57	00:02:51	480	500	13	
024_FL37	0637	00:05:12	00:10:12	480	500	13	
025_FL16	0616	00:12:53	00:17:57	480	500	13	
026_FL17	0617	00:20:14	00:25:25	480	500	13	

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL06, QC62	DATE:	9 January 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Malekula, TongaVanuatu	CLOUDS:	Clouds @ 2000ft
BASE STATION:	PPP	WIND:	10-15 kts @ 130
LIDAR DRIVE:	HE4X-07	RCD DRIVE:	RCD-05
ENGINE START:	20:12	ENGINE OFF:	1:29
TAKEOFF:	20:31	LANDING:	1:24
		ENGINE TIME:	05:17
		AIR TIME	04:53

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR		REMARKS
027_FL15	0615	00:28:13	00:33:08	480	500	13	
028_FL14	0614	00:35:35	00:40:19	480	500	13	
029_FL13	0613	00:42:45	00:47:31	480	500	13	
030_FL40	0640	00:50:08	00:55:07	480	500	13	
031_FL41	0641	00:57:00	01:01:46	480	500	13	
		01:24:00					Landing
		01:26:00					Static close out at Santo airport

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL05, QC62	DATE:	10 January 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Malekula, TongaVanuatu	CLOUDS:	Clouds @ 3000ft
BASE STATION:	PPP	WIND:	15-20 kts @ 170
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-05
ENGINE START:	2:25	ENGINE OFF:	6:05
TAKEOFF:	2:42	LANDING:	5:58
		ENGINE TIME:	03:40
		AIR TIME	03:16

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		02:42:00				Takeoff
		03:07:10				DS: BL05_20220110_030710
000_FL44	0544	03:07:10	03:08:15	480	500 13	
001_FL43	0543	03:09:54	03:11:11	480	500 13	
002_FL42	0542	03:13:11	03:14:38	480	500 13	
003_FL41	0541	03:16:46	03:18:06	480	500 13	
004_FL40	0540	03:20:03	03:21:42	480	500 13	
005_FL39	0539	03:23:37	03:25:01	480	500 13	
006_FL38	0538	03:26:46	03:28:11	480	500 13	
007_FL37	0537	03:30:05	03:31:30	480	500 13	
008_FL36	0536	03:33:23	03:34:41	480	500 13	
009_FL35	0535	03:36:46	03:38:00	480	500 13	
010_FL1	0501	03:40:25	03:41:36	480	500 13	
011_FL2	0502	03:43:23	03:44:47	480	500 13	
012_FL3	0503	03:46:41	03:48:20	480	500 13	
013_FL4	0504	03:49:47	03:51:42	480	500 13	
014_FL5	0505	03:53:51	03:55:54	480	500 13	
015_FL6	0506	03:57:26	03:59:34	480	500 13	
016_FL7	0507	04:01:42	04:03:59	480	500 13	
017_FL8	0508	04:05:21	04:09:12	480	500 13	
018_FL9	0509	04:10:38	04:14:46	480	500 13	
019_FL10	0510	04:16:13	04:20:52	480	500 13	
020_FL11	0511	04:22:44	04:27:35	480	500 13	
021_FL12	0512	04:28:57	04:33:57	480	500 13	
022_FL13	0513	04:36:09	04:41:19	480	500 13	
023_FL14	0514	04:42:59	04:48:33	480	500 13	
024_FL15	0515	04:50:45	04:56:17	480	500 13	
025_FL16	0516	04:58:03	05:03:46	480	500 13	
026_FL17	0517	05:06:03	05:12:07	480	500 13	
027_FL18	0518	05:14:16	05:20:41	480	500 13	
028_FL19	0519	05:52:18	05:52:24	480	500 13	

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL05, QC62	DATE:	10 January 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Malekula, TongaVanuatu	CLOUDS:	Clouds @ 3000ft
BASE STATION:	PPP	WIND:	15-20 kts @ 170
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-05
ENGINE START:	2:25	ENGINE OFF:	6:05
TAKEOFF:	2:42	LANDING:	5:58
		ENGINE TIME:	03:40
		AIR TIME	03:16

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		05:54:37				DS: QC62_20220110_055437
000_FL1	6201	05:54:37	05:56:14	480	500 13	
		05:58:00				Landing
		06:00:00				Static close out at Port Vila airport

PROJECT NAME: 2021-81155 SPC TongaVanuatu
LOCATION / AREA: Tonga and Vanuatu / BL05, QC61, QC62
AIRCRAFT: ZK-XLF
SYSTEM: Hawkeye 4X

BASE AIRPORT: Port Vila (VLI)
DATE: 10 January 2022
PILOT: Jabin L.
OPERATOR: Brent H.

MISSION ID: Malekula, TongaVanuatu
BASE STATION: PPP
LIDAR DRIVE: HE4X-06

CLOUDS: Clouds @ 1600ft
WIND: 5-10 kts @ 150
RCD DRIVE: RCD-04

ENGINE START: 20:40 **ENGINE OFF:** 0:32 **ENGINE TIME:** 03:52
TAKEOFF: 21:03 **LANDING:** 0:24 **AIR TIME:** 03:21

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		21:03:00				Takeoff
		21:06:39				DS: QC61_20220110_210639
000_FL1	6101	21:06:39	21:08:28	480	500 13	
		21:32:27				DS: BL05_20220110_213227
000_FL20	0520	21:32:27	21:39:11	480	500 13	BAD: Rain
001_FL34	0534	21:40:38	21:45:55	480	500 13	
002_FL33	0533	21:47:50	21:53:05	480	500 13	
003_FL32	0532	21:54:47	22:00:26	480	500 13	
004_FL31	0531	22:01:54	22:07:28	480	500 13	
005_FL30	0530	22:09:24	22:15:05	480	500 13	
006_FL29	0529	22:16:38	22:22:31	480	500 13	
007_FL28	0528	22:24:29	22:30:28	480	500 13	
008_FL27	0527	22:32:15	22:38:06	480	500 13	
009_FL26	0526	22:40:04	22:46:33	480	500 13	
010_FL25	0525	22:47:56	22:54:14	480	500 13	
011_FL24	0524	22:56:09	23:02:34	480	500 13	
012_FL23	0523	23:04:14	23:10:36	480	500 13	
013_FL22	0522	23:12:37	23:19:15	480	500 13	
014_FL21	0521	23:20:49	23:27:15	480	500 13	
015_FL20	0520	23:29:13	23:35:55	480	500 13	
016_FL19	0519	23:37:32	23:43:50	480	500 13	
017_FL45	0595	23:48:07	23:50:38	480	500 13	
		00:19:50				DS: QC62_20220111_001950
000_FL1	6201	00:19:50	00:21:37	480	500 13	
		00:24:00				Landing
		00:26:00				Static close out at Port Vila airport

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL01, BL03, BL04, QC61, QC62	DATE:	11 January 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Tanna, TongaVanuatu	CLOUDS:	Clouds @ 1500ft
BASE STATION:	PPP	WIND:	15-20 kts @ 180
LIDAR DRIVE:	HE4X-01	RCD DRIVE:	RCD-03
ENGINE START:	20:00	ENGINE OFF:	0:27
ENGINE TIME:		ENGINE TIME:	04:27
TAKEOFF:	20:19	LANDING:	0:20
AIR TIME		AIR TIME	04:01

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		20:19:00				Takeoff
		20:23:18				DS: QC61_20220111_202318
000_FL1	6101	20:23:18	20:23:42	480	500 13	BAD: Eye safe, clouds
		20:27:01				DS: QC62_20220111_202701
000_FL1	6201	20:27:01	20:28:53	480	500 13	
		21:10:23				DS: BL04_20220111_211023
000_FL17	0417	21:10:23	21:13:02	480	500 13	
001_FL16	0416	21:15:28	21:17:25	480	500 13	
002_FL15	0415	21:19:55	21:20:13	480	500 13	BAD: Eye safe, clouds
003_FL15	0415	21:23:09	21:24:34	480	500 13	
004_FL14	0414	21:27:07	21:28:43	480	500 13	
005_FL13	0413	21:28:49	21:29:05	480	500 13	BAD: Eye safe, clouds
006_FL13	0413	21:30:02	21:30:09	480	500 13	BAD: Eye safe, clouds
007_FL13	0413	21:30:54	21:32:04	480	500 13	
008_FL13	0413	21:34:40	21:36:37	480	500 13	
009_FL12	0412	21:37:58	21:39:00	480	500 13	BAD: Eye safe, clouds
010_FL11	0411	21:41:44	21:43:25	480	500 13	
011_FL12	0412	21:45:50	21:47:57	480	500 13	
012_FL10	0410	21:52:03	21:53:47	480	500 13	
013_FL9	0409	21:56:54	21:57:24	480	500 13	BAD: Eye safe, volcanic ash
014_FL8	0408	22:00:32	22:02:21	480	500 13	
015_FL7	0407	22:04:46	22:06:52	480	500 13	
016_FL6	0406	22:09:05	22:10:58	480	500 13	
017_FL9	0409	22:12:57	22:14:45	480	500 13	
018_FL5	0405	22:18:45	22:20:15	480	500 13	
019_FL3	0403	22:21:56	22:21:57	480	500 13	BAD: Eye safe, clouds
020_FL3	0403	22:24:07	22:25:54	480	500 13	
021_FL1	0401	22:27:49	22:28:51	480	500 13	
022_FL2	0402	22:30:22	22:31:36	480	500 13	
023_FL4	0404	22:34:04	22:36:40	480	500 13	
		22:38:53				DS: BL01_20220111_223853

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL01, BL03, BL04, QC61, QC62	DATE:	11 January 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Tanna, TongaVanuatu	CLOUDS:	Clouds @ 1500ft
BASE STATION:	PPP	WIND:	15-20 kts @ 180
LIDAR DRIVE:	HE4X-01	RCD DRIVE:	RCD-03
ENGINE START:	20:00	ENGINE OFF:	0:27
ENGINE TIME:			04:27
TAKEOFF:	20:19	LANDING:	0:20
AIR TIME			04:01

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
000_FL76	0173	22:38:53	22:40:06	480	500 13	
		22:43:45				DS: BL03_20220111_224345
000_FL45	0345	22:43:45	22:44:52	620	440 14	
001_FL46	0346	22:47:14	22:48:24	620	440 14	
002_FL47	0347	22:50:05	22:50:07	620	440 14	BAD: Eye safe, clouds
003_FL57	0357	22:56:01	22:56:24	620	440 14	BAD: Eye safe, clouds
004_FL50	0350	23:10:10	23:11:20	620	440 14	
005_FL49	0349	23:12:54	23:14:15	620	440 14	
006_FL48	0348	23:17:40	23:18:59	620	440 14	
007_FL47	0347	23:21:06	23:22:17	620	440 14	
008_FL51	0351	23:28:20	23:29:58	620	440 14	BAD: Low flight level, clouds
009_FL52	0352	23:30:39	23:31:40	620	440 14	BAD: Low flight level, clouds
010_FL55	0355	23:32:44	23:34:06	620	440 14	BAD: Low flight level, clouds
011_FL56	0356	23:35:50	23:37:02	620	440 14	BAD: Low flight level, clouds
012_FL57	0357	23:37:50	23:38:08	620	440 14	BAD:
		00:16:15				DS: QC62_20220112_001615
000_FL1	6201	00:16:15	00:18:09	480	500 13	
		00:20:00				Landing
		00:22:00				Static close out at Port Vila airport

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL17, BL18, BL21, BL22, BL23	DATE:	13 January 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Efate	CLOUDS:	Clouds @ 3000ft
BASE STATION:	AAMPri	WIND:	5-10 kts @ 270
LIDAR DRIVE:	HE4X-08	RCD DRIVE:	RCD-05
ENGINE START:	20:44	ENGINE OFF:	0:03
TAKEOFF:	21:28	LANDING:	23:56
		ENGINE TIME:	03:19
		AIR TIME	02:28

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		21:28:00				Takeoff
		21:32:25				DS: BL23_20220113_213225
000_FL35	2335	21:32:25	21:32:47	620	440 14	BAD:
001_FL34	2334	21:34:51	21:35:39	620	440 14	
002_FL33	2333	21:37:08	21:37:58	620	440 14	
003_FL31	2331	21:46:32	21:46:49	620	440 14	BAD:
		21:55:15				DS: BL23_20220113_215515
000_FL31	2331	21:55:15	21:56:13	620	440 14	
		21:58:14				DS: BL21_20220113_215814
000_FL77	2175	21:58:14	21:59:14	620	440 14	
001_FL73	2171	22:01:12	22:02:15	620	440 14	
002_FL72	2170	22:03:48	22:04:57	620	440 14	
003_FL76	2174	22:07:14	22:08:06	620	440 14	
004_FL74	2172	22:09:43	22:10:37	620	440 14	
005_FL75	2173	22:12:39	22:13:33	620	440 14	
006_FL78	2176	22:15:55	22:16:43	620	440 14	
007_FL87	2180	22:20:21	22:21:03	620	440 14	
		22:30:11				DS: BL22_20220113_223011
000_FL33	2233	22:30:11	22:30:13	620	440 14	BAD:
001_FL32	2232	22:31:15	22:32:17	620	440 14	
002_FL31	2231	22:34:05	22:35:17	620	440 14	
003_FL33	2233	22:37:30	22:38:05	620	440 14	BAD:
		22:44:13				DS: BL23_20220113_224413
000_FL32	2332	22:44:13	22:45:20	620	440 14	
		22:50:31				DS: BL18_20220113_225031
000_FL93	1886	22:50:31	22:51:45	480	500 13	
		22:59:23				DS: BL17_20220113_225923
000_FL41	1740	22:59:23	23:00:43	480	500 13	
		23:05:52				DS: BL23_20220113_230552
000_FL35	2335	23:05:52	23:06:50	620	440 14	
		23:11:42				DS: BL21_20220113_231142

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL17, BL18, BL21, BL22, BL23	DATE:	13 January 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Efate	CLOUDS:	Clouds @ 3000ft
BASE STATION:	AAMPri	WIND:	5-10 kts @ 270
LIDAR DRIVE:	HE4X-08	RCD DRIVE:	RCD-05
ENGINE START:	20:44	ENGINE OFF:	0:03
TAKEOFF:	21:28	LANDING:	23:56
		ENGINE TIME:	03:19
		AIR TIME	02:28

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
000_FL81	2179	23:11:42	23:12:19	620	440 14	
		23:18:34				DS: BL22_20220113_231834
000_FL33	2233	23:18:34	23:19:48	620	440 14	BAD:
001_FL33	2233	23:21:49	23:22:43	620	440 14	
		23:26:44				DS: BL21_20220113_232644
000_FL79	2177	23:26:44	23:27:41	620	440 14	BAD:
001_FL88	2181	23:30:07	23:30:54	620	440 14	
002_FL79	2177	23:37:42	23:38:25	620	440 14	BAD:
		23:45:50				DS: BL21_20220113_234550
000_FL79	2177	23:45:50	23:47:18	620	440 14	
		23:56:00				Landing
		23:58:00				Static close out at Port Vila airport

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL03, BL21, QC61, QC62	DATE:	16 January 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.

MISSION ID:	Efate, Tanna, TongaVanuatu	CLOUDS:	Clouds @ 3000ft
BASE STATION:	AAMPri and PPP	WIND:	5-10 kts @
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-04

ENGINE START:	19:17	ENGINE OFF:	22:10	ENGINE TIME:	02:53
TAKEOFF:	19:42	LANDING:	22:04	AIR TIME	02:22

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		19:42:00				Takeoff
		19:50:14				DS: BL21_20220116_195014
000_FL80	2178	19:50:14	19:50:47	620	440 14	
001_FL80	2178	19:52:50	19:53:27	620	440 14	
		20:00:08				DS: QC61_20220116_200008
000_FL1	6101	20:00:08	20:01:50	480	500 13	
		20:04:00				AAMPri base check-out
		20:44:02				DS: BL03_20220116_204402
000_FL57	0357	20:44:02	20:45:55	620	440 14	
001_FL56	0356	20:47:01	20:48:11	620	440 14	
002_FL55	0355	20:49:55	20:51:21	620	440 14	
003_FL54	0354	20:53:00	20:54:38	620	440 14	
004_FL53	0353	20:56:26	20:57:48	620	440 14	
005_FL51	0351	20:59:50	21:01:24	620	440 14	
006_FL52	0352	21:03:55	21:04:53	620	440 14	
		21:44:55				DS: QC62_20220116_214455
000_FL1	6201	21:44:55	21:46:42	480	500 13	
		21:49:00				AAMPri base check-in
		22:04:00				Landing
		22:06:00				Static close out at Port Vila airport

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL08, BL12, QC61, QC62	DATE:	21 January 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Malekula, TongaVanuatu	CLOUDS:	Clouds @ 3000ft
BASE STATION:	PPP	WIND:	5 kts @ 10
LIDAR DRIVE:	HE4X-01	RCD DRIVE:	RCD-01
ENGINE START:	23:36	ENGINE OFF:	5:36
TAKEOFF:	0:07	LANDING:	5:27
		ENGINE TIME:	06:00
		AIR TIME	05:20

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		00:07:00				Takeoff
		00:12:20				DS: QC61_20220121_001220
000_FL1	6101	00:12:20	00:14:05	480	500 13	
		00:39:28				DS: BL12_20220121_003928
000_FL1	1201	00:39:28	00:40:58	620	440 14	
001_FL2	1202	00:43:43	00:45:36	620	440 14	
002_FL3	1203	00:47:36	00:49:28	620	440 14	
003_FL4	1204	00:51:25	00:53:14	620	440 14	
004_FL5	1205	00:58:57	01:00:52	620	440 14	
005_FL6	1206	01:02:24	01:03:48	620	440 14	
006_FL7	1207	01:07:13	01:08:25	620	440 14	
007_FL8	1208	01:09:58	01:11:12	620	440 14	
008_FL19	1219	01:11:34	01:12:11	620	440 14	BAD: False start
009_FL9	1209	01:13:09	01:14:17	620	440 14	
010_FL10	1210	01:15:49	01:17:01	620	440 14	
011_FL12	1212	01:19:09	01:20:24	620	440 14	
012_FL11	1211	01:22:21	01:23:39	620	440 14	
013_FL13	1213	01:25:17	01:26:17	620	440 14	
014_FL14	1214	01:27:26	01:28:51	620	440 14	
015_FL15	1215	01:30:24	01:31:30	620	440 14	
016_FL16	1216	01:33:02	01:34:03	620	440 14	
017_FL17	1217	01:35:49	01:37:11	620	440 14	
018_FL18	1218	01:38:19	01:39:21	620	440 14	
019_FL19	1219	01:41:01	01:42:04	620	440 14	
020_FL20	1220	01:43:51	01:44:49	620	440 14	
021_FL21	1221	01:46:38	01:47:33	620	440 14	
022_FL22	1222	01:49:12	01:50:09	620	440 14	
023_FL23	1223	01:52:12	01:53:03	620	440 14	
024_FL24	1224	01:54:12	01:55:38	620	440 14	
025_FL25	1225	01:57:48	01:59:08	620	440 14	
026_FL26	1226	02:00:38	02:01:44	620	440 14	

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL08, BL12, QC61, QC62	DATE:	21 January 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Malekula, TongaVanuatu	CLOUDS:	Clouds @ 3000ft
BASE STATION:	PPP	WIND:	5 kts @ 10
LIDAR DRIVE:	HE4X-01	RCD DRIVE:	RCD-01
ENGINE START:	23:36	ENGINE OFF:	5:36
TAKEOFF:	0:07	LANDING:	5:27
		ENGINE TIME:	06:00
		AIR TIME	05:20

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
027_FL27	1227	02:04:12	02:05:06	620	440 14	
028_FL28	1228	02:07:16	02:08:12	620	440 14	
029_FL29	1229	02:08:48	02:10:10	620	440 14	
030_FL44	1244	02:11:25	02:13:05	620	440 14	
031_FL43	1243	02:15:09	02:16:22	620	440 14	
032_FL42	1242	02:16:33	02:17:55	620	440 14	
033_FL41	1241	02:18:36	02:19:53	620	440 14	
034_FL40	1240	02:21:00	02:22:04	620	440 14	
035_FL50	1250	02:24:15	02:25:36	620	440 14	
036_FL30	1230	02:27:51	02:29:24	620	440 14	
037_FL35	1235	02:29:47	02:31:52	620	440 14	
038_FL31	1231	02:33:53	02:36:00	620	440 14	
039_FL34	1234	02:37:53	02:39:28	620	440 14	
040_FL36	1236	02:39:55	02:40:52	620	440 14	
041_FL49	1249	02:42:58	02:44:12	620	440 14	
042_FL32	1232	02:45:43	02:46:47	620	440 14	
043_FL39	1239	02:48:37	02:49:38	620	440 14	
044_FL37	1237	02:51:51	02:53:35	620	440 14	
045_FL33	1233	02:55:15	02:56:16	620	440 14	
046_FL48	1248	02:57:20	02:58:23	620	440 14	
047_FL47	1247	02:59:52	03:00:53	620	440 14	
		03:17:32				DS: BL12_20220121_031732
000_FL46	1246	03:17:32	03:18:48	620	440 14	
001_FL45	1245	03:20:09	03:21:24	620	440 14	
002_FL52	1252	03:23:14	03:24:13	620	440 14	
003_FL56	1256	03:25:14	03:26:24	620	440 14	
004_FL38	1238	03:29:25	03:30:40	620	440 14	
005_FL53	1253	03:31:53	03:32:48	620	440 14	
006_FL51	1251	03:34:44	03:36:08	620	440 14	
007_FL55	1255	03:38:17	03:39:44	620	440 14	
008_FL54	1254	03:41:40	03:42:58	620	440 14	

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL08, BL12, QC61, QC62	DATE:	21 January 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Malekula, TongaVanuatu	CLOUDS:	Clouds @ 3000ft
BASE STATION:	PPP	WIND:	5 kts @ 10
LIDAR DRIVE:	HE4X-01	RCD DRIVE:	RCD-01
ENGINE START:	23:36	ENGINE OFF:	5:36
TAKEOFF:	0:07	LANDING:	5:27
		ENGINE TIME:	06:00
		AIR TIME	05:20

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
009_FL57	1257	03:44:33	03:45:34	620	440 14	
010_FL58	1258	03:47:10	03:48:09	620	440 14	
011_FL59	1259	03:49:59	03:52:20	620	440 14	
012_FL60	1260	03:53:48	03:55:05	620	440 14	
013_FL61	1261	03:56:31	03:58:01	620	440 14	
014_FL62	1262	03:59:44	04:01:15	620	440 14	
015_FL63	1263	04:02:43	04:03:59	620	440 14	
016_FL64	1264	04:05:42	04:07:03	620	440 14	
017_FL65	1265	04:09:17	04:10:22	620	440 14	
018_FL66	1266	04:12:18	04:13:44	620	440 14	
019_FL68	1268	04:15:51	04:16:55	620	440 14	
020_FL72	1272	04:17:15	04:18:22	620	440 14	
021_FL71	1271	04:20:26	04:21:29	620	440 14	
022_FL67	1267	04:23:24	04:24:30	620	440 14	
023_FL69	1269	04:27:10	04:28:36	620	440 14	
024_FL70	1270	04:31:02	04:32:16	620	440 14	
		04:34:22				DS: BL08_20220121_043422
000_FL59	0857	04:34:22	04:35:23	480	500 13	
001_FL58	0856	04:37:16	04:38:19	480	500 13	
002_FL57	0855	04:38:48	04:39:51	480	500 13	
003_FL56	0854	04:42:00	04:43:19	480	500 13	
		05:19:17				DS: QC62_20220121_051917
000_FL1	6201	05:19:17	05:19:42	480	500 13	BAD: Eye safety
001_FL1	6201	05:22:38	05:24:15	480	500 13	
		05:27:00				Landing
		05:29:00				Static close out at Port Vila airport

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / 400m, 500m, 800m, BL04, BL2C	DATE:	21 January 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Efate, Tanna, TongaVanuatu	CLOUDS:	Clouds @ 3500ft
BASE STATION:	AAMPri and PPP	WIND:	5-10 kts @ 310
LIDAR DRIVE:	HE4X-01	RCD DRIVE:	RCD-05
ENGINE START:	20:48	ENGINE OFF:	1:02
TAKEOFF:	21:07	LANDING:	0:54
		ENGINE TIME:	04:14
		AIR TIME	03:47

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		21:07:00				Takeoff
		21:14:03				DS: BL21_20220121_211403
000_FL91	2184	21:14:03	21:14:59	620	440 14	
001_FL90	2183	21:15:46	21:16:52	620	440 14	
002_FL89	2182	21:18:51	21:19:52	620	440 14	
		21:24:07				DS: BL22_20220121_212407
000_FL34	2234	21:24:07	21:25:10	620	440 14	
001_FL35	2235	21:27:18	21:28:21	620	440 14	
002_FL36	2236	21:29:53	21:30:14	620	440 14	BAD:
		21:36:18				DS: BL20_20220121_213618
000_FL2	2040	21:36:18	21:37:40	480	500 13	
001_FL1	2041	21:39:33	21:41:06	480	500 13	
		21:47:00				Check-out AAMPri
		21:50:00				DS: QC62_20220121_215000
000_FL1	6201	21:50:00	21:52:02	480	500 13	
		22:30:29				DS: BL04_20220121_223029
000_FL5	0405	22:30:29	22:32:13	480	500 13	
001_FL6	0406	22:34:33	22:36:31	480	500 13	
002_FL7	0407	22:37:59	22:40:12	480	500 13	
003_FL5	0405	22:41:50	22:43:26	480	500 13	
		23:20:03				DS: QC61_20220121_232003
000_FL1	6101	23:20:03	23:22:01	480	500 13	
		23:24:00				Check-in AAMPri
		23:28:27				DS: 800m_20220121_232827
000_FL1	6381	23:28:27	23:30:35	800	330 30	
001_FL2	6382	23:32:21	23:34:31	800	330 30	
002_FL3	6383	23:37:01	23:38:56	800	330 30	
003_FL4	6384	23:40:55	23:42:45	800	330 30	
004_FL5	6385	23:44:44	23:46:38	800	330 30	
005_FL6	6386	23:50:07	23:51:58	800	330 30	
		23:54:22				DS: 500m_20220121_235422

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / 400m, 500m, 800m, BL04, BL2C	DATE:	21 January 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Efate, Tanna, TongaVanuatu	CLOUDS:	Clouds @ 3500ft
BASE STATION:	AAMPri and PPP	WIND:	5-10 kts @ 310
LIDAR DRIVE:	HE4X-01	RCD DRIVE:	RCD-05
ENGINE START:	20:48	ENGINE OFF:	1:02
TAKEOFF:	21:07	LANDING:	0:54
		ENGINE TIME:	04:14
		AIR TIME	03:47

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR		REMARKS
000_FL1	6351	23:54:22	23:56:14	490	500	13	
001_FL2	6352	23:58:06	23:59:57	490	500	13	
002_FL3	6353	00:02:08	00:03:45	490	500	13	
003_FL4	6354	00:05:58	00:07:29	490	500	13	
004_FL5	6355	00:09:10	00:10:46	490	500	13	
005_FL6	6356	00:14:16	00:15:47	490	500	13	
		00:18:19					DS: 400m_20220122_001819
000_FL1	6341	00:18:19	00:20:11	400	300	10	
001_FL2	6342	00:22:11	00:24:06	400	300	10	
002_FL3	6343	00:26:35	00:28:01	400	300	10	
003_FL4	6344	00:29:47	00:31:14	400	300	10	
004_FL5	6345	00:33:17	00:34:45	400	300	10	
005_FL6	6346	00:38:06	00:39:30	400	300	10	
		00:45:10					DS: BL22_20220122_004510
000_FL36	2236	00:45:10	00:46:05	620	440	14	
		00:54:00					Landing
		00:56:00					Static close out at Port Vila airport

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL05, BL07, BL08, BL09, BL10, E	DATE:	22 January 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Malekula, TongaVanuatu	CLOUDS:	Clouds @ 2500ft
BASE STATION:	PPP	WIND:	5-10 kts @ 220
LIDAR DRIVE:	HE4X-06	RCD DRIVE:	RCD-05
ENGINE START:	1:41	ENGINE OFF:	6:39
TAKEOFF:	1:58	LANDING:	6:32
		ENGINE TIME:	04:58
		AIR TIME	04:34

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		01:58:00				Takeoff
		02:35:14				DS: BL13_20220122_023514
000_FL1	1301	02:35:14	02:36:50	620	440 14	
001_FL2	1302	02:38:43	02:39:40	620	440 14	
002_FL3	1303	02:41:05	02:42:12	620	440 14	
003_FL4	1304	02:43:52	02:44:53	620	440 14	
004_FL5	1305	02:47:07	02:48:37	620	440 14	
005_FL6	1306	02:50:40	02:51:49	620	440 14	
006_FL7	1307	02:53:45	02:54:56	620	440 14	
007_FL8	1308	02:57:25	02:58:41	620	440 14	
008_FL9	1309	03:00:35	03:01:40	620	440 14	
009_FL23	1323	03:04:22	03:05:37	620	440 14	
010_FL10	1310	03:08:38	03:09:55	620	440 14	
011_FL11	1311	03:13:12	03:14:09	620	440 14	
012_FL12	1312	03:16:04	03:17:13	620	440 14	
013_FL13	1313	03:19:06	03:20:31	620	440 14	
014_FL16	1316	03:26:11	03:27:06	620	440 14	
015_FL18	1318	03:29:14	03:30:16	620	440 14	
016_FL17	1317	03:32:39	03:33:42	620	440 14	
017_FL19	1319	03:35:16	03:36:17	620	440 14	
018_FL20	1320	03:39:05	03:40:29	620	440 14	
019_FL21	1321	03:41:09	03:42:40	620	440 14	
020_FL22	1322	03:44:57	03:46:34	620	440 14	
		03:48:34				DS: BL10_20220122_034834
000_FL72	1069	03:48:34	03:49:49	480	500 13	
001_FL73	1070	03:52:11	03:53:26	480	500 13	
002_FL76	1073	03:55:28	03:57:16	480	500 13	
003_FL75	1072	03:59:25	04:01:07	480	500 13	
004_FL74	1071	04:03:26	04:04:35	480	500 13	
005_FL71	1068	04:07:02	04:07:54	480	500 13	
006_FL70	1067	04:09:47	04:10:38	480	500 13	

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL05, BL07, BL08, BL09, BL10, E	DATE:	22 January 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Malekula, TongaVanuatu	CLOUDS:	Clouds @ 2500ft
BASE STATION:	PPP	WIND:	5-10 kts @ 220
LIDAR DRIVE:	HE4X-06	RCD DRIVE:	RCD-05
ENGINE START:	1:41	ENGINE OFF:	6:39
TAKEOFF:	1:58	LANDING:	6:32
		ENGINE TIME:	04:58
		AIR TIME	04:34

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		04:16:32				DS: BL13_20220122_041632
000_FL14	1314	04:16:32	04:17:30	620	440 14	
001_FL15	1315	04:19:53	04:21:15	620	440 14	
		04:25:34				DS: BL09_20220122_042534
000_FL55	0953	04:25:34	04:26:51	480	500 13	
001_FL54	0952	04:28:37	04:30:01	480	500 13	
002_FL53	0951	04:31:31	04:32:30	480	500 13	
		04:38:12				DS: BL08_20220122_043812
000_FL55	0853	04:38:12	04:39:21	480	500 13	
001_FL54	0852	04:40:54	04:42:22	480	500 13	
002_FL53	0851	04:44:21	04:45:47	480	500 13	
003_FL52	0850	04:47:46	04:49:16	480	500 13	
004_FL51	0849	04:51:18	04:52:32	480	500 13	
		04:54:52				DS: BL07_20220122_045452
000_FL23	0723	04:54:52	05:01:25	480	500 13	
001_FL53	0752	05:03:45	05:05:20	480	500 13	
002_FL52	0751	05:07:26	05:09:08	480	500 13	
003_FL51	0750	05:11:04	05:12:45	480	500 13	
004_FL50	0749	05:14:44	05:16:35	480	500 13	
005_FL49	0748	05:18:38	05:20:29	480	500 13	
		05:24:56				DS: BL05_20220122_052456
000_FL46	0545	05:24:56	05:28:28	480	500 13	
001_FL47	0546	05:30:03	05:33:37	480	500 13	
002_FL48	0547	05:35:24	05:38:39	480	500 13	
003_FL49	0548	05:40:18	05:43:47	480	500 13	
004_FL50	0549	05:45:42	05:48:55	480	500 13	
005_FL51	0550	05:50:28	05:53:18	480	500 13	
006_FL52	0551	05:55:09	05:57:26	480	500 13	
007_FL53	0552	05:59:05	06:00:56	480	500 13	
		06:27:03				DS: QC61_20220122_062703
000_FL1	6101	06:27:03	06:28:47	480	500 13	

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL05, BL07, BL08, BL09, BL10, E	DATE:	22 January 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Malekula, TongaVanuatu	CLOUDS:	Clouds @ 2500ft
BASE STATION:	PPP	WIND:	5-10 kts @ 220
LIDAR DRIVE:	HE4X-06	RCD DRIVE:	RCD-05
ENGINE START:	1:41	ENGINE OFF:	6:39
TAKEOFF:	1:58	LANDING:	6:32
		ENGINE TIME:	04:58
		AIR TIME	04:34

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		06:32:00				Landing
		06:34:00				Static close out at Port Vila airport

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL06, BL07, BL08, BL09, BL12, E	DATE:	16 February 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Efate, Malekula, TongaVanuatu	CLOUDS:	Clouds @ 2000ft
BASE STATION:	AAMPri and PPP	WIND:	5-10 kts @ 50
LIDAR DRIVE:	HE4X-08	RCD DRIVE:	RCD-04
ENGINE START:	22:13	ENGINE OFF:	2:56
TAKEOFF:	22:27	LANDING:	2:45
		ENGINE TIME:	04:43
		AIR TIME	04:18

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		22:27:00				Takeoff
		22:32:22				DS: QC61_20220216_223222
000_FL1	6101	22:32:22	22:34:10	480	500 13	
		22:41:51				DS: BL21_20220216_224151
000_FL89	2182	22:41:51	22:42:43	620	440 14	Eye safe at end
001_FL90	2183	22:45:26	22:45:44	620	440 14	BAD: Clouds
002_FL90	2183	22:47:56	22:48:09	620	440 14	BAD: Clouds
		22:57:00				Check-out AAMPri Base
003_FL90	2183	23:24:47	23:25:00	620	440 14	BAD: Clouds
		23:28:13				DS: BL12_20220216_232813
000_FL73	1273	23:28:13	23:29:09	620	440 14	
001_FL74	1274	23:30:34	23:31:54	620	440 14	
002_FL75	1275	23:34:19	23:35:21	620	440 14	
003_FL77	1277	23:37:05	23:38:01	620	440 14	
004_FL76	1276	23:39:56	23:41:09	620	440 14	
005_FL78	1278	23:44:21	23:45:32	620	440 14	
006_FL79	1279	23:47:59	23:49:10	620	440 14	
		23:54:00				DS: BL13_20220216_235400
000_FL24	1324	23:54:00	23:55:06	620	440 14	
001_FL25	1325	00:00:39	00:01:44	620	440 14	
002_FL26	1326	00:05:41	00:06:49	620	440 14	
003_FL27	1327	00:08:19	00:09:27	620	440 14	
004_FL28	1328	00:11:04	00:12:05	620	440 14	
005_FL29	1329	00:13:35	00:14:24	620	440 14	
006_FL30	1330	00:16:12	00:16:42	620	440 14	BAD: Clouds
007_FL30	1330	00:19:51	00:20:49	620	440 14	Eye safe at end
		00:27:40				DS: BL09_20220217_002740
000_FL54	0952	00:27:40	00:29:09	480	500 13	
001_FL53	0951	00:33:43	00:34:47	480	500 13	
		00:40:17				DS: BL08_20220217_004017
000_FL57	0855	00:40:17	00:41:18	480	500 13	

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL06, BL07, BL08, BL09, BL12, E	DATE:	16 February 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Efate, Malekula, TongaVanuatu	CLOUDS:	Clouds @ 2000ft
BASE STATION:	AAMPri and PPP	WIND:	5-10 kts @ 50
LIDAR DRIVE:	HE4X-08	RCD DRIVE:	RCD-04
ENGINE START:	22:13	ENGINE OFF:	2:56
TAKEOFF:	22:27	LANDING:	2:45
		ENGINE TIME:	04:43
		AIR TIME	04:18

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		00:47:01				DS: BL07_20220217_004701
000_FL55	0754	00:47:01	00:48:02	480	500 13	
001_FL54	0753	00:50:14	00:51:47	480	500 13	
		00:56:01				DS: BL06_20220217_005601
000_FL81	0665	00:56:01	00:59:53	480	500 13	
001_FL80	0664	01:01:54	01:06:03	480	500 13	
002_FL79	0663	01:07:57	01:12:55	480	500 13	
003_FL78	0662	01:14:43	01:19:12	480	500 13	
004_FL77	0661	01:21:16	01:26:15	480	500 13	
005_FL76	0660	01:28:07	01:33:08	480	500 13	
006_FL75	0659	01:34:59	01:39:58	480	500 13	
007_FL74	0658	01:42:00	01:46:45	480	500 13	
008_FL73	0657	01:48:32	01:53:25	480	500 13	
009_FL72	0656	01:55:10	01:59:59	480	500 13	
010_FL71	0655	02:02:18	02:07:11	480	500 13	
011_FL56	0654	02:09:38	02:12:20	480	500 13	
012_FL55	0653	02:14:13	02:16:43	480	500 13	
013_FL54	0652	02:18:29	02:21:00	480	500 13	
		02:45:00				Landing
		02:46:00				Static close out at Santo airport

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL11, BL21, BL22, QC62	DATE:	17 February 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Efate, Malekula, TongaVanuatu	CLOUDS:	Clouds @ 2000ft
BASE STATION:	AAMPri and PPP	WIND:	10-15 kts @ 50
LIDAR DRIVE:	HE4X-07	RCD DRIVE:	RCD-04
ENGINE START:	3:16	ENGINE OFF:	6:48
TAKEOFF:	3:29	LANDING:	6:39
		ENGINE TIME:	03:32
		AIR TIME	03:10

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		03:29:00				Takeoff
		03:42:07				DS: BL11_20220217_034207
000_FL40	1195	03:42:07	03:45:19	480	500 13	
001_FL39	1139	03:48:18	03:50:17	480	500 13	
002_FL38	1138	03:51:48	03:54:00	480	500 13	
003_FL37	1137	03:55:54	03:58:04	480	500 13	
004_FL36	1136	03:59:35	04:03:19	480	500 13	
005_FL35	1135	04:05:21	04:09:12	480	500 13	
006_FL34	1134	04:10:51	04:14:50	480	500 13	
007_FL33	1133	04:16:58	04:20:59	480	500 13	
008_FL32	1132	04:23:21	04:27:35	480	500 13	
009_FL19	1119	04:30:00	04:35:28	480	500 13	
010_FL20	1120	04:37:18	04:42:55	480	500 13	
011_FL21	1121	04:44:46	04:50:09	480	500 13	
012_FL22	1122	04:52:03	04:57:27	480	500 13	
013_FL23	1123	04:59:24	05:04:33	480	500 13	
014_FL24	1124	05:06:35	05:11:35	480	500 13	
015_FL25	1125	05:12:52	05:17:44	480	500 13	
016_FL18	1118	05:19:45	05:22:00	480	500 13	BAD: Clouds and rain
		05:59:22				DS: BL22_20220217_055922
000_FL34	2234	05:59:22	06:00:33	620	440 14	Prior to base check-in
001_FL35	2235	06:02:17	06:03:27	620	440 14	Prior to base check-in
		06:11:00				Check-in AAMPri Base
002_FL34	2234	06:18:33	06:19:24	620	440 14	BAD: Clouds
003_FL35	2235	06:21:31	06:22:11	620	440 14	BAD: Clouds
		06:25:13				DS: BL21_20220217_062513
000_FL90	2183	06:25:13	06:25:52	620	440 14	BAD: Clouds
		06:33:55				DS: QC62_20220217_063355
000_FL1	6201	06:33:55	06:35:50	480	500 13	
		06:39:00				Landing
		06:42:00				Static close out at Port Vila airport

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL11, QC61, QC62	DATE:	17 February 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Malekula, TongaVanuatu	CLOUDS:	Clouds @ 1800ft
BASE STATION:	PPP	WIND:	10-15 kts @ 110
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-05
ENGINE START:	22:21	ENGINE OFF:	2:36
TAKEOFF:	22:36	LANDING:	2:29
		ENGINE TIME:	04:15
		AIR TIME	03:53

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		22:36:00				Takeoff
		22:41:57				DS: QC61_20220217_224157
000_FL1	6101	22:41:57	22:43:52	480	500 13	
		23:18:05				DS: BL11_20220217_231805
000_FL31	1131	23:18:05	23:22:33	480	500 13	
001_FL30	1130	23:24:26	23:29:02	480	500 13	
002_FL29	1129	23:30:29	23:35:07	480	500 13	
003_FL28	1128	23:36:30	23:41:08	480	500 13	
004_FL27	1127	23:43:01	23:47:44	480	500 13	
005_FL26	1126	23:49:34	23:54:15	480	500 13	
006_FL18	1118	23:55:48	00:01:13	480	500 13	
007_FL17	1117	00:02:45	00:08:00	480	500 13	
008_FL16	1116	00:09:27	00:14:38	480	500 13	
009_FL15	1115	00:16:07	00:21:15	480	500 13	
010_FL14	1114	00:23:27	00:28:32	480	500 13	
011_FL13	1113	00:30:07	00:35:03	480	500 13	
012_FL12	1112	00:36:51	00:41:44	480	500 13	
013_FL11	1111	00:43:33	00:48:18	480	500 13	
014_FL10	1110	00:50:14	00:54:53	480	500 13	
015_FL9	1109	00:56:19	01:00:52	480	500 13	
016_FL8	1108	01:02:34	01:07:04	480	500 13	
017_FL7	1107	01:08:28	01:12:50	480	500 13	
018_FL6	1106	01:14:46	01:18:58	480	500 13	
019_FL5	1105	01:20:50	01:24:54	480	500 13	
020_FL4	1104	01:27:21	01:31:21	480	500 13	
021_FL3	1103	01:33:35	01:37:23	480	500 13	
022_FL2	1102	01:39:17	01:42:41	480	500 13	
023_FL1	1101	01:44:10	01:47:01	480	500 13	
		02:23:47				DS: QC62_20220218_022347
000_FL1	6201	02:23:47	02:25:43	480	500 13	
		02:29:00				Landing

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL11, QC61, QC62	DATE:	17 February 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.
MISSION ID:	Malekula, TongaVanuatu	CLOUDS:	Clouds @ 1800ft
BASE STATION:	PPP	WIND:	10-15 kts @ 110
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-05
ENGINE START:	22:21	ENGINE OFF:	2:36
TAKEOFF:	22:36	LANDING:	2:29
		ENGINE TIME:	04:15
		AIR TIME	03:53

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		02:30:00				Static close out at Port Vila airport

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL11, BL14, QC61, QC62	DATE:	22 February 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.

MISSION ID:	Malekula, TongaVanuatu	CLOUDS:	Clouds @ 2300ft
BASE STATION:	PPP	WIND:	15-20 kts @ 60
LIDAR DRIVE:	HE4X-01	RCD DRIVE:	RCD-03

ENGINE START:	22:00	ENGINE OFF:	0:48	ENGINE TIME:	02:48
TAKEOFF:	22:18	LANDING:	0:38	AIR TIME	02:20

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		22:18:00				Takeoff
		22:30:04				DS: QC61_20220222_223004
000_FL1	6101	22:30:04	22:31:48	480	500 13	
		23:07:07				DS: BL14_20220222_230707
000_FL1	1401	23:07:07	23:08:03	620	440 14	BAD: Up to 200' Low
001_FL2	1402	23:13:17	23:13:48	620	440 14	BAD: Clouds
002_FL3	1403	23:16:22	23:17:20	620	440 14	
003_FL4	1404	23:19:44	23:21:06	620	440 14	
004_FL5	1405	23:23:08	23:24:04	620	440 14	
005_FL2	1402	23:26:18	23:27:38	620	440 14	
006_FL1	1401	23:30:40	23:31:43	620	440 14	
		23:33:41				DS: BL11_20220222_233341
000_FL47	1146	23:33:41	23:34:51	480	500 13	
001_FL46	1145	23:36:32	23:37:49	480	500 13	
002_FL45	1144	23:39:45	23:41:03	480	500 13	
003_FL41	1140	23:43:16	23:44:37	480	500 13	
004_FL42	1141	23:46:41	23:48:07	480	500 13	
005_FL43	1142	23:50:06	23:51:20	480	500 13	
006_FL44	1143	23:53:44	23:54:51	480	500 13	
		00:32:04				DS: QC62_20220223_003204
000_FL1	6201	00:32:04	00:34:00	480	500 13	
		00:38:00				Landing
		00:40:00				Static close out at Port Vila airport

PROJECT NAME:	2021-81155 SPC TongaVanuatu	BASE AIRPORT:	Port Vila (VLI)
LOCATION / AREA:	Tonga and Vanuatu / BL13, BL14, QC61	DATE:	4 March 2022
AIRCRAFT:	ZK-XLF	PILOT:	Jabin L.
SYSTEM:	Hawkeye 4X	OPERATOR:	Brent H.

MISSION ID:	Malekula, TongaVanuatu	CLOUDS:	Clouds @ 2500ft
BASE STATION:	PPP	WIND:	5-10 kts @ 230
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-05

ENGINE START:	21:37	ENGINE OFF:	23:55	ENGINE TIME:	02:18
TAKEOFF:	22:00	LANDING:	23:46	AIR TIME	01:46

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		22:00:00				Takeoff
		22:35:11				DS: BL13_20220304_223511
000_FL31	1331	22:35:11	22:36:11	620	440 14	
001_FL32	1332	22:41:19	22:42:25	620	440 14	
002_FL33	1333	22:45:34	22:46:42	620	440 14	
003_FL34	1334	22:48:38	22:49:49	620	440 14	
		22:55:18				DS: BL14_20220304_225518
000_FL6	1406	22:55:18	22:56:43	620	440 14	
001_FL7	1407	22:58:33	22:59:28	620	440 14	Slightly high
002_FL8	1408	23:00:55	23:02:13	620	440 14	
003_FL9	1409	23:03:47	23:04:47	620	440 14	
004_FL7	1407	23:06:49	23:07:47	620	440 14	
		23:41:15				DS: QC61_20220304_234115
000_FL1	6101	23:41:15	23:42:37	480	500 13	Eye safe at end
		23:46:00				Landing
		23:48:00				Static close out at Port Vila airport

APPENDIX B : PROCESSING LOGS

PROJECT NAME: 81155 SPC Tonga/Vanuatu
LOCATION: Tonga and Vanuatu
AIRCRAFT: Cessna Reims 406 (ZK-XLF)
SYSTEM: Hawkeye 4X

PID	Project Name	Latitude			Longitude			Height (m)	XY Datum	Z Datum	Source	Comments
		Deg	Min	Sec	Deg	Min	Sec					
	AAMPri	S17	41	59.79551	E168	19	07.74891	82.680	ITRF2014 Epoch 2021.9	ITRF2014 Epoch 2021.9	AUSPOS	Used for Calibration / Efate

PROJECT NAME: 81155 SPC Tonga/Vanuatu
LOCATION: Tonga and Vanuatu
AIRCRAFT: Cessna Reims 406 (ZK-XLF)
SYSTEM: Hawkeye 4X

Project Name	Download Airborne Data	Create IE Project Directory	Copy Data to IE Project / Raw	Run Project Wizard	Base Station			Check Base Coordinate & Datum	Check Lever Arms	Process Time Window		Process TC	Review QC Plots	Separation			Solution Status	Comments
					Station ID	Receiver Type	Antenna (ARP) Height (m)			Start Time (GPS Week Time)	End Time (GPS Week Time)			East RMS (m)	North RMS (m)	Up RMS (m)		
2021-11-23A_CH4X	DA	DA	DA	DA	AAMPri	R10	2.1	DA	DA			DA	DA	0.019	0.008	0.044	Final	
2021-11-23A_HE4X	DA	DA	DA	DA	AAMPri	R10	2.1	DA	DA			DA	DA	0.019	0.008	0.042	Final	
2021-11-26A_CH4X	DA	DA	DA	DA	AAMPri	R10	2.1	DA	DA			DA	DA	0.022	0.011	0.030	Final	
2021-11-26A_HE4X	DA	DA	DA	DA	AAMPri	R10	2.1	DA	DA			DA	DA	0.019	0.010	0.028	Final	
2021-11-28A_CH4X	DA	DA	DA	DA	AAMPri	R10	2.1	DA	DA			DA	DA	0.010	0.008	0.022	Final	Calibration Flight
2021-11-28A_HE4X	DA	DA	DA	DA	AAMPri	R10	2.1	DA	DA			DA	DA	0.008	0.011	0.020	Final	Calibration Flight
2021-11-28B_CH4X	DA	DA	DA	DA	AAMPri	R10	2.1	DA	DA	74050	79935	DA	DA	0.007	0.007	0.017	Final	Calibration Flight (Main Project)
2021-11-28B_HE4X	DA	DA	DA	DA	AAMPri	R10	2.1	DA	DA	74050	79935	DA	DA	0.007	0.009	0.028	Final	Calibration Flight (Main Project)
2021-11-28B_CH4X	DA	DA	DA	DA	AAMPri	R10	2.1	DA	DA	79730		DA	DA	0.036	0.021	0.057	Final	Efate Production (Efate Project)
2021-11-28B_HE4X	DA	DA	DA	DA	AAMPri	R10	2.1	DA	DA	79730		DA	DA	0.038	0.020	0.041	Final	Efate Production (Efate Project)
2021-11-29A_CH4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH	157582		BH	BH	0.043	0.018	0.028	Final	Efate Production
2021-11-29A_HE4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH	157582		BH	BH	0.043	0.019	0.040	Final	Efate Production
2021-11-30A_CH4X	BH	DA	DA	DA	AAMPri	R10	2.1	DA	DA	185251		DA	DA	0.008	0.009	0.035	Final	Efate Production
2021-11-30A_HE4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH	185251		DA	DA	0.009	0.007	0.030	Final	Efate Production
2021-11-30B_CH4X	BH	BH	BH	BH	PPP	N/A	N/A	N/A	BH			BH	BH	0.014	0.012	0.088	Final	Tanna Production
2021-11-30B_HE4X	BH	BH	BH	BH	PPP	N/A	N/A	N/A	BH			BH	BH	0.012	0.011	0.088	Final	Tanna Production
2021-12-01A_CH4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH	268200		BH	BH	0.010	0.012	0.020	Final	Efate Production
2021-12-01A_HE4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH	268200		BH	BH	0.010	0.012	0.021	Final	Efate Production
2021-12-03A_CH4X	DA	DA	DA	DA	PPP	N/A	N/A	N/A	DA			DA	DA	0.017	0.015	0.091	Final	Tanna Production
2021-12-03A_HE4X	DA	DA	DA	DA	PPP	N/A	N/A	N/A	DA			DA	DA	0.014	0.017	0.079	Final	Tanna Production
2021-12-04A_CH4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH	598038		BH	BH	0.022	0.014	0.500	Final	Efate Production
2021-12-04A_HE4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH	598038		BH	BH	0.026	0.015	0.047	Final	Efate Production
2021-12-04A_CH4X	BH	BH	BH	BH	PPP	N/A	N/A	N/A	BH			BH	BH	0.031	0.013	0.085	Final	Tanna Production (Tanna Project) (Elev. Mask = 20 to reduce RMS)
2021-12-04A_HE4X	BH	BH	BH	BH	PPP	N/A	N/A	N/A	BH			BH	BH	0.034	0.017	0.084	Final	Tanna Production (Tanna Project) (Elev. Mask = 20 to reduce RMS)
2021-12-05A_CH4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH	12428		BH	BH	0.009	0.005	0.028	Final	Efate Production
2021-12-05A_HE4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH	12428		BH	BH	0.009	0.005	0.027	Final	Efate Production
2021-12-07A_CH4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH			BH	BH	0.019	0.039	0.075	Final	Efate Production
2021-12-07A_HE4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH			BH	BH	0.028	0.037	0.066	Final	Efate Production
2021-12-08A_CH4X	BH	BH	BH	BH	PPP	N/A	N/A	N/A	BH	347160	361909	DA	DA	0.023	0.015	0.103	Final	Tanna Production
2021-12-08A_HE4X	BH	BH	BH	BH	PPP	N/A	N/A	N/A	BH	347160	361909	DA	DA	0.020	0.013	0.094	Final	Tanna Production
2021-12-15A_CH4X	BH	BH	BH	BH	PPP	N/A	N/A	N/A	BH			BH	BH	0.014	0.012	0.083	Final	Tanna Production
2021-12-15A_HE4X	BH	BH	BH	BH	PPP	N/A	N/A	N/A	BH			BH	BH	0.013	0.011	0.077	Final	Tanna Production
2021-12-16A_CH4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH			BH	BH	0.070	0.010	0.034	Final	Efate Production
2021-12-16A_HE4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH			BH	BH	0.005	0.007	0.038	Final	Efate Production
2021-12-17A_CH4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH			BH	BH	0.006	0.007	0.022	Final	Efate Production

PROJECT NAME: 81155 SPC Tonga/Vanuatu
LOCATION: Tonga and Vanuatu
AIRCRAFT: Cessna Reims 406 (ZK-XLF)
SYSTEM: Hawkeye 4X

Project Name	Download Airborne Data	Create IE Project Directory	Copy Data to IE Project / Raw	Run Project Wizard	Base Station			Check Base Coordinate & Datum	Check Lever Arms	Process Time Window		Process TC	Review QC Plots	Separation			Solution Status	Comments
					Station ID	Receiver Type	Antenna (ARP) Height (m)			Start Time (GPS Week Time)	End Time (GPS Week Time)			East RMS (m)	North RMS (m)	Up RMS (m)		
2021-12-17A1_HE4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH			BH	BH	0.004	0.008	0.016	Final	Efate Production (Do not use, BAD Line)
2021-12-17A2_HE4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH			BH	BH	0.005	0.005	0.022	Final	Efate Production
2021-12-17B_CH4X	BH	BH	BH	BH	PPP	N/A	N/A	N/A	BH			BH	BH	0.013	0.017	0.095	Final	Malekula Production
2021-12-17B_HE4X	BH	BH	BH	BH	PPP	N/A	N/A	N/A	BH			BH	BH	0.014	0.018	0.088	Final	Malekula Production
2021-12-19A_CH4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH			BH	BH	0.016	0.019	0.047	Final	Efate Production
2021-12-19A_HE4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH			BH	BH	0.014	0.016	0.035	Final	Efate Production
2021-12-20A_CH4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.016	0.011	0.038	Final	Malekula Production
2021-12-20A_HE4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.022	0.013	0.050	Final	Malekula Production
2021-12-21A_CH4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.025	0.013	0.098	Final	Malekula Production
2021-12-21A_HE4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.019	0.017	0.101	Final	Malekula Production
2021-12-21B_CH4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH			BH	BH	0.006	0.003	0.017	Final	Efate Production
2021-12-21B_HE4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH			BH	BH	0.005	0.003	0.015	Final	Efate Production
2021-12-26A_CH4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.029	0.016	0.063	Final	Malekula Production (Elev. Mask = 20 to reduce RMS)
2021-12-26A_HE4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.028	0.016	0.061	Final	Malekula Production (Elev. Mask = 20 to reduce RMS)
2021-12-27A_CH4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.008	0.009	0.053	Final	Malekula Production
2021-12-27A_HE4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.008	0.008	0.053	Final	Malekula Production
2021-12-27B_CH4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.021	0.013	0.120	Final	Malekula Production
2021-12-27B_HE4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.020	0.014	0.119	Final	Malekula Production
2021-12-28A_CH4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.026	0.009	0.029	Final	Efate Production
2021-12-28A_HE4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.024	0.009	0.027	Final	Efate Production
2021-12-29A_CH4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.034	0.023	0.140	Final	Malekula Production
2021-12-29A_HE4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.021	0.022	0.096	Final	Malekula Production
2022-01-03A_CH4X	BH	BH	BH	BH	PPP	N/A	N/A	N/A	BH			BH	BH	0.049	0.031	0.081	Final	Tanna Production (Elev. Mask = 20 to reduce RMS)
2022-01-03A_HE4X	BH	BH	BH	BH	PPP	N/A	N/A	N/A	BH			BH	BH	0.043	0.020	0.074	Final	Tanna Production (Elev. Mask = 20 to reduce RMS)
2022-01-03A_CH4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH	161713		BH	BH	0.006	0.005	0.016	Final	Efate Production (Efate Project)
2022-01-03A_HE4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH	161713		BH	BH	0.007	0.004	0.015	Final	Efate Production (Efate Project)
2022-01-04A_CH4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH			BH	BH	0.027	0.026	0.046	Final	Efate Production
2022-01-04A_HE4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH			BH	BH	0.030	0.026	0.050	Final	Efate Production
2022-01-04B_CH4X																		Not processed due to no usable data being collected
2022-01-04B_HE4X																		Not processed due to no usable data being collected
2022-01-06A_CH4X	AB	AB	AB	AB	PPP	N/A	N/A	N/A	AB			AB	AB	0.017	0.017	0.040	Final	Tanna Production (Tanna Project)
2022-01-06A_HE4X	AB	AB	AB	AB	PPP	N/A	N/A	N/A	AB			AB	AB	0.018	0.017	0.036	Final	Tanna Production (Tanna Project)
2022-01-07A_CH4X	AB	AB	AB	AB	PPP	N/A	N/A	N/A	AB			AB	AB	0.020	0.014	0.052	Final	Malekula Production (Elev. Mask = 20 to reduce RMS)
2022-01-07A_HE4X	AB	AB	AB	AB	PPP	N/A	N/A	N/A	AB			AB	AB	0.022	0.016	0.055	Final	Malekula Production (Elev. Mask = 20 to reduce RMS)
2022-01-07B_CH4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH			BH	BH	0.022	0.012	0.019	Final	Efate Reflights

PROJECT NAME: 81155 SPC Tonga/Vanuatu
LOCATION: Tonga and Vanuatu
AIRCRAFT: Cessna Reims 406 (ZK-XLF)
SYSTEM: Hawkeye 4X

Project Name	Download Airborne Data	Create IE Project Directory	Copy Data to IE Project / Raw	Run Project Wizard	Base Station			Check Base Coordinate & Datum	Check Lever Arms	Process Time Window		Process TC	Review QC Plots	Separation			Solution Status	Comments
					Station ID	Receiver Type	Antenna (ARP) Height (m)			Start Time (GPS Week Time)	End Time (GPS Week Time)			East RMS (m)	North RMS (m)	Up RMS (m)		
2022-01-07B_HE4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH			BH	BH	0.011	0.006	0.014	Final	Efate Reflights
2022-01-08A_CH4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH			BH	BH	0.004	0.006	0.019	Final	Efate Reflights
2022-01-08A_HE4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH			BH	BH	0.004	0.007	0.014	Final	Efate Reflights
2022-01-09A_CH4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.019	0.011	0.033	Final	Malekula Production (non-AR)
2022-01-09A_HE4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.018	0.009	0.028	Final	Malekula Production (non-AR)
2022-01-10A_CH4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.023	0.020	0.050	Final	Malekula Production (non-AR)
2022-01-10A_HE4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.025	0.012	0.049	Final	Malekula Production (non-AR)
2022-01-10B_CH4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.007	0.007	0.039	Final	Malekula Production (non-AR)
2022-01-10B_HE4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.011	0.007	0.038	Final	Malekula Production (non-AR)
2022-01-11A_CH4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.017	0.017	0.088	Final	Malekula Production (non-AR)
2022-01-11A_HE4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.017	0.013	0.093	Final	Malekula Production (non-AR)
2022-01-13A_CH4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH			BH	BH	0.018	0.013	0.026	Final	Efate Reflights
2022-01-13A_HE4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH			BH	BH	0.009	0.017	0.016	Final	Efate Reflights
2022-01-16A_CH4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH			BH	BH	0.020	0.012	0.065	Final	Efate Reflights
2022-01-16A_HE4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH			BH	BH	0.030	0.022	0.072	Final	Efate Reflights
2022-01-16A_CH4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.022	0.011	0.049	Final	Tanna Reflights (Tanna Project) (Elev. Mask = 20 to reduce RMS)
2022-01-16A_HE4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.019	0.017	0.057	Final	Tanna Reflights (Tanna Project) (Elev. Mask = 20 to reduce RMS)
2022-01-21A_CH4X	DA	DA	DA	DA	PPP	N/A	N/A	DA	DA			DA	DA	0.016	0.016	0.069	Final	Malekula Reflights
2022-01-21A_HE4X	DA	DA	DA	DA	PPP	N/A	N/A	DA	DA			DA	DA	0.011	0.018	0.089	Final	Malekula Reflights
2022-01-21B_CH4X					AAMPri	R10	2.1										Final	Efate Reflights and Calibration (Only 17 minutes of base station data) NOT PROCESSED
2022-01-21B_HE4X					AAMPri	R10	2.1										Final	Efate Reflights and Calibration (Only 17 minutes of base station data) NOT PROCESSED
2022-01-21B_CH4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.008	0.007	0.035	Final	Tanna Reflights
2022-01-21B_HE4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.007	0.006	0.035	Final	Tanna Reflights
2022-01-22A_CH4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.016	0.007	0.055	Final	Malekula Reflights
2022-01-22A_HE4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.016	0.006	0.035	Final	Malekula Reflights
2022-02-16A_CH4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH		342589	BH	BH	0.013	0.018	0.023	Final	Efate Reflights (Efate Project)
2022-02-16A_HE4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH		342589	BH	BH	0.013	0.018	0.012	Final	Efate Reflights (Efate Project)
2022-02-16A_CH4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.010	0.008	0.109	Final	Malekula Reflights
2022-02-16A_HE4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.019	0.020	0.166	Final	Malekula Reflights
2022-02-17A_CH4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH		366803	BH	BH	0.020	0.005	0.014	Final	Efate Reflights (Efate Project)
2022-02-17A_HE4X	BH	BH	BH	BH	AAMPri	R10	2.1	BH	BH		366803	BH	BH	0.021	0.005	0.012	Final	Efate Reflights (Efate Project)
2022-02-17A_CH4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.058	0.034	0.145	Final	Malekula Production
2022-02-17A_HE4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.060	0.033	0.155	Final	Malekula Production
2022-02-17B_CH4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.022	0.011	0.105	Final	Malekula Production
2022-02-17B_HE4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.032	0.018	0.120	Final	Malekula Production

PROJECT NAME: 81155 SPC Tonga/Vanuatu
LOCATION: Tonga and Vanuatu
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Project Name	Download Airborne Data	Create IE Project Directory	Copy Data to IE Project / Raw	Run Project Wizard	Base Station			Check Base Coordinate & Datum	Check Lever Arms	Process Time Window		Process TC	Review QC Plots	Separation			Solution Status	Comments
					Station ID	Receiver Type	Antenna (ARP) Height (m)			Start Time (GPS Week Time)	End Time (GPS Week Time)			East RMS (m)	North RMS (m)	Up RMS (m)		
2022-02-22A_CH4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.032	0.015	0.069	Final	Malekula Reflights
2022-02-22A_HE4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.026	0.015	0.069	Final	Malekula Reflights
2022-03-04A_CH4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.032	0.011	0.067	Final	Malekula Reflights
2022-03-04A_HE4X	BH	BH	BH	BH	PPP	N/A	N/A	BH	BH			BH	BH	0.018	0.012	0.099	Final	Malekula Reflights